Algebraic Number Theory 2[1]

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[1] Lectured by Prof. Dr. Eugen Hellmann at Universität Münster in Summer 2025

These are my personal notes from Prof. Dr. Eugen Hellmann's Algebraic Number Theory 2 lecture course in Summer 2025 in Münster. The course covered adèles, quaternion algebras over local and global fields, properties of adèlic points of certain algebraic groups over number fields, and an introduction to automorphic forms. The course (and hence these notes) assumed knowledge from Algebraic Number Theory 1, as well as some understanding of algebraic geometry and modular forms.

These notes generally reflect the notation, content, and style of the lectures, but I have also made some of my own stylistic choices. Of note, a number of proofs are my own, either because they were omitted from the lecture, assigned as exercises, or I simply missed them at the time. In the beginning discussion on places, I have as much as possible tried not to distinguish between finite and infinite places. A few theorems (such as Hilbert reciprocity) were proven only for $F = \mathbf{Q}$ in the lecture, as we did not assume knowledge of class field theory. I have kept these proofs and added my own for the general case. All errors should be assumed to be mine.

These notes use the `tufte-handout` LaTeXdocument class based on the design of Edward Tufte's books. My thanks to Eugen for lecturing the course and to Connor Lane for many insightful discussions which have far increased my understanding of algebraic number theory.

Contents

1 Adèles

1.1 Places of a number field

Let F be a number field. Recall the following equivalent characterizations of equivalence of absolute values:

Lemma 1.1 (Equivalence of absolute values). *Let $|\cdot|_1, |\cdot|_2$ be non-trivial absolute values on F. The following are equivalent:*

1. *There exists*[2] $\alpha \in \mathbf{R}_{>0}$ *such that* $|\cdot|_1 = |\cdot|_2^{\alpha}$;
2. *There exist* $C_1, C_2 \in \mathbf{R}_{>0}$ *such that* $C_1 |\cdot|_1 \leq |\cdot|_2 \leq C_2 |\cdot|_1$;
3. $|\cdot|_1$ *and* $|\cdot|_2$ *induce the same topologies on* F.

[2] For a given absolute value $|\cdot|$ and $\alpha \in \mathbf{R}_{>0}$, $|\cdot|^{\alpha}$ need not be an absolute value. If $|\cdot|$ is non-archimedean, however, then $|\cdot|^{\alpha}$ will be an absolute value for all α.

Proof. Left as an exercise.

Definition 1.2 (Places of a number field). *Let F be a number field.*

- *A* place *of F is an equivalence class of non-trivial absolute values on F.*
- *A place v is called* infinite *or* archimedean *if it corresponds to an equivalence class of archimedean absolute values. In this case, v is the equivalence class of*[3] $|\cdot|_\infty \circ \tau$ *for some embedding*[4] $\tau \colon F \hookrightarrow \mathbf{C}$. *We say that v is a* real place *if τ factors through $\mathbf{R}$ (equiv. if the completion F_v is isomorphic to $\mathbf{R}$). and v is a* complex place *otherwise (equiv. if $F_v \simeq \mathbf{C}$).*
- *A place v is called* finite *or* non-archimedean *if it corresponds to the equivalence class of a non-archimedean absolute value. v is called a* p-adic place *if v contains an extension of $|\cdot|_p$ for some (necessarily unique) prime p.*

[3] Here and forevermore $|\cdot|_\infty$ denotes the standard absolute value on $\mathbf{C}$.
[4] This is a result of Ostrowski [Lan02, Cor. XII.2.4] which follows from the Gelfand–Mazur theorem [Rud73, Thm. 10.14].

We establish some notation which will be used for the rest of the lecture:

- We write $V = V_F$ for the set of places of F;
- For $v \in V_F$, we write F_v for the completion of F with respect to some choice[5] of absolute value in v;
- We write $V_\infty \subset V_F$ for the set of infinite places;
- We write $V_f \subset V_F$ for the set of finite places;
- We write $V_p \subset V_F$ for the set of p-adic places.

[5] Note that equivalent absolute values yield completions which are isomorphic as topological fields.

Lemma 1.3. *Let F be a number field. Let p be a (possibly infinite) place of $\mathbf{Q}$ and let $|\cdot|_p$ denote the unique extension of the absolute value on $\mathbf{Q}_p$ to $\overline{\mathbf{Q}_p}$. Assume we have two embeddings $\tau_1, \tau_2 \colon F \hookrightarrow \overline{\mathbf{Q}_p}$. Then $|\cdot|_1 := |\cdot|_p \circ \tau_1$ is equivalent to $|\cdot|_2 := |\cdot|_p \circ \tau_2$ if and only if $\tau_1 = \sigma \circ \tau_2$ for some $\sigma \in \mathrm{Gal}_{\mathbf{Q}_p} = \mathrm{Gal}(\overline{\mathbf{Q}_p}/\mathbf{Q}_p)$.*

Proof. Recall that $|\cdot|_p$ is extended to $\overline{\mathbf{Q}_p}$ by setting

$$|x|_p := \sqrt[n]{\left|\mathrm{Nm}_{\mathbf{Q}_p(x)/\mathbf{Q}_p}(x)\right|_p} \tag{1}$$

where $n = [\mathbf{Q}_p(x) \colon \mathbf{Q}_p]$. From this it is clear that $\mathrm{Gal}_{\mathbf{Q}_p}$ is isometric with respect to $|\cdot|_p$, and hence if $\tau_1 = \sigma \circ \tau_2$ for some $\sigma \in \mathrm{Gal}_{\mathbf{Q}_p}$ we have $|\cdot|_1 = |\cdot|_2$.

Conversely, suppose $|\cdot|_1$ is equivalent to $|\cdot|_2$ i.e. there exists $\alpha \in \mathbf{R}_{>0}$ such that $|\cdot|_1 = |\cdot|_2^\alpha$. As $\mathbf{Q}$ is fixed by all embeddings $F \hookrightarrow \overline{\mathbf{Q}_p}$, we have $|\cdot|_p = |\cdot|_p^\alpha$ on $\mathbf{Q}$. Hence $\alpha = 1$ and $|\cdot|_1 = |\cdot|_2$. Now $\tau_1 \circ \tau_2^{-1}$ is an isomorphism $\tau_2(F) \xrightarrow{\sim} \tau_1(F)$ which is furthermore isometric. Thus we may extend the isomorphism to the closures of $\tau_2(F)$ and $\tau_1(F)$ in $\overline{\mathbf{Q}_p}$; call these $\widehat{F}_2$ and $\widehat{F}_1$, respectively. As $\mathbf{Q}$ is preserved by τ_1, τ_2, we find via continuity that $\mathbf{Q}_p$ is preserved by the isomorphism $\widehat{F}_2 \xrightarrow{\sim} \widehat{F}_1$. As these are both finite extensions of $\mathbf{Q}_p$, we can extend the isomorphism to an automorphism $\sigma \in \mathrm{Gal}_{\mathbf{Q}_p}$. Finally, restricting to F, we find $\tau_1 \circ \tau_2^{-1} = \sigma$ or $\tau_1 = \sigma \circ \tau_2$.

Corollary 1.4. *For $K/\mathbf{Q}_p$ a local field, we have a bijection between $\left\{v \in V_p \colon F_v \simeq K\right\}$ and $\left\{\tau \colon F \hookrightarrow \overline{\mathbf{Q}_p} \colon \tau(F)\mathbf{Q}_p \simeq K\right\} / \mathrm{Gal}_{\mathbf{Q}_p}$. In particular, real places are in bijection with $\{\tau \colon F \hookrightarrow \mathbf{R}\}$ and complex places are in bijection with $\{\tau \colon F \hookrightarrow \mathbf{C} \colon \tau(F) \not\subset \mathbf{R}\} / \mathrm{Gal}_{\mathbf{R}}$.*

Lemma 1.5. *Let v be a finite place of a number field F. Then there exists a unique prime ideal $\mathfrak{p} \subset \mathcal{O}_F$ such that v is the equivalence class containing*

$$|\cdot|_{\mathfrak{p}} := q^{-\mathrm{val}_{\mathfrak{p}}(\cdot)}$$

where $q = |\kappa(\mathfrak{p})| = |\mathcal{O}_F/\mathfrak{p}|$ and $\mathrm{val}_{\mathfrak{p}} \colon F \to \mathbf{Z} \cup \{\infty\}$ is the normalized discrete valuation corresponding to $\mathfrak{p}$.

In particular, we have a bijection

$$V_p \xrightarrow{\sim} \{\mathfrak{p} \in \mathrm{MaxSpec}(\mathcal{O}_F) \colon \mathfrak{p} \mid p\}.$$

Proof. The uniqueness follows from

$$\mathfrak{p} = \left\{x \in \mathcal{O}_F \colon |x|_{\mathfrak{p}} < 1\right\} \tag{2}$$

and the fact that, for any $\alpha \in \mathbf{R}_{>0}$, $|x|_{\mathfrak{p}} < 1$ if and only if $|x|_{\mathfrak{p}}^\alpha < 1$.

Now for existence, let $|\cdot| \in v$ and set

$$\mathfrak{p} := \{x \in \mathcal{O}_F \colon |x| < 1\}.$$

Evidently $\mathfrak{p}$ is a subgroup of $\mathcal{O}_F$; to establish that $\mathfrak{p}$ is a proper ideal, it suffices to show that $|x| \le 1$ for all $x \in \mathcal{O}_F$. Assuming these, it is clear that $\mathfrak{p}$ is prime since $x \notin \mathfrak{p}$ if and only if $|x| = 1$ and $|\cdot|$ is multiplicative. To establish that $|x| \le 1$ for all $x \in \mathcal{O}_F$, we first note that this is clearly true for all $x \in \mathbf{Z}$. Now assume $x \in \mathcal{O}_F$ is such that $|x| > 1$ and let

$$f(y) = y^n + a_{n-1}y^{n-1} + \cdots + a_1 y + a_0 \in \mathbf{Z}[y]$$

be the minimal polynomial of x. We have $|a_i| \leq 1$ for all i and hence

$$1 < |x|^n = \left|a_{n-1}x^{n-1} + \cdots + a_0\right| \leq \max_{0 \leq i \leq n-1} |a_i| \, |x|^i \leq \max_{0 \leq i \leq n-1} |x|^i = |x|^{n-1}.$$

But then $|x| \leq 1$, which contradicts the assumption that $|x| > 1$ so $|x| \leq 1$ for all $x \in \mathcal{O}_F$.

Now we want to establish that $|\cdot|_{\mathfrak{p}} \in v$ or that $|\cdot| = |\cdot|_{\mathfrak{p}}^{\alpha}$ for some $\alpha \in \mathbf{R}_{>0}$. To that end, let $\pi \in \mathfrak{p} \setminus \mathfrak{p}^2$ and set

$$\alpha := -\log_q |\pi|.$$

For $x \in F^\times$, let $n = \mathrm{val}_{\mathfrak{p}}(x)$ so that

$$\left|\frac{x}{\pi^n}\right|_{\mathfrak{p}} = 1.$$

Then x/π^n can be written as y/z where $y, z \in \mathcal{O}_K$ and $z \notin \mathfrak{p}$. But then $y \notin \mathfrak{p}$ and hence

$$1 = \left|\frac{y}{z}\right| = \left|\frac{x}{\pi^n}\right| = |x| \, |\pi|^{-n} = |x| \, q^{n\alpha}.$$

Thus $|x| = q^{-\mathrm{val}_{\mathfrak{p}}(x)\alpha} = |x|_{\mathfrak{p}}^{\alpha}$ for all $x \in F^\times$.

Proposition 1.6. *Let A be a Dedekind domain and $\mathfrak{p} \subset A$ a prime ideal. Let $K = \mathrm{Frac}(A)$, L/K a finite separable extension, and B the integral closure of A in L. Then there is a canonical isomorphism*

$$\widehat{A_{\mathfrak{p}}} \otimes_A B \xrightarrow{\sim} \prod_{\mathfrak{q} | \mathfrak{p}} \widehat{B_{\mathfrak{q}}}$$

given by $a \otimes b \mapsto (ab)_{\mathfrak{q}}$, where $\mathfrak{q}$ ranges over the prime ideals of B lying above $\mathfrak{p}$ and $\widehat{A_{\mathfrak{p}}} = \varprojlim A/\mathfrak{p}^n$ is the $\mathfrak{p}$-adic completion of A.

Proof. Note that B is a finite projective A-module of rank $n = [L \colon K]$, so $\widehat{A_{\mathfrak{p}}} \otimes_A B$ is a finite projective[6] $\widehat{A_{\mathfrak{p}}}$-module of rank n. Moreover, $\widehat{B_{\mathfrak{q}}}$ is a finite projective $\widehat{A_{\mathfrak{p}}}$-module of rank $e_{\mathfrak{q}} f_{\mathfrak{q}}$, and the formula

$$n = \sum_{\mathfrak{q} | \mathfrak{p}} e_{\mathfrak{q}} f_{\mathfrak{q}} \tag{3}$$

shows that the product

$$\prod_{\mathfrak{q} | \mathfrak{p}} \widehat{B_{\mathfrak{q}}}$$

is a finite projective $\widehat{A_{\mathfrak{p}}}$-module of rank n. Hence to show the given map is an isomorphism, it suffices to show surjectivity. Per Nakayama's lemma, it suffices to check that the map is surjective after tensoring with $\widehat{A_{\mathfrak{p}}}/\mathfrak{p}\widehat{A_{\mathfrak{p}}}$. But then the multiplication map gives isomorphisms

$$\widehat{A_{\mathfrak{p}}}/\mathfrak{p}\widehat{A_{\mathfrak{p}}} \otimes_{A_{\mathfrak{p}}} (\widehat{A}_{\mathfrak{p}} \otimes_A B) \simeq \widehat{A_{\mathfrak{p}}}/\mathfrak{p}\widehat{A_{\mathfrak{p}}} \otimes_A B \simeq A/\mathfrak{p}A \otimes_A B \simeq B/\mathfrak{p}B$$

[6] actually free, by Nakayama's lemma or more generally Kaplansky's theorem

and

$$B/\mathfrak{p}B \simeq \prod_{\mathfrak{q}|\mathfrak{p}} B/\mathfrak{q}^{e_\mathfrak{q}}B = \prod_{\mathfrak{q}|\mathfrak{p}} \widehat{B_\mathfrak{q}}/\mathfrak{q}^{e_\mathfrak{q}}\widehat{B_\mathfrak{q}} \simeq \prod_{\mathfrak{q}|\mathfrak{p}} \widehat{B_\mathfrak{q}}/\mathfrak{p}\widehat{B_\mathfrak{q}} \simeq \prod_{\mathfrak{q}|\mathfrak{p}} \widehat{B_\mathfrak{q}} \otimes_{\widehat{A_\mathfrak{p}}} \widehat{A_\mathfrak{p}}/\mathfrak{p}\widehat{A_\mathfrak{p}} \simeq \left(\prod_{\mathfrak{q}|\mathfrak{p}} \widehat{B_\mathfrak{q}}\right) \otimes_{\widehat{A_\mathfrak{p}}} \widehat{A_\mathfrak{p}}/\mathfrak{p}\widehat{A_\mathfrak{p}}.$$

Proposition 1.7. *Let F be a number field.*

1. *For p a finite rational prime, there is a canonical isomorphism*

$$F \otimes_{\mathbf{Q}} \mathbf{Q}_p \xrightarrow{\sim} \prod_{\mathfrak{p}|p} F_\mathfrak{p}$$

given by $a \otimes b \mapsto (ab)_\mathfrak{p}$. Here $F_\mathfrak{p} = \mathrm{Frac}(\mathcal{O}_{F_\mathfrak{p}})$ and $\mathcal{O}_{F_\mathfrak{p}} := \varprojlim \mathcal{O}_F/\mathfrak{p}^n$ is the $\mathfrak{p}$-adic completion of $\mathcal{O}_F$.

2. *For $p \in V_{\mathbf{Q}}$ (not necessarily finite), there is a canonical isomorphism*

$$F \otimes_{\mathbf{Q}} \mathbf{Q}_p \xrightarrow{\sim} \prod_{v \in V_p} F_v$$

given by $a \otimes b \mapsto (ab)_v$.

Proof.

1. By proposition 1.6, we have

$$\mathcal{O}_F \otimes_{\mathbf{Z}} \mathbf{Z}_p \simeq \prod_{\mathfrak{p}|p} \mathcal{O}_{F_\mathfrak{p}}.$$

Rationalising, we obtain

$$F \otimes_{\mathbf{Q}} \mathbf{Q}_p \simeq \prod_{\mathfrak{p}|p} F_\mathfrak{p}.$$

2. We note first that

$$F \otimes_{\mathbf{Q}} \mathbf{Q}_p = (F \otimes_{\mathbf{Q}} \overline{\mathbf{Q}_p})^{\mathrm{Gal}_{\mathbf{Q}_p}},$$

where $\mathrm{Gal}_{\mathbf{Q}_p}$ acts on the tensor product by acting on the right factor and stabilising F. Now the multiplication map gives us an isomorphism

$$F \otimes_{\mathbf{Q}} \overline{\mathbf{Q}_p} \simeq \prod_{\tau\colon F \hookrightarrow \overline{\mathbf{Q}_p}} \overline{\mathbf{Q}_p}.$$

Under this isomorphism, the action of $\mathrm{Gal}_{\mathbf{Q}_p}$ translates to

$$(\tau(x)z)_\tau \mapsto (\tau(x)\sigma(z))_\tau = \sigma(\sigma^{-1}\tau(x)z)_\tau = \sigma(\tau(x)z)_{\sigma^{-1}\tau}.$$

Hence the action of the Galois group first acts on the embeddings $F \hookrightarrow \overline{\mathbf{Q}_p}$, then acts on the product as normal. Within each Galois orbit of embeddings, it is clear that choosing one coordinate of a Galois-invariant element determines the other coordinates. Hence we have

$$F \otimes_{\mathbf{Q}} \mathbf{Q}_p \simeq \prod_{[\tau]} \tau(F)\mathbf{Q}_p$$

where the product runs over Galois orbits of embeddings $\tau \colon F \hookrightarrow \overline{Q_p}$. Now by corollary 1.4, the Galois orbits of embeddings with $\tau(F)\mathbf{Q}_p \simeq K$ is in bijection with the places with $F_v \simeq K$. Hence we have the desired isomorphism

$$F \otimes_{\mathbf{Q}} \mathbf{Q}_p \simeq \prod_{v \in V_p} F_v.$$

Lemma 1.8. *Let $x \in F$ and $p \in V_{\mathbf{Q}}$; then*

$$\mathrm{Nm}_{F/\mathbf{Q}}(x) = \prod_{v \in V_p} \mathrm{Nm}_{F_v/\mathbf{Q}_p}(x).$$

Proof. This follows from the previous proposition along with the fact that the norm can be defined as

$$\mathrm{Nm}_{L/K}(x) = \prod_{\tau \colon L \hookrightarrow C} \tau(x)$$

for any algebraically closed field C into which L embeds.

Given a place $v \in V_F$, we will generally use the following choice of absolute value $|\cdot|_v$ to represent v:

If v is real: there exists an embedding $\tau \colon F \hookrightarrow \mathbf{R}$ such that $|\cdot|_\infty \circ \tau \in v$. Take this to be $|\cdot|_v$.

If v is complex: there exists an embedding $\tau \colon F \hookrightarrow \mathbf{C}$ such that $\tau(F) \not\subset \mathbf{R}$ and $|\cdot|_\infty \circ \tau \in v$. Set[7]

$$|\cdot|_v := (|\cdot|_\infty \circ \tau)(|\cdot|_\infty \circ \bar{\tau}) = (|\cdot|_\infty \circ \tau)^2.$$

[7] This is technically *not* an absolute value: it violates the triangle inequality. Nevertheless it will be convenient to use this in place of complex absolute values.

If v is finite: by lemma 1.5 there exists a unique prime $\mathfrak{p} \subset \mathcal{O}_F$ such that $q^{-\mathrm{val}_{\mathfrak{p}}(\cdot)} \in v$, where $q = |\mathcal{O}_F/\mathfrak{p}|$ and $\mathrm{val}_{\mathfrak{p}} \colon F \to \mathbf{Z} \cup \{\infty\}$ is the normalized discrete valuation. Take[8] $|\cdot|_v := q^{-\mathrm{val}_{\mathfrak{p}}(\cdot)}$.

[8] Note that for $p \in V_{\mathbf{Q},f}$ such that $\mathfrak{p} \mid p$, the chosen absolute value here is not necessarily an extension of the p-adic absolute value to F.

Lemma 1.9 (Product Formula). *For F any number field,*

1. *If $p \in V_{\mathbf{Q}}$, $v \in V_p$, and $x \in F_v$ we have*

$$|x|_v = \left|\mathrm{Nm}_{F_v/\mathbf{Q}_p}(x)\right|_p.$$

2. *For any $x \in F^\times$, $|x|_v = 1$ for all but finitely many $v \in V_F$ and*

$$\prod_{v \in V_F} |x|_v = 1.$$

Proof.

1. is clear for $p = \infty$ so assume p is finite. Recall that the extension of the p-adic absolute value to F_v is given by

$$|x|'_{F_v} = \sqrt[n]{\left|\mathrm{Nm}_{F_v/\mathbf{Q}_p}(x)\right|_p}.$$

Moreover, if $\mathrm{val}_v \colon F \to \mathbf{Z} \cup \{\infty\}$ is the *normalized* discrete valuation on F and $q = |\mathcal{O}_{F_v}/\pi|$ is the residue cardinality, we can also write

$$|x|'_{F_v} = p^{-\frac{1}{e}\mathrm{val}_v(x)}$$

where e is the ramification index of $F_v/\mathbf{Q}_p$. Now

$$\left|\mathrm{Nm}_{F_v/\mathbf{Q}_p}(x)\right|_p = p^{-\frac{n}{e}\mathrm{val}_v(x)} = p^{-d\,\mathrm{val}_v(x)}$$

where d is the inertial index of $F_v/\mathbf{Q}_p$. But then $q = p^d = |\mathcal{O}_{F_v}/\pi| = |\mathcal{O}_F/\mathfrak{p}|$ is the residue cardinality of the prime $\mathfrak{p} \subset \mathcal{O}_F$ which defines $|\cdot|_v$, so

$$\left|\mathrm{Nm}_{F_v/\mathbf{Q}_p}(x)\right|_p = q^{-\mathrm{val}_{\mathfrak{p}}(x)} = |x|_v\,.$$

2. By 1. and lemma 1.8, we have

$$\prod_{v\in V_F}|x|_v = \prod_{p\in V_{\mathbf{Q}}}\prod_{v\in V_p}|x|_v = \prod_{p\in V_{\mathbf{Q}}}\prod_{v\in V_p}\left|\mathrm{Nm}_{F_v/\mathbf{Q}_p}(x)\right|_p = \prod_{p\in V_{\mathbf{Q}}}\left|\mathrm{Nm}_{F/\mathbf{Q}}(x)\right|_p.$$

Hence it suffices to prove the product formula for $F = \mathbf{Q}$. To that end, let $x \in \mathbf{Q}^\times$ and write

$$x = \varepsilon\prod_i p_i^{a_i}$$

where $\varepsilon \in \{\pm 1\}$, p_i are distinct finite rational primes, and $a_i \in \mathbf{Z}$. Then $|x|_{p_i} = p_i^{-a_i}$ and $|x|_p = 1$ for finite primes $p \neq p_i$. Moreover

$$|x|_\infty = \prod_i p_i^{a_i},$$

so

$$\prod_{p\in V_{\mathbf{Q}}}|x|_p = \left(\prod_i p_i^{-a_i}\right)\left(\prod_i p_i^{a_i}\right) = 1.$$

1.2 Restricted Products and the Adèles

Definition 1.10. *Let $(X_i)_{i\in I}$ be a family of topological spaces and for all but finitely many*[9] *$i \in I$, let $\mathcal{O}_i \subset X_i$ be a fixed open subspace. The restricted product of the X_i with respect to the open subspaces $\mathcal{O}_i$ is defined as*[10]

$$\prod_{i\in I}{}' X_i := \left\{ (x_i)_{i\in I} \in \prod_{i\in I} X_i \colon x_i \in \mathcal{O}_i \text{ for all but finitely many } i \in I \right\}.$$

The restricted product is equipped with the topology[11] *generated by sets of the form $\prod_{i\in I} U_i$, where $U_i \subset X_i$ is open and $U_i = \mathcal{O}_i$ for all but finitely many $i \in I$.*

[9] For the purpose of defining the restricted product, we could just as well insist that $\mathcal{O}_i$ be defined for all i: for the finitely many i such that $\mathcal{O}_i$ is undefined, set $\mathcal{O}_i = X_i$.

[10] Some authors also use the notation

$$\prod_{i\in I}{}'(X_i, \mathcal{O}_i)$$

to keep track of the open subspaces $\mathcal{O}_i$.

[11] This is also the final topology induced from writing

$$\prod_{i\in I}{}' X_i = \varinjlim_S \left(\prod_{i\in S} X_i\right) \times \left(\prod_{i\notin S} \mathcal{O}_i\right)$$

where the colimit runs over finite subsets $S \subset I$ such that $\mathcal{O}_i$ is defined for all $i \notin S$. Here the products in the RHS are all given the product topology.

Remark 1.11. ***Caution:*** *The topology on the restricted product is* not *the subspace topology from the product topology on $\prod_{i\in I} X_i$. For example, consider the sequence of rational finite adèles $x_p \in \mathbf{A}_{\mathbf{Q}}^{\infty}$, which consist of $\frac{1}{p}$ in the place p and 0 in all other places. Then in the product topology, $x_p \to 0$ as convergence is pointwise. However, in $\mathbf{A}_{\mathbf{Q}}^{\infty}$ the basic open neighbourhoods of 0 are of the form*

$$\prod_{i=1}^{n} U_i \times \prod \mathbf{Z}_p,$$

and clearly no such neighbourhood can contain infinitely many x_p.

Lemma 1.12. *Assume each X_i is locally compact, and that each $\mathcal{O}_i \subset X_i$ is compact. Then the restricted product $\prod_{i\in I}{}' X_i$ is locally compact.*

Proof. For any finite $S \subset I$, the product

$$X_S = \prod_{i\in S} X_i \times \prod_{i\notin S} \mathcal{O}_i$$

is locally compact as Tychonoff's theorem gives compactness of the product $\prod_{i\notin S} \mathcal{O}_i$ and finite products of locally compact spaces are locally compact. Then since

$$\prod_{i\in I}{}' X_i = \varinjlim_S X_S$$

it follows that the restricted product is locally compact.

Lemma 1.13. *If each X_i is a topological group (resp. a topological ring) and $\mathcal{O}_i$ is a subgroup (resp. subring) then the restricted product $\prod_{i\in I}{}' X_i$ is a topological group (resp. topological ring).*

Proof. The product topology makes each X_S a topological group (resp. topological ring), and the direct limit endows $\prod_{i\in I}{}' X_i$ with the natural continuous operations.

Definition 1.14 (Adèles and Idèles)**.** *Let F be a number field.*

1. *The adèles over F are*

$$\mathbf{A}_F := \prod_{v\in V_F}{}' F_v$$

where the restricted product is taken with respect to $\mathcal{O}_{F_v} \subset F_v$ for v finite. Since each F_v is a locally compact Hausdorff topological field and each $\mathcal{O}_{F_v} \subset F_v$ (for v finite) is a compact open subring, $\mathbf{A}_F$ is a locally compact Hausdorff topological ring.

2. *The ring of finite adèles is*

$$\mathbf{A}_{F,f} = \mathbf{A}_F^\infty := \prod_{v \in V_f}{}' F_v \simeq \mathbf{A}_F / \prod_{v \in V_\infty} F_v.$$

More generally, for $S \subset V_F$ finite, we write

$$\mathbf{A}_{F,S} := \prod_{v \in S} F_v$$

for the adèles at S and

$$\mathbf{A}_F^S := \prod_{v \in V_F \setminus S}{}' F_v \simeq \mathbf{A}_F / \mathbf{A}_{F,S} \tag{4}$$

for the adèles away from S.

3. *The group of idèles of F is*[12]

$$\mathbf{A}_F^\times := \prod_{v \in V_F}{}' F_v^\times$$

where the restricted product is taken with respect to $\mathcal{O}_{F_v}^\times \subset F_v^\times$ for v finite. Since each $F_v^\times$ is a locally compact Hausdorff topological group and $\mathcal{O}_{F_v}^\times \subset F_v^\times$ is a compact open subgroup, $\mathbf{A}_F^\times$ is a locally compact Hausdorff topological group. Just as with the adèles, we define $\mathbf{A}_{F,f}^\times$, $\mathbf{A}_{F,S}^\times$, $A_F^{S\times}$.

[12] Some authors prefer $\mathbf{I}_F$ or $\mathcal{I}_F$ to denote the idèles, to signify that the idèles do not carry the subspace topology from $\mathbf{A}_F$. However, there is little risk of confusion since we will have no need for the subspace topology beyond mentioning how poorly behaved it is.

Remark 1.15. *$\mathbf{A}_F^\times$ is the group of units of $\mathbf{A}_F$, but the restricted product topology is* not *the subspace topology from $\mathbf{A}_F$. There are two issues with the subspace topology:*

It is not locally compact: The subspace topology from $\mathbf{A}_F$ corresponds to taking the restricted product $\prod_{v \in V_F}{}' F_v^\times$ with respect to the subsets $\mathcal{O}_{F_v} \cap F_v^\times \subset F_v^\times$, which are not compact.

It is not a topological group: The subsets $\mathcal{O}_{F_v} \cap F_v^\times \subset F_v^\times$ are not subgroups, so the resulting topology will not make $\mathbf{A}_F^\times$ a topological group. More specifically, we can see that inversion is not continuous by considering the sequence of idèles $x_p \in \mathbf{A}_{\mathbf{Q}}^\times$ which have p in the place p and 1 in all other places. This converges to 1 in the subspace topology, but the sequence x_p^{-1} fails to converge.

The topology on $\mathbf{A}_F^\times$ agrees[13] *with the subspace topology from the closed embedding $\mathbf{A}_F^\times \hookrightarrow \mathbf{A}_F \times \mathbf{A}_F \colon x \mapsto (x, x^{-1})$.*

[13] This comes from the closed embedding $\mathbf{G}_m \hookrightarrow \mathbf{A}^2$ recognizing $\mathbf{G}_m = \mathrm{GL}_1$ as a finite-type affine scheme. This will become relevant later when we topologize the adèlic points of linear algebraic groups over F.

Exercise 1. *$\mathbf{A}_F^n$ with the product topology is isomorphic as a topological ring to the restricted product $\prod_{v \in V_F}{}' F_v^n$ with respect to $\mathcal{O}_{F_v}^n \subset F_v^n$.*

Lemma 1.16. *Let $Y_i \subset X_i$ be a subspace for each $i \in I$; then the subspace topology on $\prod_{i \in I}{}' Y_i \subset \prod_{i \in I}{}' X_i$ is the restricted product topology $\prod_{i \in I}{}' Y_i$ taken with respect to $Y_i \cap \mathcal{O}_i \subset Y_i$. Moreover, if each X_i is locally compact, each $\mathcal{O}_i$ is compact, and each $Y_i \subset X_i$ is closed, then the restricted product $\prod_{i \in I}{}' Y_i$ is a locally compact closed subspace of $\prod_{i \in I}{}' X_i$.*

Proof. The first part is clear, and the second follows from the fact that closed subspaces of (locally) compact spaces are (locally) compact.

Example 1.17. *Let F be a number field.*

a. *For E/F a finite extension, we have an isomorphism of topological rings*

$$\mathbf{A}_E \simeq \mathbf{A}_F \otimes_F E.$$

b. *There is an isomorphism of groups*

$$\mathrm{GL}_n(\mathbf{A}_F) \simeq \prod_{v \in V_F}{}' \mathrm{GL}_n(F_v)$$

where the RHS is taken with respect to $\mathrm{GL}_n(\mathcal{O}_{F_v}) \subset \mathrm{GL}_n(F_v)$. This makes $\mathrm{GL}_n(\mathbf{A}_F)$ a locally compact Hausdorff topological group. As noted in the $n = 1$ case, $\mathrm{GL}_n(\mathbf{A}_F)$ does not carry the subspace topology from the matrix ring $M_n(\mathbf{A}_F)$. Instead it embeds as a closed subspace of $M_n(\mathbf{A}_F) \times \mathbf{A}_F$ by $g \mapsto (g, \det g^{-1})$ and as a closed subspace of $M_n(\mathbf{A}_F) \times M_n(\mathbf{A}_F)$ by $g \mapsto (g, g^{-1})$.

Let G/F be a linear algebraic group i.e. a smooth[14] affine group scheme of finite type over F. We want to topologize $G(\mathbf{A}_F)$ such that it is a locally compact topological group. The natural solution is to write

$$G(\mathbf{A}_F) := \prod_{v \in V_F}{}' G(F_v)$$

where the restricted product is taken with respect to $G(\mathcal{O}_{F_v}) \subset G(F_v)$. **Problem:** $G(\mathcal{O}_{F_v})$ is not necessarily defined!

[14] Since F is a number field and has characteristic 0, the smoothness assumption is superfluous.

Definition 1.18 (Smooth models). *Let $A = S^{-1}\mathcal{O}_F$ for some finite set $S \subset \mathcal{O}_F$. A smooth model of G over A is a smooth group scheme $\mathcal{G}/A$ such that*

$$\mathcal{G} \times_{\operatorname{Spec} A} \operatorname{Spec} F \simeq G.$$

Lemma 1.19. *Let G be a linear algebraic group over F.*

1. *There exists a finite set of primes S such that G admits a smooth model over $S^{-1}\mathcal{O}_F$.*

2. *Let S_1, S_2 be finite sets of primes and let $\mathcal{G}_1, \mathcal{G}_2$ be smooth models of G over $S_1^{-1}\mathcal{O}_F$ and $S_2^{-1}\mathcal{O}_F$, respectively. Then there exists a finite set of primes $S \supset S_1 \cup S_2$ such that*

$$\mathcal{G}_1 \times_{\operatorname{Spec} S_1^{-1}\mathcal{O}_F} \operatorname{Spec} S^{-1}\mathcal{O}_F \simeq \mathcal{G}_2 \times_{\operatorname{Spec} S_2^{-1}\mathcal{O}_F} \operatorname{Spec} S^{-1}\mathcal{O}_F.$$

Proof.

1. Pick a closed embedding $G \hookrightarrow \mathrm{GL}_n$ defined by polynomials $P_1, \ldots, P_m \in F[T_{11}, \ldots, T_{nn}, \det^{-1}]$. There exists a finite set of primes S_1 such that all $P_i \in S_1^{-1}\mathcal{O}_F[T_{ij}, \det^{-1}]$, so we get a model $\mathcal{G}_1 \hookrightarrow \mathrm{GL}_n$ over $S_1^{-1}\mathcal{O}_F$. The smoothness of G implies that the ideal generated by the P_i and their Jacobian

determinants generate all of $F[T_{ij}, \det^{-1}]$; this implies that there is a finite set of primes $S_2 \supset S_1$ such that

$$\mathcal{G}_2 := \mathcal{G}_1 \times_{\operatorname{Spec} S_1^{-1}\mathcal{O}_F} \operatorname{Spec} S_2^{-1}\mathcal{O}_F$$

is smooth.

2. Let f be an isomorphism of F-group schemes

$$\mathcal{G}_1 \times_{\operatorname{Spec} S_1^{-1}\mathcal{O}_F} \operatorname{Spec} F \xrightarrow{\sim} \mathcal{G}_2 \times_{\operatorname{Spec} S_2^{-1}\mathcal{O}_F} \operatorname{Spec} F.$$

Since $F = \varinjlim_S S^{-1}\mathcal{O}_F$, [GW20, Cor. 10.64] implies that there exists a finite $S_3 \supset S_1 \cup S_2$ and an isomorphism

$$f_3\colon \mathcal{G}_1 \times_{\operatorname{Spec} S_1^{-1}\mathcal{O}_F} \operatorname{Spec} S_3^{-1}\mathcal{O}_F \xrightarrow{\sim} \mathcal{G}_2 \times_{\operatorname{Spec} S_2^{-1}\mathcal{O}_F} \operatorname{Spec} S_3^{-1}\mathcal{O}_F$$

such that

$$f = f_3 \times_{\operatorname{Spec} S_3^{-1}\mathcal{O}_F} \operatorname{Spec} F.$$

Definition 1.20. *Let G/F be a linear algebraic group. The adèlic points of G are the topological group*

$$G(\mathbf{A}_F) := \prod_{v \in V_F}{}' G(F_v)$$

taken with respect to $\mathcal{G}(\mathcal{O}_{F_v}) \subset \mathcal{G}(F_v) = G(F_v)$ for some smooth model $\mathcal{G}$ of G over $S^{-1}\mathcal{O}_F$.

Remark 1.21.

- *$G(F_v)$ is a closed subgroup of $\mathrm{GL}_n(F_v)$ and hence locally compact.*
- *$\mathcal{G}$ as constructed in lemma 1.19 1. is a closed subgroup scheme of GL_n, hence $\mathcal{G}(\mathcal{O}_{F_v}) \subset \mathrm{GL}_n(\mathcal{O}_{F_v})$ is a closed subgroup and thus compact.*
- *By the previous two points, the restricted product topology on $G(\mathbf{A}_F)$ makes $G(\mathbf{A}_F)$ a locally compact Hausdorff topological group.*
- *By lemma 1.19 2., the definition of $G(\mathbf{A}_F)$ does not depend on the choice of smooth model $\mathcal{G}$.*
- *$G(\mathbf{A}_F)$ as defined above agrees as a group with the $\mathbf{A}_F$-points $G(\mathbf{A}_F) = \operatorname{Hom}_{\mathsf{Sch}/F}(\operatorname{Spec} \mathbf{A}_F, G)$.*

Now, let's return to studying $\mathbf{A}_F$. Note that the embeddings $F \hookrightarrow F_v$ induce a diagonal embedding $F \hookrightarrow \mathbf{A}_F$ since any $x \in F$ is in $\mathcal{O}_{F_v}$ for all but finitely many $v \in V_F$.

The embeddings $F \hookrightarrow F_v$ by definition have dense image, so we might expect something similar to happen for the embedding $F \hookrightarrow \mathbf{A}_F$, or even for $G(F) \hookrightarrow G(\mathbf{A}_F)$ when G is a linear algebraic group. What happens is something quite interesting.

Theorem 1.22. *Let F be a number field.*

1. *The diagonal embedding $F \hookrightarrow \mathbf{A}_F$ identifies F as a closed discrete subgroup of $\mathbf{A}_F$.*
2. *The quotient $\mathbf{A}_F/F$ is compact.*

Proof. We have a commutative diagram of topological groups

$$\begin{tikzcd} \mathbf{A}_F \arrow[r, "\sim"] & \mathbf{A}_{\mathbf{Q}}^{[F:\,\mathbf{Q}]} \\ F \arrow[u, hook] \arrow[r, "\sim"] & \mathbf{Q}^{[F:\,\mathbf{Q}]} \arrow[u, hook] \end{tikzcd}$$

where the top and bottom morphisms are isomorphisms of topological groups. Hence $\mathbf{A}_F/F \simeq (\mathbf{A}_{\mathbf{Q}}/\mathbf{Q})^{[F:\,\mathbf{Q}]}$ and it suffices to prove both claims for $F = \mathbf{Q}$.

1. To show $\mathbf{Q}$ is discrete it suffices to find an open neighbourhood $U \subset \mathbf{A}_{\mathbf{Q}}$ of 0 such that $U \cap \mathbf{Q} = \{0\}$, as the homogeneity of topological groups means the same is true for all $x \in \mathbf{Q}$. Then $\mathbf{Q}$ is also closed since discrete subgroups of Hausdorff groups are always closed. Define

   $$U := (-1,1) \times \prod_{p \in V_{\mathbf{Q},f}} \mathbf{Z}_p.$$

   If $x \in \mathbf{Q} \cap U$, then looking at the finite places we see that $x \in \mathbf{Q} \cap \mathbf{Z}_p$ for all p. Hence $x \in \mathbf{Z}$, and by construction $x \in \mathbf{Z} \cap (-1,1) = \{0\}$. Thus $x = 0$.

2. We will prove that

   $$\left[-\frac{1}{2}, \frac{1}{2}\right] \times \prod_{p \in V_{\mathbf{Q},f}} \mathbf{Z}_p$$

   surjects onto $\mathbf{A}_{\mathbf{Q}}/\mathbf{Q}$. As this is compact by Tychonoff's theorem, it will follow that $\mathbf{A}_{\mathbf{Q}}/\mathbf{Q}$ is compact. Let $(a_p)_p \in \mathbf{A}_{\mathbf{Q}}$ be an adèle; then $a_p \in \mathbf{Z}_p$ for all but finitely many p. For such p which violate this, let $a_p = u_p p^{-n_p}$ for $u_p \in \mathbf{Z}_p^\times$ and let $b_p \in \mathbf{Z}$ such that $u_p \equiv b_p \bmod p^{n_p}$. Then

   $$\left|a_p - b_p p^{-n_p}\right|_p = p^{n_p}\left|u_p - b_p\right|_p \le p^{n_p} p^{-n_p} = 1.$$

   Moreover, for $x \in \mathbf{Z}_l$ ($l \neq p$), we have

   $$\left|x - b_p p^{-n_p}\right|_l \le \max\left\{|x|_l, \left|b_p p^{-n_p}\right|_l\right\} \le \max\left\{1, \left|b_p\right|_l\right\} = 1.$$

   Hence defining

   $$q := \sum b_p p^{-n_p} \in \mathbf{Q},$$

   where the sum runs over those p such that $a_p \notin \mathbf{Z}_p$, we have $a_p - q \in \mathbf{Z}_p$ for all $p \in V_{\mathbf{Q},f}$. Finally, there exists $n \in \mathbf{Z}$ such that $a_\infty - q - n \in [-1/2, 1/2]$, so $(a_p)_p = (q+n) + (z_p)_p$ where $(z_p)_p \in [-1/2, 1/2] \times \prod_p \mathbf{Z}_p$.

Remark 1.23. *We can see from the previous proof that $[-1/2, 1/2] \times \widehat{\mathbf{Z}}$ is a fundamental domain for the action of $\mathbf{Q}$ on $\mathbf{A}_{\mathbf{Q}}$.*

We had hoped that $F \hookrightarrow \mathbf{A}_F$ might be dense, but the previous theorem says this is rather not the case. Despite this, what we hoped for is very close to true:

Theorem 1.24 (Strong approximation for $\mathbf{G}_a$). *Let $w \in V_F$ and $V_F = \{w\} \sqcup S \sqcup T$, where S is finite. Let $a_v \in F_v$ and $\varepsilon_v > 0$ for all $v \in S$. Then there exists $x \in F$ such that*

$$|x - a_v|_v < \varepsilon_v \text{ for all } v \in S \quad \text{and} \quad |x|_v \leq 1 \text{ for all } v \in T.$$

Corollary 1.25. *For any $w \in V_F$, the image of $F \hookrightarrow \mathbf{A}_F^{\{w\}} = \prod_{v \in V_F \setminus \{w\}}{}' F_v$ is dense.*

Remark 1.26. *For $a = (a_v)_v \in \mathbf{A}_F$ we have $|a_v|_v \leq 1$ for all but finitely many $v \in V_F$. Hence we can define the* adèlic norm

$$|a| := \prod_{v \in V_F} |a_v|_v .$$

Either $|a_v|_v = 1$ for all but finitely many v, in which case this product is a finite product, or $|a_v| \leq \frac{1}{2}$ for infinitely many v, in which case the product converges to 0. This shows that $|a| \neq 0$ if and only if $a \in \mathbf{A}_F^\times$.

Lemma 1.27 (Blichfeldt–Minkowski). *There exists[15] $B \in \mathbf{R}_{>0}$ such that for any $a \in \mathbf{A}_F$ with $|a| > B$, there exists $x \in F^\times$ so that*

$$|x|_v \leq |a_v|_v \text{ for all } v \in V_F.$$

[15] *A priori* B depends on F, but with an appropriate choice of normalisations for the Haar measures on F_v, we can get B independent of F [Sut15, Lem. 22.13].

Proof of Strong Approximation for $\mathbf{G}_a$. Let

$$W := \prod_{v \in V_F} \{x \in F_v \colon |x|_v \leq 1\} ;$$

this is compact and surjects onto $\mathbf{A}_F/F$, so $\mathbf{A}_F = F + W$. Then for any $u \in F^\times \subset \mathbf{A}_F^\times$ we have $\mathbf{A}_F = u\mathbf{A}_F = uF + uW = F + uW$. Let B be as in the Blichfeldt–Minkowski lemma and let $z = (z_v)_v \in \mathbf{A}_F$ be such that

$$0 < |z_v|_v < \varepsilon_v \text{ for } v \in S, \quad |z_v| = 1 \text{ for } v \in T, \quad |z_w|_w > B \prod_{v \neq w} |z_v|_v^{-1} .$$

Define the adèle $a = (a_v)_v$ by taking a_v for $v \in S$ as given in the formulation of Strong Approximation and $a_v = 0$ for $v \notin S$. By design, we have $|z| > B$ so by the Blichfeldt–Minkowski lemma there exists $u \in F^\times$ such that $|u|_v \leq |z_v|_v$ for all $v \in V_F$. As $\mathbf{A}_F = F + uW$, we can write $a = x + uy$ for $x \in F$ and $y \in W$; we claim this x satisfies the conditions in the statement of Strong Approximation. For $v \in T$, we have

$$|x|_v = |a_v - uy_v|_v = |u|_v |y_v|_v \leq |u|_v \leq |z_v|_v = 1.$$

For $v \in S$, we have

$$|x - a_v|_v = |uy_v|_v = |u|_v |y_v|_v \leq |u|_v \leq |z_v|_v < \varepsilon_v.$$

To prove the Blichfeldt–Minkowski lemma, we need to use Haar measure. See [Coh13, Ch. 9] for more, including the proof of Haar's theorem.

Definition 1.28. *Let G be a topological group. A left (resp. right) Haar measure on G is a nonzero measure $\mu\colon \mathcal{B}(G) \to \mathbf{R}_{\geq 0} \cup \{\infty\}$ on the Borel σ-algebra $\mathcal{B}(G)$ such that*

μ is outer regular: For all Borel sets $B \in \mathcal{B}(G)$,

$$\mu(B) = \inf\{\mu(U) \colon U \supset B \text{ open}\};$$

μ is inner regular: For all open sets $U \subset G$,

$$\mu(U) = \sup\{\mu(K) \colon K \subset U \text{ compact}\};$$

μ is finite on compacts: For all compact sets $K \subset G$,

$$\mu(K) < \infty;$$

μ is left (resp. right) translation-invariant: For all Borel sets $B \in \mathcal{G}$ and all $g \in G$,

$$\mu(gB) = \mu(B) \quad \text{resp.} \quad \mu(Bg) = \mu(B).$$

Theorem 1.29 (Haar)**.** *Let G be a locally compact Hausdorff topological group. There exists a left (equiv. right) Haar measure μ on G. Moreover this measure is unique in the sense that if μ, ν are two left (resp. right) Haar measures on G then there exists $r \in \mathbf{R}_{>0}$ such that*

$$\mu(B) = r\nu(B)$$

for all $B \in \mathcal{B}(G)$.

Example 1.30.

- *On any commutative topological group, left Haar measures are right Haar measures and vice versa.*
- *On $\mathbf{R}$, the 1-dimensional Lebesgue measure $\mu_{\mathbf{R}}$ is a Haar measure, and we take the usual normalization so that $\mu_{\mathbf{R}}([0,1]) = 1$.*
- *On $\mathbf{C}$, the 2-dimensional Lebesgue measure is a Haar measure, and we normalize so that*
$$\mu_{\mathbf{C}}(\{x \in \mathbf{C} \colon |x|_\infty \leq 1\}) = 1.$$
- *Let $K = \mathrm{Frac}(A)$, where A is a complete discrete valuation ring. Let $\varpi \in A$ be a uniformizer and assume the residue field A/ϖ is finite with order q. Then there is a unique Haar measure μ_K such that $\mu_A(A) = 1$ and more generally*
$$\mu_A(a + \varpi^n A) = q^{-n}.$$

Using these normalized Haar measures, we can define a Haar measure $\mu_{\mathbf{A}_F}$ on $\mathbf{A}_F$ by writing

$$\mathbf{A}_F = \varinjlim_{S \supset V_\infty \text{ finite}} \prod_{v \in S} F_v \times \prod_{v \notin S} \mathcal{O}_{F_v}$$

and taking the product of the Haar measures on each subspace

$$X_S := \prod_{v \in S} F_v \times \prod_{v \notin S} \mathcal{O}_{F_v}.$$

Proof of lemma 1.27. Let $W \subset \mathbf{A}_F$ be a compact fundamental domain for the action of F on $\mathbf{A}_F$. Define

$$b_1 := \mu_{\mathbf{A}_F}\left(\left\{z \in \mathbf{A}_F \colon \left|z_v\right|_v \leq 1 \ \forall v \in V_F,\ \left|z_v\right|_v \leq \frac{1}{4} \ \forall v \in V_\infty\right\}\right).$$

We claim $B := \frac{\mu_{\mathbf{A}_F}(W)}{b_1}$ satisfies the Blichfeldt–Minkowski lemma. To see this, let $a \in \mathbf{A}_F$ with $|a| > B$ and set

$$M := \left\{z \in \mathbf{A}_F \colon \left|z_v\right|_v \leq \left|a_v\right|_v \ \forall v \in V_F,\ \left|z_v\right|_v \leq \frac{1}{4}\left|a_v\right|_v \ \forall v \in V_\infty\right\}.$$

Then

$$\mu_{\mathbf{A}_F}(M) = |a|\, b_1 > Bb_1 = \mu_{\mathbf{A}_F}(W).$$

Hence there exist $y_1, y_2 \in M$ such that $x := y_1 - y_2 \in F^\times$. Then for $v \in V_f$,

$$|x|_v = \left|y_{1,v} - y_{2,v}\right|_v \leq \max\left\{\left|y_{1,v}\right|_v, \left|y_{2,v}\right|_v\right\} \leq \left|a_v\right|_v.$$

For $v \in V_\infty$ real,

$$|x|_v \leq \left|y_{1,v}\right|_v + \left|y_{2,v}\right|_v \leq 2 \cdot \frac{1}{4}\left|a_v\right|_v \leq \left|a_v\right|_v.$$

For $v \in V_\infty$ complex[16],

$$|x|_v = \left(\left|y_{1,v} - y_{2,v}\right|_\infty\right)^2 \leq \left(\left|y_{1,v}\right|_\infty + \left|y_{2,v}\right|_\infty\right)^2 \leq \left(2 \cdot \frac{1}{2}\left|a_v\right|_\infty\right)^2 = \left|a_v\right|_v.$$

[16] Remember, our chosen normalization of the complex absolute value does not satisfy the triangle inequality. Its square root does, however.

1.3 *Finiteness of Class Numbers ($\mathbf{G}_m$-case)*

We would like to prove similar statements as above but for $\mathbf{A}_F^\times$. We might view the previous statements as being about the adèlic points of the linear algebraic group $\mathbf{G}_a$, and now we wish to make them work for $\mathbf{G}_m$.

Theorem 1.31. *Let F be a number field.*

1. *$F^\times \hookrightarrow \mathbf{A}_F^\times$ is closed and discrete.*
2. *$\mathbf{A}_F^1 / F^\times$ is compact, where $\mathbf{A}_F^1 = \left\{x \in \mathbf{A}_F^\times \colon |x| = 1\right\}$.*

Remark 1.32. *Strong Approximation fails for $\mathbf{G}_m$: for example, if we consider $\mathbf{Q}^\times \hookrightarrow \mathbf{A}_{\mathbf{Q},f}^\times$, the subgroup $\prod_p \mathbf{Z}_p^\times$ is open. Its preimage in $\mathbf{Q}^\times$ is $\mathbf{Z}^\times = \{\pm 1\}$, which is clearly not dense in $\prod_p \mathbf{Z}_p^\times$.*

In $\mathbf{A}_F^\times$ we only have Weak Approximation*, which is that*

$$F^\times \hookrightarrow \mathbf{A}_{F,S}^\times = \prod_{v \in S} F_v^\times$$

is dense for any finite $S \subset V_F$. This follows easily from Strong Approximation for $\mathbf{G}_a$.

Remark 1.33. *The adèlic norm*

$$|\cdot| : \mathbf{A}_F \to \mathbf{R}_{\geq 0}$$

is not continuous *as* $\mathbf{A}_F^\times = \{x \in \mathbf{A}_F : |x| \neq 0\}$ *is not open. However, the following hold:*

a *For all finite* $S \subset V_F$*, the partial norm* $\mathbf{A}_F \to \mathbf{R}_{\geq 0}$ *given by* $(x_v)_v \mapsto \prod_{v \in S} |x_v|_v$ *is continuous.*

b $|\cdot|$ *is continuous on* $\mathbf{A}_F^\times$*, since* $\mathbf{A}_F^\times$ *is covered by basic open sets of the form*

$$\prod_{v \in S} F_v^\times \times \prod_{v \notin S} \mathcal{O}_{F_v}^\times$$

for $S \supset V_\infty$ *finite, and by the previous point the norm is continuous on these open subsets.*

Lemma 1.34. *Let* $b \geq c \in \mathbf{R}_{\geq 0}$*.*

1. *The subspace*

$$\{x \in \mathbf{A}_F : |x| \geq c\} \subset \mathbf{A}_F$$

is closed. Moreover, the subspace topologies inherited from $\mathbf{A}_F$ *and* $\mathbf{A}_F^\times$ *agree on this set.*

2. $\{x \in \mathbf{A}_F : b \geq |x| \geq c\} \subset \mathbf{A}_F$ *is closed.*

Proof of theorem 1.31. 1. follows from the corresponding statement for $\mathbf{G}_a$ so it remains to prove 2. Let $C_0 \subset \mathbf{A}_F$ be any compact set with $\mu_{\mathbf{A}_F}(C_0) > \mu_{\mathbf{A}_F}(\mathbf{A}_F/F)$. Let

$$C_1 := \{y - z : y, z \in C_0\} \quad \text{and} \quad C := C_1 \cap \mathbf{A}_F^1.$$

C_1 is compact as is the image of $C_0 \times C_0$ under $(y, z) \mapsto y - z$; by lemma 1.34, the subspace topologies from $\mathbf{A}_F^\times$ and $\mathbf{A}_F$ coincide on $\mathbf{A}_F^1$. Moreover, $\mathbf{A}_F^1$ is closed in $\mathbf{A}_F$ so C is closed in C_1 and hence compact. Now we show that C surjects onto $\mathbf{A}_F^1/F^\times$. For $x \in \mathbf{A}_F^1$ we have $\mu(x^{-1}C_0) = |x|^{-1}\mu(C_0) > \mu(\mathbf{A}_F/F)$ so there exist $y, z \in C_0$ so that

$$u := x^{-1}(y - z) \in F^\times.$$

As $|x| = |u| = 1$, we get that $|y - z| = 1$ so $y - z \in C$.

Before we prove lemma 1.34, we demonstrate an application of theorem 1.31. Recall that the ideal class group of a number field F is defined as

$$\mathrm{Cl}_F := \mathcal{I}_F / \mathcal{P}_F$$

where $\mathcal{I}_F$ is the group of invertible fractional ideals of $\mathcal{O}_F$ and $\mathcal{P}_F$ is the subgroup of principal fractional ideals. This is the cokernel of the map

$$F^\times \to \mathrm{Div}(\mathcal{O}_F) = \bigoplus_{v \in V_f} \mathbf{Z}$$

given by $x \mapsto (\mathrm{val}_v(x))_v$.

Theorem 1.35.
$$\mathrm{Cl}_F \simeq F^\times \backslash \mathbf{A}_{F,f}^\times / \prod_{v \in V_f} \mathcal{O}_{F_v}^\times.$$
Hence, Cl_F *is finite.*

Corollary 1.36. *For all* $K \subset \mathbf{A}_{F,f}^\times$ *compact open the quotient* $F^\times \backslash \mathbf{A}_{F,f}^\times / K$ *is finite.*

Proof. We may replace K by a smaller compact open subgroup
$$K' := K \cap \prod_{v \in V_f} \mathcal{O}_{F_v}^\times.$$
But $\prod_{v \in V_f} \mathcal{O}_{F_v}^\times$ is compact and K' is open, hence $\prod_{v \in V_f} \mathcal{O}_{F_v}^\times / K'$ is finite. The result now follows from theorem 1.35.

Proof of theorem 1.35. We have a commutative diagram
$$\begin{tikzcd} F^\times \arrow[r] \arrow[d, equal] & \bigoplus_{v \in V_f} \mathbf{Z} \\ F^\times \arrow[r] & \mathbf{A}_{F,f}^\times / \prod_{v \in V_f} \mathcal{O}_{F_v}^\times . \arrow[u, "\wr"'] \end{tikzcd}$$
Hence
$$\mathrm{Cl}_F \simeq F^\times \backslash \mathbf{A}_{F,f}^\times / \prod_{v \in V_f} \mathcal{O}_{F_v}^\times.$$
To show this quotient is finite, we note that
$$F^\times \cdot \prod_{v \in V_f} \mathcal{O}_{F_v}^\times = \bigcup_{x \in F^\times} x \left(\prod_{v \in V_f} \mathcal{O}_{F_v}^\times \right) \subset \mathbf{A}_{F,f}^\times$$
is open, so
$$F^\times \backslash \mathbf{A}_{F,f}^\times / \prod_{v \in V_f} \mathcal{O}_{F_v}^\times$$
is discrete. Hence it suffices to show the quotient is compact. To do so, we show $\mathbf{A}_F^1 / F^\times$ surjects onto it via the sequence of maps
$$\mathbf{A}_F^1 / F^\times \hookrightarrow \mathbf{A}_F^\times / F^\times \twoheadrightarrow \mathbf{A}_{F,f}^\times / \left(F^\times \cdot \textstyle\prod_{v \in V_f} \mathcal{O}_{F_v}^\times \right).$$
The surjection $\mathbf{A}_F^\times / F^\times \twoheadrightarrow \mathrm{Cl}_F$ ignores infinite places and the absolute values from $\mathbf{R}$ and $\mathbf{C}$ surject onto $\mathbf{R}_{\geq 0}$. Hence, given an idèle class $[x] \in \mathbf{A}_F^\times / F^\times$, we can freely modify the components in the infinite places to get an idèle with norm 1 which has the same image in Cl_F.

Proof of lemma 1.34. 2. follows from 1. since $|\cdot| : \mathbf{A}_F^\times \to \mathbf{R}_{>0}$ is continuous and hence the preimage of $[c, b]$ is closed in $\mathbf{A}_F^\times$. To prove 1., we first show the subspace
$$X_c := \{x \in \mathbf{A}_F : \ |x| \geq c\} \subset \mathbf{A}_F$$

is closed. Let $x \in \mathbf{A}_F$ such that $|x| < c$; let $S \subset V_F$ be a finite set of places satisfying the following:

- $V_\infty \subset S$;
- $x_v \in \mathcal{O}_{F_v}$ for all $v \notin S$;
- $\prod\limits_{v \in S} |x_v|_v < c$.

Such a finite set exists since one of the following scenarios must occur:

- $|x_v|_v = 1$ for all but finitely many v — take S to be the infinite places along with those v which do not satisfy this;
- $|x_v|_v = 0$ for some v — take S to include this v;
- $\prod\limits_{v \in V_F} |x_v|_v \to 0$ — by the definition of convergence, for any $c > 0$ there exists a finite set $S \subset V_F$ such that
$$\prod_{v \in S} |x_v|_v < c.$$

Since $a \mapsto \prod_{v \in S} |a_v|_v$ is a continuous map $\mathbf{A}_F \to \mathbf{R}_{\geq 0}$, the set

$$U_1 := \left\{ a \in \mathbf{A}_F \colon \prod_{v \in S} |a_v|_v < c \right\}$$

is open. Now
$$U := U_1 \cap \left(\prod_{v \in S} F_v \times \prod_{v \notin S} \mathcal{O}_{F_v} \right)$$
is also open since S is finite, and clearly any $a \in U$ satisfies $|a| < c$. Moreover, $x \in U$ since we chose S so that $\prod_{v \in S} |x_v|_v < c$ and $x_v \in \mathcal{O}_{F_v}$ for $v \notin S$.

It remains to show that the subspace topologies from $\mathbf{A}_F$ and $\mathbf{A}_F^\times$ agree on X_c. Let
$$V = \prod_{v \in S} V_v \times \prod_{v \notin S} \mathcal{O}_{F_v} \subset \mathbf{A}_F$$
be a basic open set i.e. $S \supset S_\infty$ finite and $V_v \subset F_v$ open for each $v \in S$. Since $X_c \subset \mathbf{A}_F^\times$, we have that for any $x \in X_c \cap V$, there exists a finite set $S' \supset S$ such that $x_v \in \mathcal{O}_{F_v}^\times$ for $v \notin S'$. Define

$$W := \prod_{v \in S} (V_v \setminus \{0\}) \times \prod_{v \in S' \setminus S} (\mathcal{O}_{F_v} \setminus \{0\}) \times \prod_{v \notin S'} \mathcal{O}_{F_v}^\times;$$

then $x \in W$, $W \subset \mathbf{A}_F^\times$ is open, and $X_c \cap W \subset X_c \cap V$. Hence $X_c \cap V$ is open in the subspace topology induced from $\mathbf{A}_F^\times$.

Conversely, let
$$V = \prod_{v \in S} V_v \times \prod_{v \notin S} \mathcal{O}_{F_v}^\times$$

be a basic open set in $\mathbf{A}_F^\times$. We need to show that for every $x \in V \cap X_c$, there exists an open $U \subset \mathbf{A}_F$ such that

$$x \in U \cap X_c \subset V \cap X_c.$$

To that end, let $S' \supset V_\infty$ be finite such that $x_v \in \mathcal{O}_{F_v}^\times$ for all $v \notin S'$, and let $r \in \mathbf{R}_{>0}$ such that

$$r > \prod_{v \in S'} |x_v|_v .$$

Define

$$U := \left\{ y \in \mathbf{A}_F \colon \prod_{v \in S'} |y_v|_v < r \right\} \cap \left(\prod_{v \in S'} F_v \times \prod_{v \notin S'} \mathcal{O}_{F_v} \right);$$

note that $x \in U$ and $U \subset \mathbf{A}_F$ is open since $y \mapsto \prod_{v \in S'} |y_v|_v$ is a continuous map $\mathbf{A}_F \to \mathbf{R}_{\geq 0}$. Define the finite set

$$T := S' \cup \left\{ v \in V_{F,f} \colon |\kappa(v)| < rc^{-1} \right\}$$

where $\kappa(v) := \mathcal{O}_{F_v}/\mathfrak{m}_v$ is the residue field at v. Also define

$$Y_T := \prod_{v \in T} F_v^\times \times \prod_{v \notin T} \mathcal{O}_{F_v}^\times.$$

Claim: $X_c \cap U \subset Y_T$.

Proof of Claim. For $y \in X_c \cap U$, we need to show that $y_v \in \mathcal{O}_{F_v}^\times$ for all $v \notin T$. Recall that we already know $y_v \in \mathcal{O}_{F_v}$ for all $v \notin S'$ (and thus also for $v \notin T$). Hence for $v \notin T$ we have

$$c \leq |y| = \prod_{w \in V_F} |y_w|_w \leq |y_v|_v \prod_{w \in S'} |y_w|_w < r\, |y_v|_v .$$

Then (recalling that $|\kappa(v)| \geq rc^{-1}$ for $v \notin T$),

$$1 \geq |y_v|_v > cr^{-1} \geq |\kappa(v)|^{-1} = |\pi_v|_v ,$$

where $\pi_v \in \mathcal{O}_{F_v}$ is a uniformizer. Hence $|y_v|_v = 1$ i.e. $y_v \in \mathcal{O}_{F_v}^\times$ as required.

Now to finish the proof, note that the product topology on Y_T coincides with the subspace topologies from both $\mathbf{A}_F$ and $\mathbf{A}_F^\times$. In particular, $V \cap Y_T$ is open in the subspace topology from $\mathbf{A}_F$, so there exists an open set $W \subset \mathbf{A}_F$ such that $V \cap Y_T = W \cap Y_T$. Now $U \cap W \subset \mathbf{A}_F$ is open and

$$x \in U \cap W \cap X_c = U \cap W \cap X_c \cap Y_T = U \cap X_c \cap V \cap Y_T = U \cap X_c \cap V \subset V \cap X_c.$$

Theorem 1.37 (Dirichlet's Unit Theorem)**.** *Let S be a finite set of places of F such that $V_\infty \subset S$ and define the S-integers*

$$\mathcal{O}_{F,S} := \{ x \in F \colon |x|_v \leq 1 \ \forall v \notin S \} .$$

Then

$$\mathcal{O}_{F,S}^\times \simeq \mathbf{Z}^{|S|-1} \oplus \mu(F)$$

where $\mu(F)$ is the finite group of roots of unity in F. In particular,

$$\mathcal{O}_F^\times \simeq \mathbf{Z}^{r+s-1} \oplus \mu(F)$$

where r is the number of real embeddings of F and s is the number of complex conjugate pairs of complex embeddings.

Proof. Let

$$Y_S := \prod_{v \in S} F_v^\times \times \prod_{v \notin S} \mathcal{O}_{F_v}^\times \subset \mathbf{A}_F^\times$$

and note that the image of the diagonal embedding $\mathcal{O}_{F,S}^\times \hookrightarrow \mathbf{A}_F^\times$ lands in Y_S. Define the map $\log\colon Y_S \to \mathbf{R}^S$ by

$$x = (x_v)_v \mapsto (\log |x_v|_v)_{v \in S}.$$

Since $|x_v|_v = 1$ for all $v \notin S$, the product formula implies that the image of log is contained in the trace-zero hyperplane $(\mathbf{R}^S)_0 \subset \mathbf{R}^S$. By Kronecker's lemma, the kernel of $\log\colon \mathcal{O}_{F,S}^\times \to (\mathbf{R}^S)_0$ is precisely $\mu(F)$.

Lemma 1.38 (Kronecker)**.** *Let $x \in F$ such that $|x|_v = 1$ for all $v \in V_F$. Then $x \in \mu(F)$.*

If $U \subset (\mathbf{R}^S)_0$ is bounded, then any $x \in \log(\mathcal{O}_{F,S}^\times) \cap U$ has all of its absolute values bounded. The discreteness of $F^\times$ in $\mathbf{A}_F^\times$ implies that this set must be finite and hence $\log \mathcal{O}_{F,S}^\times$ is discrete in $(\mathbf{R}^S)_0$ and is a finite free $\mathbf{Z}$-module.

Let V be the span of $\log \mathcal{O}_{F,S}^\times$ in $(\mathbf{R}^S)_0$; if we can show that $V = (\mathbf{R}^S)_0$ then $\log \mathcal{O}_{F,S}^\times$ is a lattice of rank $\dim(\mathbf{R}^S)_0 = |S| - 1$ and hence we will have an exact sequence

$$1 \longrightarrow \mu(F) \longrightarrow \mathcal{O}_{F,S}^\times \xrightarrow{\log} \mathbf{Z}^{|S|-1} \longrightarrow 0$$

which must split. To check that $V = (\mathbf{R}^S)_0$, we may first enlarge S. This is since for $S \subset T$, $\mathcal{O}_{F,S}^\times \subset \mathcal{O}_{F,T}^\times$, and any $\mathbf{Z}$-basis of $\log \mathcal{O}_{F,S}^\times$ can be extended to a $\mathbf{Z}$-basis of $\log \mathcal{O}_{F,T}^\times$. Then given $x \in (\mathbf{R}_0^S)$, we may consider $(x, 0, \ldots, 0) \in (\mathbf{R}^T)_0$ and see that if it is in the span of an extended $\mathbf{Z}$-basis from $\log \mathcal{O}_{F,S}^\times$ then $x \in V$. Hence, we will enlarge S as follows: recall that

$$\mathrm{Cl}_F \simeq F^\times \backslash \mathbf{A}_{F,f}^\times / \prod_{v \in V_f} \mathcal{O}_{F_v}^\times$$

is finite. Let $x_1, \ldots, x_h \in \mathbf{A}_{F,f}^\times$ lift the classes in the class group; then clearly there exists a finite set $T \supset S$ such that $x_1, \ldots, x_h \in Y_T$. Then it follows that $\mathbf{A}_F^\times = Y_T F^\times$, so the composition

$$Y_T \hookrightarrow \mathbf{A}_F^\times \twoheadrightarrow \mathbf{A}_F^\times / F^\times$$

is surjective and its kernel is $\mathcal{O}_{F,T}^\times$. Using the induced isomorphism

$$Y_T / \mathcal{O}_{F,T}^\times \xrightarrow{\sim} \mathbf{A}_F^\times / F^\times,$$

we get a continuous homomorphism $\mathbf{A}_F^1 / F^\times \to (\mathbf{R}^T)_0 / V$ whose image generates $(\mathbf{R}^T)_0 / V$. But by theorem 1.31 $\mathbf{A}_F^1 / F^\times$ is compact and hence so is its image; however, the only compact subgroup of $\mathbf{R}^n$ is 0 so $V = (\mathbf{R}^T)_0$ as desired.

2 Quaternion Algebras

2.1 Central Simple Algebras

Definition 2.1 (Central Simple Algebras)**.** *Let K be a field.*

1. *A central simple algebra (CSA) over K is a (not necessarily commutative[17]) finite dimensional K-algebra D such that*
   - *$Z(D) = K$ i.e. the centre of D is K;*
   - *D is a simple algebra i.e. D has no non-trivial two-sided ideals.*
2. *A quaternion algebra is a CSA of dimension 4 over K.*

[17] Actually, never commutative unless $D = K$.

Example 2.2.

- *The algebra of $d \times d$ matrices $M_d(K)$ is a CSA of dimension d^2 over K.*
- *The Hamilton quaternion algebra over $\mathbf{R}$ is defined as*

  $$\mathbf{H} := \mathbf{R} \oplus \mathbf{R}i \oplus \mathbf{R}j \oplus \mathbf{R}k,$$

  with multiplication defined by

  $$i^2 = j^2 = -1, \quad ij = -ji = k.$$

  This is a division algebra over $\mathbf{R}$ and hence a quaternion algebra.

Proposition 2.3. *Let D be a finite dimensional K-algebra.*

1. *D is a CSA if and only if $D \otimes_K \bar{K} \simeq M_d(\bar{K})$. Hence the dimension of a CSA is always a square.*
2. *When $\operatorname{char} K \neq 2$, define for $a, b \in K^\times$ the quaternion algebra*

   $$\left(\frac{a, b}{K}\right) := K \oplus Ki \oplus Kj \oplus Kk$$

   with multiplication given by

   $$i^2 = a,\ j^2 = b,\ ij = -ji = k.$$

   Then D is a quaternion algebra if and only if $D \simeq \left(\frac{a,b}{K}\right)$ for some $a, b \in K^\times$. Moreover, when D is a quaternion algebra, D is either a division algebra or isomorphic to $M_2(K)$.

Proof.

1. Note that for any K-algebra D, $Z(D) \otimes_K \bar{K} = Z(D \otimes_K \bar{K})$ and $I \otimes_K \bar{K}$ is a non-trivial two-sided ideal of $D \otimes_K \bar{K}$ for any non-trivial two-sided ideal $I \subset D$. Hence if D is not central simple, neither is $D \otimes_K \bar{K}$. Conversely, if D is a CSA then $D \otimes_K \bar{K}$ is a CSA and hence isomorphic[18] to $M_d(\tilde{D})$ for some

[18] This follows from the Artin–Wedderburn theorem: any semisimple K-algebra is isomorphic to one of the form

$$M_{d_1}(D_1) \times M_{d_2}(D_2) \times \cdots M_{d_k}(D_k)$$

where each D_i is a division algebra over K. The simplicity of D implies there can only be one factor $M_d(\tilde{D})$ in such a decomposition.

finite-dimensional $\bar{K}$-division algebra $\tilde{D}$. Since $\bar{K}$ is algebraically closed we must have $\tilde{D} = \bar{K}$.

Alternatively, if $\operatorname{char} K \neq 2$ and $\dim_K D = 4$, we can use the classification from 2. Let $a, b \in K^\times$ such that $D \simeq \left(\frac{a,b}{K}\right)$. Then clearly

$$D \otimes_K \bar{K} \simeq \left(\frac{a,b}{\bar{K}}\right);$$

moreover it is clear that

$$\left(\frac{\mu^2 a, b}{K}\right) \simeq \left(\frac{a,b}{K}\right)$$

so

$$D \otimes_K \bar{K} \simeq \left(\frac{1,b}{\bar{K}}\right).$$

But then $\left(\frac{1,b}{K}\right) \simeq M_2(K)$ via

$$i \mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad j \mapsto \begin{pmatrix} 0 & 1 \\ b & 0 \end{pmatrix}.$$

2. As shown above, any algebra of the form $\left(\frac{a,b}{K}\right)$ is a CSA since it is isomorphic to $M_2(\bar{K})$ over $\bar{K}$. D is either a division algebra or $M_2(K)$ as a consequence of the Artin–Wedderburn theorem[19]. More explicitly, assume D is not a division algebra, so there exists $x \in D \setminus 0$ such that $Dx \neq D$. Then D contains a non-trivial left ideal; let I be a minimal such ideal. Then $d = \dim_K I \in \{1, 2, 3\}$. We can consider the K-algebra map $D \to \operatorname{End}_K(I) \simeq M_d(K)$ given by $x \mapsto (y \mapsto yx)$. Since D is simple this must be injective, and hence $\dim_K I \in \{2, 3\}$. If $\dim_K I = 2$ then by dimensionality the map must be an isomorphism so $D \simeq M_2(K)$. If $\dim_K I = 3$ then every non-trivial left ideal has dimension 3. By dimension considerations, the intersection of any two dimension 3 left ideals must be a non-trivial left ideal, so I is the unique left ideal of D. But then Ix is a left ideal for all $x \in D$, so I is also a right ideal; but this contradicts the simplicity of D.

   Now we show that $D \simeq \left(\frac{a,b}{K}\right)$ for some $a, b \in K^\times$. Without loss of generality we may assume D is a division algebra. For $x \in D \setminus K$, we know that $L = K(x)$ is a degree 2 field extension of K, and D is a 2-dimensional L-vector space. Since L/K is quadratic and $\operatorname{char} K \neq 2$, there exist $a \in K^\times$, $i \in L^\times$ such that $L = K(i)$ and $i^2 = a$. Consider the map $f \colon D \to D$ defined by $z \mapsto izi^{-1}$; this satisfies $f^2 = \mathrm{id}$ so its eigenvalues can only be 1 or -1. The eigenspace for 1 is $C_D(i) = C_D(L) = L$, so the eigenspace for -1 is non-trivial. In particular, there exists $j \in D$ such that $ij = -ji$. Then $i^2 j = j^2 i$ so $j^2 \in L$, and hence $j^2 = ci + b$ for $b, c \in K$. If $c \neq 0$ then $K(i) \subsetneq K(j)$ which is impossible, hence $c = 0$ and $j^2 = b \in K^\times$. Thus we have an injective map $\left(\frac{a,b}{K}\right) \to D$, which must be an isomorphism because of dimension.

[19] Artin–Wedderburn also shows the same is true for D a CSA of dimension p^2, where p is a prime.

Remark 2.4. *If K is a finite field, then every quaternion algebra over K is isomorphic to $M_2(K)$ since Wedderburn's little theorem says any finite dimensional division algebra over K is a field.*

Exercise 2. *Show that every K-algebra isomorphism $\varphi\colon M_2(K) \to M_2(K)$ is inner i.e. there exists $g \in \mathrm{GL}_2(K)$ such that $\varphi(x) = gxg^{-1}$.*

Solution. This fact for $M_n(K)$ is an immediate consequence of the Skolem–Noether theorem:

Theorem 2.5 (Skolem–Noether)**.** *Let A, B be simple K-algebras with B a CSA, and suppose $f, g\colon A \to B$ are K-algebra homomorphisms. Then there exists $x \in B^\times$ such that $f(a) = xg(a)x^{-1}$ for all $a \in A$.*

We prove this using *Morita equivalence*. For R a (possibly noncommutative) ring, we define functors

$$\begin{aligned} M_n(R)\text{-}\mathsf{Mod} &\underset{G}{\overset{F}{\rightleftarrows}} R\text{-}\mathsf{Mod} \\ M &\longmapsto R^n \otimes_{M_n(R)} M \\ \mathrm{Hom}_R(R^n, V) &\longleftarrow\!\shortmid V \end{aligned}$$

where R-Mod and $M_n(R)$-Mod are the respective categories of *left* modules and R^n in the definition of each functor is considered as a *right* $M_n(R)$-module[20].

[20] We could also do this for the right R- and $M_n(R)$-modules and consider R^n as a left $M_n(R)$-module. The important thing is that these actions are on opposite sides so the functors F and G make sense.

Theorem 2.6. *The functors F and G are mutually inverse equivalences between the categories $M_n(R)$-Mod and R-Mod. In this situation where the categories of modules are equivalent, we say R and $M_n(R)$ are* ***Morita equivalent****.*

Proof. We define natural transformations

$$\eta\colon \mathrm{id}_{M_n(R)\text{-}\mathsf{Mod}} \Rightarrow GF \quad \text{and} \quad \varepsilon\colon FG \Rightarrow \mathrm{id}_{R\text{-}\mathsf{Mod}}$$

by

$$\eta_M\colon m \mapsto (y \mapsto m \otimes y) \quad \text{and} \quad \varepsilon_V\colon \varphi \otimes y \mapsto \varphi(y).$$

It is easy to see these are isomorphisms for $V = R$ and $M = M_n(R)$. Now we note that since $\otimes$ is a left adjoint, it preserves colimits and hence there is a natural isomorphism

$$F\left(\bigoplus_{i \in I} M_i\right) \simeq \bigoplus_{i \in I} F(M_i).$$

Moreover, as R^n is a finite free R-module, we also have natural isomorphisms

$$G\left(\bigoplus_{i \in I} M_i\right) \simeq \bigoplus_{i \in I} G(M_i).$$

Thus since ε_R and $\eta_{M_n(R)}$ are isomorphisms, it follows that ε_V and η_M are isomorphisms for all free R-modules V and free $M_n(R)$-modules M. Now we can also see that F and G are both right exact – F since it is a tensor product and G since R^n is a projective R-module. Hence given an R-module V, we may take a presentation

$$R^{\oplus I} \longrightarrow R^{\oplus J} \longrightarrow V \longrightarrow 0$$

and use right exactness of F and G to obtain the commutative diagram with exact rows

$$\begin{tikzcd} FG(R^{\oplus I}) \arrow[r] \arrow[d, "\varepsilon_{R^{\oplus I}}"] & FG(R^{\oplus J}) \arrow[r] \arrow[d, "\varepsilon_{R^{\oplus J}}"] & FG(V) \arrow[r] \arrow[d, "\varepsilon_V"] & 0 \arrow[r, equal] \arrow[d, equal] & 0 \arrow[d, equal] \\ R^{\oplus I} \arrow[r] & R^{\oplus J} \arrow[r] & V \arrow[r] & 0 \arrow[r, equal] & 0 \end{tikzcd}.$$

Since we already know ε is an isomorphism for free R-modules, the Five Lemma implies ε_V is also an isomorphism. Similarly, for a left $M_n(R)$-module M, we take a presentation and use right exactness of G, F plus the fact that η is an isomorphism for free modules to deduce η_M is an isomorphism.

Under this equivalence, we must clearly have simple R-modules mapping to simple $M_n(R)$-modules and vice versa. In particular, for $R = D$ a division ring there is a unique simple module in each category – D in D-Mod and D^n in $M_n(D)$-Mod. As Morita equivalence is additive, we further deduce that in this case all modules over $M_n(D)$ are direct sums of D^n and are classified up to isomorphism by their rank over D. Now let A, B, f, g be as in the Skolem–Noether theorem; simplicity of B implies $B \simeq M_n(D)$ for some finite-dimensional central division algebra D/K. Simplicity of A also implies the morphisms f, g must be injective and hence A is a finite-dimensional K-algebra. We may endow D^n with two A-module structures via f, g. As D is central in $M_n(D)$, we can extend this to two module structures of D^n over the algebra $A \otimes_K D$ by

$$(a \otimes d) \cdot_f v = f(a)dv \quad \text{and} \quad (a \otimes d) \cdot_g v = g(a)dv.$$

Since D is central over K and A, D are simple, the tensor product $A \otimes_K D$ is also a finite-dimensional simple K-algebra. By comparing dimensions over K, we see that the two module structures on D^n are isomorphic; hence there is $\varphi \in \mathrm{GL}(D^n)$ such that $\varphi(f(a)y) = g(a)\varphi(y)$ for all $a \in A, y \in D^n$. Passing back through the isomorphism $B \simeq M_n(D)$, we see there exists $x \in B$ such that $xf(a) = g(a)x$.

Definition 2.7. *Let $D = \left(\frac{a,b}{K}\right)$ be a quaternion algebra.*

1. *The map ${}^*: D \to D$ defined by*

$$x = \alpha + \beta i + \gamma j + \delta k \mapsto x^* = \alpha - \beta i - \gamma j - \delta k$$

   is a K-linear anti-involution i.e. $(x^*)^* = x$ and $(xy)^* = y^* x^*$.

2. *The* reduced norm *of D is a map* $\mathrm{Nrd}_D \colon D \to K$ *defined by*

$$x \mapsto xx^*.$$

The trace *of D is a map* $\mathrm{Tr}_D \colon D \to K$ *defined by*

$$x \mapsto x + x^*.$$

Example 2.8. *Consider the isomorphism*

$$M_n(K) \simeq \left(\frac{1,b}{K}\right), \quad i \mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad j \mapsto \begin{pmatrix} 0 & b \\ 1 & 0 \end{pmatrix}.$$

Then under this isomorphism,

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}^* = \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix}.$$

In particular,

$$\mathrm{Nrd}_{M_2(K)} = \det \quad \textit{and} \quad \mathrm{Tr}_{M_2(K)} = \mathrm{tr}\,.$$

The maps $()^*, \mathrm{Nrd}_{M_2(K)}, \mathrm{Tr}_{M_2(K)}$ *do not depend on the choice of isomorphism* $M_2(K) \simeq \left(\frac{1,b}{K}\right)$. *Indeed, any other isomorphism is given by precomposition with an automorphism of $M_2(K)$ which must be inner. It is easy to see any inner automorphism preserves* $()^*$, *hence* $\mathrm{Nrd}_{M_2(K)}$ *and* $\mathrm{Tr}_{M_2(K)}$ *are also preserved.*

As a consequence, we see that for any quaternion algebra D/K, the maps $()^*, \mathrm{Nrd}_D, \mathrm{Tr}_D$ *do not depend on the choice of isomorphism* $D \simeq \left(\frac{a,b}{K}\right)$. *Indeed, after choosing such an isomorphism we get*

$$M_2(\bar{K}) \simeq D \otimes_K \bar{K} \simeq \left(\frac{a,b}{\bar{K}}\right)$$

and $()^*, \mathrm{Nrd}, \mathrm{Tr}$ *agree with* $()^*, \det, \mathrm{tr}$ *on* $M_2(\bar{K})$, *independent of the choice of a, b and the isomorphism* $D \simeq \left(\frac{a,b}{K}\right)$. *As* $()^*, \mathrm{Nrd}_D, \mathrm{Tr}_D$ *are the restriction to D of the corresponding maps on* $D \otimes_K \bar{K} \simeq M_2(\bar{K})$, *these maps are independent of the choice of isomorphism* $D \simeq \left(\frac{a,b}{K}\right)$.

Exercise 3. *Assume* $\bar{K} = K^{\mathrm{sep}}$. *By taking* Gal_K *invariants, show directly that*

$$D \hookrightarrow D \otimes_K \bar{K} \xrightarrow[\sim]{\dagger} M_2(\bar{K}) \xrightarrow{\det} \bar{K}$$

takes values in K and is independent of the choice of isomorphism $\dagger$.

2.2 Quaternion Algebras over Local Fields

Let $K = F_v$, where F is a number field and $v \in V_F$ is a fixed place. We want to understand quaternion algebras over K.

Proposition 2.9.

1. *Let $K = \mathbf{C}$; then $M_2(\mathbf{C})$ is (up to isomorphism) the unique quaternion algebra over $\mathbf{C}$.*
2. *Let $K = \mathbf{R}$; then $M_2(\mathbf{R})$ and $\mathbf{H} = \left(\frac{-1,-1}{\mathbf{R}}\right)$ are (up to isomorphism) the only quaternion algebras over $\mathbf{R}$.*

Proof.

1. Let $D/\mathbf{C}$ be a quaternion algebra. Then
$$D \simeq \left(\frac{a,b}{\mathbf{C}}\right) = \left(\frac{z^2,b}{\mathbf{C}}\right) \simeq \left(\frac{1,b}{\mathbf{C}}\right) \simeq M_2(\mathbf{C}).$$
2. Let $D/\mathbf{R}$ be a quaternion algebra. Then
$$D \simeq \left(\frac{a,b}{\mathbf{R}}\right) = \left(\frac{\varepsilon_1\alpha^2,\varepsilon_2\beta^2}{\mathbf{R}}\right) \simeq \left(\frac{\varepsilon_1,\varepsilon_2}{\mathbf{R}}\right)$$
where $\varepsilon_1, \varepsilon_2 \in \{\pm 1\}$. If one of $\varepsilon_1, \varepsilon_2$ is 1 then $D \simeq M_2(\mathbf{R})$. Else if $\varepsilon_1 = \varepsilon_2 = -1$ we have $D \simeq \mathbf{H}$ and $\mathbf{H} \not\simeq M_2(\mathbf{R})$ as $\mathbf{H}$ is a division algebra.

Remark 2.10. *Let $K \in \{\mathbf{R}, \mathbf{C}\}$ and D a quaternion algebra over K. Then D carries a natural topology that makes it into a topological K-algebra. In particular, the unit group $D^\times$ is a topological group. If $D = M_2(K)$ then $D^\times = \mathrm{GL}_2(K)$.*

Lemma 2.11. *Let*
$$\mathbf{H}^1 := \left\{x \in \mathbf{H}^\times : \ \mathrm{Nrd}_{\mathbf{H}}(x) = 1\right\}.$$
Then $\mathbf{H}^1$ is a closed and compact subgroup of $\mathbf{H}^\times$. In fact, it is the unique maximal compact subgroup of $\mathbf{H}^\times$. Moreover, $\mathbf{H}^\times/\mathbf{R}^\times$ is compact.

Proof. Clearly $\mathbf{H}^1$ is a closed subgroup, as $\mathrm{Nrd}_{\mathbf{H}}$ is a continuous homomorphism $\mathbf{H}^\times \to \mathbf{R}^\times$. Moreover, writing $x = a + bi + cj + dk$, we compute
$$\mathrm{Nrd}_{\mathbf{H}}(x) = (a + bi + cj + dk)(a - bi - cj - dk) = a^2 + b^2 + c^2 + d^2.$$
Hence we can identify $\mathbf{H}^1$ with the unit sphere in $\mathbf{R}^4$, which is compact.

Now let $H \subset \mathbf{H}^\times$ be a compact subgroup; then $\mathrm{Nrd}_{\mathbf{H}}(H) \subset \mathbf{R}_{>0}$ is compact. If $\gamma \in \mathrm{Nrd}_{\mathbf{H}}(H)$ such that $1 \neq \gamma$, then the powers γ^i for $i \in \mathbf{Z}$ are also in $\mathrm{Nrd}_{\mathbf{H}}(H)$. But these must be unbounded and $\mathrm{Nrd}_{\mathbf{H}}(H)$ would not be compact. Hence $H \subset \mathbf{H}^1$.

To show $\mathbf{H}^\times/\mathbf{R}^\times$ is compact, we note that
$$\mathbf{H}^1 \hookrightarrow \mathbf{H}^\times \twoheadrightarrow \mathbf{H}^\times/\mathbf{R}^\times$$
is surjective. Indeed, for $x \in \mathbf{H}^\times$, there exists $y \in \mathbf{R}_{>0}$ such that $y^2 = \mathrm{Nrd}_{\mathbf{H}}(x)$. Then $y^{-1}x \in \mathbf{H}^1$ and maps to the same class as x in $\mathbf{H}^\times/\mathbf{R}^\times$.

Remark 2.12. *Let $D = M_2(K)$, where $K = \mathbf{R}$ or $\mathbf{C}$. Define*
$$D^1 := \{x \in M_2(K) : \ \det(x) = 1\} = \mathrm{SL}_2(K).$$

Then D^1 is a closed subgroup of $\mathrm{GL}_2(K)$ but it is not compact. To see this, consider the embedding

$$K^\times \hookrightarrow \mathrm{SL}_2(K), \quad a \mapsto \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}.$$

The image is a closed subgroup of $\mathrm{SL}_2(K)$, but $K^\times$ is not compact. Moreover, the quotient

$$\mathrm{PGL}_2(K) := \mathrm{GL}_2(K)/K^\times$$

is also not compact.

Proposition 2.13. *Let $K/\mathbf{Q}_p$ be a finite extension. There are, up to isomorphism, exactly two quaternion algebras over K:*

- *$M_2(K)$;*
- *The division algebra $\left(\frac{a,\pi}{K}\right)$, where*
  - *$\pi \in K$ is a uniformizer and*
  - *$a \in \mathcal{O}_K^\times$ is such that $K(\sqrt{a})$ is the unique degree 2 unramified extension of K.*

The proof of this proposition proceeds in 3 steps: Fixing D/K a quaternion algebra which is a division algebra,

1. Extend the normalized valuation $v\colon K \to \mathbf{Z} \cup \{\infty\}$ to D.
2. Show K^{unr}, the unique degree 2 unramified extension of K, embeds into D.
3. Deduce the proposition i.e. $D \simeq \left(\frac{a,\pi}{K}\right)$.

Definition 2.14. *Let D/K be a quaternion algebra which is a division algebra. A (discrete) valuation on D is a map $w\colon D \to \mathbf{Z} \cup \{\infty\}$ satisfying*

1. *$w(x) = \infty$ if and only if $x = 0$;*
2. *$w(xy) = w(x) + w(y)$;*
3. *$w(x+y) \geq \min\{w(x), w(y)\}$.*

Note that as with valuations on fields, the third condition also implies that $w(x+y) = \min\{w(x), w(y)\}$ if $w(x) \neq w(y)$. An element $\pi \in D^\times$ is called a uniformizer *if $w(\pi) \in \mathbf{Z}_{>0}$ is minimal.*

Lemma 2.15. *The normalized discrete valuation $v\colon K \to \mathbf{Z} \cup \{\infty\}$ extends to a discrete valuation $w\colon D \to \mathbf{Z} \cup \{\infty\}$.*

Proof. Set $w := \frac{1}{2} v \circ \mathrm{Nrd}_D$. We claim this does the job[21]:

1. $w(x) = \infty \iff \mathrm{Nrd}_D(x) = 0 \iff x = 0$ since D is a division algebra.
2. $w(xy) = v(\mathrm{Nrd}_D(xy)) = v(\mathrm{Nrd}_D(x)\,\mathrm{Nrd}_D(y)) = v(\mathrm{Nrd}(x)) + v(\mathrm{Nrd}(y)) = w(x) + w(y)$.

[21] Technically, this valuation lands in $\frac{1}{2}\mathbf{Z}$ instead of $\mathbf{Z}$ but this won't make a difference.

3. To show the ultrametric inequality, it suffices to show $w(x+1) \geq \min\{w(x), w(1)\}$. We know that $K(x) \subset D$ is a subfield, and

$$\mathrm{Nrd}_D|_{K(x)} = \mathrm{Nm}_{K(x)/K}.$$

But $\frac{1}{2}v \circ \mathrm{Nm}_{K(x)/K}$ is an extension of v to $K(x)$. The claimed inequality follows.

The fact that w extends v follows from the same fact that for each $x \in D$, $w|_{K(x)}$ is the valuation on $K(x)$ extending v.

Lemma 2.16. *Let $w\colon D \to \mathbf{Z} \cup \{\infty\}$ be a discrete valuation. Then*

- *$\mathcal{O}_D := \{x \in D\colon w(x) \geq 0\}$ is a subring of D.*
- *$\mathfrak{m}_D = \{x \in D\colon w(x) > 0\}$ is the unique (two-sided) maximal ideal of $\mathcal{O}_D$.*
- *$\mathcal{O}_D^\times = \{x \in D\colon w(x) = 0\} = \mathcal{O}_D \setminus \mathfrak{m}_D$.*

Proof. Identical to the commutative case.

Lemma 2.17. *The left ideals in $\mathcal{O}_D$ are of the form*

$$\{x \in \mathcal{O}_D\colon w(x) \geq i\} \tag{5}$$

for $i \in \mathbf{Z}_{\geq 0}$. The right ideals in $\mathcal{O}_D$ are of the same form. Hence any left (or right) ideal in $\mathcal{O}_D$ is a two-sided ideal.

Definition 2.18. *Let D/K be a quaternion algebra. An* order *$\mathcal{O} \subset D$ is an $\mathcal{O}_K$-lattice in D (a finitely generated $\mathcal{O}_K$-submodule $\mathcal{O} \subset D$ that generates D as a K-vector space) which is alo a subring.*

Lemma 2.19. *Let D be a quaternion algebra over K that is a division algebra. Then $\mathcal{O}_D \subset D$ is an order, and it is the unique maximal order of D.*

Proof. It is clear that $\mathcal{O}_D$ is an $\mathcal{O}_K$-submodule of D which contains a K-basis of D. To see that it is finitely generated, note that given a K-basis $\{e_1, e_2, e_3, e_4\}$ of D we can cover D by the open $\mathcal{O}_K$-submodules

$$\pi^k \bigoplus_{i=1}^{4} \mathcal{O}_K e_i$$

for $k \in \mathbf{Z}$. w makes D a normed K-vector space and since K is locally compact it follows that $\mathcal{O}_D$ (the unit ball) is compact. Hence

$$\mathcal{O}_D \subset \pi^k \bigoplus_{i=1}^{4} \mathcal{O}_K e_i$$

for some $k \in \mathbf{Z}$ and hence is finitely generated as $\mathcal{O}_K$ is Noetherian.

To show that $\mathcal{O}_D$ is the unique maximal order, let $\mathcal{O} \subset D$ be an order. For any $x \in \mathcal{O}$, we recall that $K(x) \subset D$ is a field, $w|_{K(x)}$ is an extension of v to $K(x)$, and $\mathcal{O}_{K(x)} \subset K(x)$ is the unique maximal order of $K(x)$. But then $\mathcal{O} \cap K(x)$ is an order in $K(x)$ so $\mathcal{O} \cap K(x) \subset \mathcal{O}_{K(x)}$ and hence $w(x) \geq 0$. As this holds for all $x \in \mathcal{O}$ we have $\mathcal{O} \subset \mathcal{O}_D$.

Exercise 4. *Show that $M_n(\mathcal{O}_K) \subset M_n(K)$ is a maximal order, but not unique. Show that every maximal order in $M_n(K)$ is of the form $gM_n(\mathcal{O}_K)g^{-1}$ for some $g \in \mathrm{GL}_n(K)$.*

Proof. Recall $M_n(K) = \mathrm{End}_K(K^n)$; for any $\mathcal{O}_K$-lattice $\Lambda \subset K^n$, we also note $\mathrm{End}_{\mathcal{O}_K}(\Lambda) \subset \mathrm{End}_K(K^n)$. We first show that every order is contained in $\mathrm{End}_{\mathcal{O}_K}(\Lambda)$ for some lattice Λ. Fix the standard lattice

$$\Lambda_0 := \mathcal{O}_K^n$$

and for an order $\mathcal{O} \subset M_n(K)$ define

$$\Lambda := \{v \in \Lambda_0 \colon \mathcal{O}v \subset \Lambda_0\}.$$

Clearly Λ is a finitely generated $\mathcal{O}_K$-submodule of Λ_0; moreover it is also clear that $\mathcal{O} \subset \mathrm{End}_{\mathcal{O}_K}(\Lambda)$. It remains to check that Λ is a lattice i.e. that it has rank n. For this, note that after choosing an $\mathcal{O}_K$-basis of $\mathcal{O}$, there exists $r \in \mathbf{Z}$ such that the elements of $\mathcal{O}$ are of the form $p^{-r}\varphi$ for $\varphi \in M_n(\mathcal{O}_K)$. From this it follows that $p^r\Lambda_0 \subset \Lambda$ and hence Λ must have rank n.

Now we claim that if $\mathrm{End}_{\mathcal{O}_K}(\Lambda) \subset \mathrm{End}_{\mathcal{O}_K}(\Lambda')$ then there exists $x \in K^\times$ such that $\Lambda = x\Lambda'$.[22] For this, note that by choosing K-bases for K^n in Λ and Λ', there exists $\varphi \in \mathrm{GL}_n(K)$ such that $\Lambda = \varphi\Lambda'$. It is clear moreover that we may write

$$\Lambda = \bigoplus_{i=1}^n \mathcal{O}_K v_i \quad \text{and} \quad \Lambda' = \bigoplus_{i=1}^n \pi^{k_i}\mathcal{O}_K v_i$$

for some (unique) integers k_i. Every $\psi \in \mathrm{GL}(\Lambda)$ which permutes the basis vectors v_i hence permutes the factors $\pi^{k_i}\mathcal{O}_K v_i$ of Λ'. Since we assumed $\mathrm{End}_{\mathcal{O}_K}(\Lambda) \subset \mathrm{End}_{\mathcal{O}_K}(\Lambda')$, it follows that all k_i are equal, and hence $\Lambda' = \pi^k\Lambda$.

We now finish showing the initial claims. We know that any order containing $M_n(\mathcal{O}_K)$ must be contained in $\mathrm{End}_{\mathcal{O}_K}(\Lambda)$ for some $\mathcal{O}_K$-lattice $\Lambda \subset K^n$; however this implies that $\Lambda = \pi^k\mathcal{O}_K^n$ for some $k \in \mathbf{Z}$. As

$$\mathrm{End}_{\mathcal{O}_K}(g\Lambda) = g\,\mathrm{End}_{\mathcal{O}_K}(\Lambda)g^{-1}$$

for all lattices Λ and $g \in \mathrm{GL}_n(K)$, it follows that $M_n(\mathcal{O}_K) = \mathrm{End}_{\mathcal{O}_K}(\Lambda)$ and hence $M_n(\mathcal{O}_K)$ is maximal. The same argument shows that $\mathrm{End}_{\mathcal{O}_K}(\Lambda)$ is a maximal order of $M_n(K)$ for all lattices $\Lambda \subset K^n$, and since each such Λ may be written as $g\mathcal{O}_K$ for some $g \in \mathrm{GL}_n(K)$, we obtain that all maximal orders of $M_n(K)$ are $\mathrm{GL}_n(K)$-conjugates of $M_n(\mathcal{O}_K)$. Finally, to see the non-uniqueness of the maximal order, it suffices to find a lattice $\Lambda \in K^n$ which is non-homothetic to $\mathcal{O}_K^n$;

$$\Lambda = \pi\mathcal{O}_K \oplus \mathcal{O}_K^{n-1}$$

is such a lattice.

[22] We say Λ and Λ' are *homothetic*.

Lemma 2.20. *Let K^{unr}/K be the unique unramified extension of degree 2, and let D be a quaternion algebra over K which is a division algebra. Then there is an embedding of K-algebras $K^{\mathrm{unr}} \hookrightarrow D$.*

Proof. Suppose for contradiction there is no such embedding i.e. that $K(x)/K$ is totally ramified for all $x \in D \setminus K$. Fix a uniformizer $\pi_D \in \mathcal{O}_D$; the residue fields of K and $K(\pi_D)$ coincide so for any $x \in \mathcal{O}_D \setminus \mathcal{O}_K$ there exist $a_0 \in \mathcal{O}_K$ and $a_1 \in \mathcal{O}_D$ such that

$$x = a_0 + \pi_D a_1.$$

We may recursively iterate this to find that for any $n \in \mathbf{Z}_{\geq 0}$,

$$x = a_0 + \pi_D a_1 + \cdots + \pi_D^n a_n$$

where $a_i \in \mathcal{O}_K$ for $i < n$ and $a_n \in \mathcal{O}_D$. We may thus express x as the limit of the truncated sums

$$a_0 + \pi_D a_1 + \cdots + \pi_D^{n-1} a_{n-1},$$

where now each $a_i \in \mathcal{O}_K$. But this implies that $\mathcal{O}_D \subset K(\pi_D)$, and hence that D is commutative.

Proposition 2.21. *Let $K/\mathbf{Q}_p$ be finite, and let D/K be a quaternion algebra which is a division algebra. Then*

$$D \simeq K^{\text{unr}} \oplus K^{\text{unr}} j$$

where $j^2 = \pi_K$ is a uniformizer in K and

$$xj = j\bar{x} \quad \text{for all } x \in K^{\text{unr}},$$

where $x \mapsto \bar{x}$ is the unique non-trivial automorphism in $\mathrm{Gal}(K^{\text{unr}}/K)$. Moreover, any two such D are isomorphic, and if $\pi_D \in \mathcal{O}_D$ is a uniformizer and $\mathfrak{m}_D = \mathcal{O}_D \pi_D = \pi_D \mathcal{O}_D$, then

$$\mathfrak{m}_D^2 = \pi_K \mathcal{O}_D \quad \text{and} \quad \mathcal{O}_D = \mathcal{O}_{K^{\text{unr}}} \oplus \mathcal{O}_{K^{\text{unr}}} \pi_D.$$

Proof. We know already that $K^{\text{unr}} = K(\sqrt{a})$ embeds into D and hence[23]

$$D = K^{\text{unr}} \oplus K^{\text{unr}} j$$

for some j such that $ij = -ji$, $i^2 = a$, $j^2 = b \in K^\times$. We note that $b \notin \mathrm{Nm}_{K^{\text{unr}}/K}(K^{\text{unr}\times})$; indeed if $b = x\bar{x}$ for some $x \in K^{\text{unr}}$, then $y = x^{-1}j$ satisfies $y \notin \{\pm 1\}$ and $y^2 = 1$, which is a contradiction as D is a division algebra. Moreover, we have that

$$\mathrm{Nm}_{K^{\text{unr}}/K}(K^{\text{unr}\times}) = \mathcal{O}_K^\times \cdot \pi_K^{2\mathbf{Z}}$$

so we can find $\mu \in K^{\text{unr},\times}$ such that

$$\mathrm{Nm}_{K^{\text{unr}}/K}(\mu) b = \pi_K.$$

Consequently $(\mu j)^2 = \pi_K$ and we still have

$$D = K^{\text{unr}} \oplus K^{\text{unr}} \mu j.$$

Replacing j with μj, note that $j \in D$ is a uniformizer, as $\mathrm{Nm}_D(j) = -\pi_K$. Thus

$$\mathfrak{m}_D = \mathcal{O}_D j \quad \text{and} \quad \mathfrak{m}_D^2 = \mathcal{O}_D j^2 = \pi_K \mathcal{O}_D.$$

[23] c.f. the proof of the general classification that $D \simeq \left(\frac{a,b}{K}\right)$.

Finally, let

$$x = y + zj \in K^{\mathrm{unr}} \oplus K^{\mathrm{unr}} j = D$$

with $w(x) \geq 0$. Then

$$w(y) = \frac{1}{2} v(\mathrm{Nm}_D(y)) = v(y\bar{y}) \in \mathbf{Z}$$

and

$$w(zj) = w(z) + w(j) \in \frac{1}{2} + \mathbf{Z}.$$

In particular,

$$w(x) = \min\left\{w(y), \frac{1}{2} + w(z)\right\}$$

so $w(y), w(z) \geq 0$ and

$$x \in \mathcal{O}_{K^{\mathrm{unr}}} \oplus \mathcal{O}_{K^{\mathrm{unr}}} j.$$

Remark 2.22.

- *$D^\times$ is a topological group.*
- *$M_2(K)$ is a topological K-vector space with the neighborhood basis*

  $$\pi_K^i M_n(\mathcal{O}_K) \subset M_n(K)$$

  at 0.
- *$\mathrm{GL}_n(K)$ is a topological group; the groups $\mathrm{GL}_n(\mathcal{O}_K)$ and $1 + \pi_K^i M_n(\mathcal{O}_K)$ are compact open subgroups that form a neighborhood basis of* 1.

Proposition 2.23. *Let D/K be a quaternion algebra which is a division algebra. Then*

1. *$\mathcal{O}_D^\times \subset D^\times$ is the unique maximal compact subgroup.*
2. *$D^\times / K^\times$ is compact.*

Proof.

1. Compactness of $\mathcal{O}_D^\times$ is clear as $\mathcal{O}_D^\times = \ker(w)$ is a closed subspace of the compact space $\mathcal{O}_D$. For any compact subgroup $K \subset D^\times$, the continuity of w implies $w(K)$ must be a compact subgroup of $\mathbf{R}$ and hence $w(K) = 0$. It follows that $K \subset \mathcal{O}_D^\times$.
2. Since $\mathcal{O}_D^\times$ is compact, so is $\pi_D \mathcal{O}_D^\times$. We claim that the composite

   $$\mathcal{O}_D^\times \cup \pi\mathcal{O}_D^\times \hookrightarrow D^\times \twoheadrightarrow D^\times / K^\times$$

   is a surjection. For $x \in D^\times$, we have $w(x) \in \mathbf{Z}$ or $w(x) \in \frac{1}{2} + \mathbf{Z}$. In either case, since $w(K^\times) = v(K^\times) = \mathbf{Z}$ there exists $y \in K^\times$ such that $x/y \in \mathcal{O}_D^\times \cup \pi\mathcal{O}_D^\times$.

Exercise 5. $\mathrm{GL}_n(\mathcal{O}_K) \subset \mathrm{GL}_n(K)$ *is a maximal compact subgroup. Moreover any maximal compact subgroup of* $\mathrm{GL}_n(K)$ *is conjugate to* $\mathrm{GL}_n(\mathcal{O}_K)$.

Solution. This can be done with the Bruhat–Tits building. We describe this construction in the case $n = 2$.

Let $K/\mathbf{Q}_p$ be finite; consider the homothety classes of $\mathcal{O}_K$-lattices in K^2:

$$\left\{\mathcal{O}_K\text{-lattices } \Lambda \subset K^2\right\} / \sim .$$

We can make a graph with vertices these equivalence classes and edges between $[\Lambda_1]$ and $[\Lambda_2]$ if $\Lambda_1 \subsetneq \Lambda_2 \subsetneq \varpi^{-1}\Lambda_1$ where $\varpi \in \mathcal{O}_K$ is a uniformizer. This graph is a $(q+1)$-regular tree, i.e. there are exactly $q+1$ edges out of every vertex and there are no cycles (here $q := |\mathcal{O}_K/\varpi|$). This tree contains many copies of $\mathbf{R}$ (where an edge corresponds to $[0,1]$). To describe these, fix a basis e_1, e_2 of K^2; this gives rise to lattices $\varpi^i\mathcal{O}_K e_1 \oplus \varpi^j\mathcal{O}_K e_2$, which are equivalent to $\mathcal{O}_K e_1 \oplus \varpi^{j-i}\mathcal{O}_K e_2$, for $i, j \in \mathbf{Z}$. This gives a copy of $\mathbf{R}$ with the vertices $[\langle e_1, \varpi^i e_2\rangle]$ for $i \in \mathbf{Z}$. Call this $\mathcal{A}_{(e_1,e_2)}$; these copies of $\mathbf{R}$ are called the apartments of T. We note that

$$T = \bigcup_{(e_1,e_2)\text{ basis}} \mathcal{A}_{(e_1,e_2)}$$

and for vertices $[\Lambda_1], [\Lambda_2] \in T$, there exists a basis (e_1, e_2) such that $[\Lambda_1], [\Lambda_2] \in \mathcal{A}_{(e_1,e_2)}$.

There is a canonical action of $\mathrm{GL}_2(K)$ on T given by $g \cdot [\Lambda] := [g\Lambda]$. Clearly if $[\Lambda_1], [\Lambda_2]$ are neighbouring lattices, then so are $[g\Lambda_1], [g\Lambda_2]$ so this action is an action on the tree (not just the set of vertices). This action is transitive on vertices, edges, and apartments. If $[\Lambda]$ is a vertex, write $\Lambda = g\mathcal{O}_K^2$; then the stabilizer of $[\Lambda]$ is $g(Z\,\mathrm{GL}_2(\mathcal{O}_K))g^{-1}$, where Z is the centre of $\mathrm{GL}_2(K)$. Up to the contribution from Z, these are exactly the maximal compact subgroups of $\mathrm{GL}_2(K)$. If $\Lambda_1 \subsetneq \Lambda_2 \subsetneq \varpi^{-1}\Lambda_1$ is a chain of lattices corresponding to an edge, if we write $\Lambda_1 = g\mathcal{O}_K^2$ and $\Lambda_2 = g(\mathcal{O}_K \oplus \varpi^{-1}\mathcal{O}_K)$, then the stabilizer of this lattice chain is gIg^{-1}, where $I \subset \mathrm{GL}_2(\mathcal{O}_K)$ is the Iwahori subgroup. The stabilizer of the corresponding edge is then $gZIg^{-1}$. For an apartment $\mathcal{A}_{(e_1,e_2)}$, the stabilizer is precisely the diagonal matrices when e_1, e_2 is the standard basis of K^2. In general, the stabilizers are conjugates of this subgroup, and the conjugates of the diagonal matrices are in bijection with the apartments. These are precisely the "K-maximal split tori" of GL_2 i.e. subgroups isomorphic to $K^\times \times K^\times$.

3 Interlude on Quadratic Forms

Let K be a field with $\mathrm{char}(K) \neq 2$.

Definition 3.1. *Let V be a finite-dimensional K-vector space. A* quadratic form *on V is a map*

$$q\colon V \to K$$

such that

- $q(ax) = a^2 q(x)$ *for all* $a \in K$, $x \in V$;
- *The map* $V \times V \to K$ *defined by* $(x, y) \mapsto q(x+y) - q(x) - q(y)$ *is a bilinear form.*

Two quadratic forms (V, q) and (V', q') are called isomorphic *if there exists a K-linear isomorphism $f\colon V \xrightarrow{\sim} V'$ such that*

$$q'(f(x)) = q(x) \quad \textit{for all } x \in V.$$

Example 3.2.

1. *The map $K^n \to K$ defined by $(x_1, \ldots, x_n) \mapsto x_1^2 + \cdots + x_n^2$ is a quadratic form. More generally*
$$(x_1, \ldots, x_n) \mapsto \sum_{i \leq j} a_{ij} x_i x_j$$
for $a_{ij} \in K$ is a quadratic form.

2. *Let $D \simeq \left(\frac{a,b}{K}\right)$ be a quaternion algebra; then $\mathrm{Nrd}_D\colon D \to K$ that maps*
$$x = \alpha + \beta i + \gamma j + \delta k \mapsto xx^* = \alpha^2 - a\beta^2 - b\gamma^2 + ab\delta^2$$
is a quadratic form.

Lemma 3.3. *There is a bijection*

$$\{\textit{quadratic forms } q\colon V \to K\} \xrightarrow{\sim} \{\textit{symmetric bilinear forms } \beta\colon V \times V \to K\}.$$

$$q \longmapsto \beta_q(x, y) = \tfrac{1}{2}(q(x+y) - q(x) - q(y))$$

$$q_\beta(x) = \beta(x, x) \longleftarrow\!\shortmid \beta$$

Let $q\colon V \to K$ be a quadratic form with associated bilinear form $\beta = \beta_q$. Let $\{e_1, \ldots, e_n\}$ be a basis of V. The matrix of q with respect to the basis $\{e_i\}$ is defined as

$$A := (\beta(e_i, e_j))_{i,j} \in M_n(K).$$

For any vector $x = \sum_{i=1}^n x_i e_i$ we have

$$q(x) = \sum_{i,j} \beta(e_i, e_j) x_i x_j = x^t A x.$$

If $\{b_1, \ldots, b_n\}$ is another basis of V such that $b = X \cdot e$ with $X \in \mathrm{GL}_n(K)$, then the matrix of q with respect to b is $X^t A X$.

Definition 3.4.

1. *A quadratic form (V, q) is called* non-degenerate *if and only if its associated bilinear form is non-degenerate. This happens if and only if A as defined above is in $\mathrm{GL}_n(K)$ i.e.* $\det(A) \neq 0$.

2. *The* discriminant *of a non-degenerate quadratic form (V, q) is defined as*

$$\operatorname{disc} q := \det(A) \in K^\times / (K^\times)^2,$$

where A is the matrix of q with respect to some chosen basis.

Example 3.5. *The quadratic form $\mathrm{Nrd}_D \colon D \to K$ for a quaternion algebra D is non-degenerate and has discriminant 1. Writing $D \simeq \left(\frac{a,b}{K}\right)$ for $a, b \in K^\times$, the matrix of Nrd_D in this basis is*

$$\begin{pmatrix} 1 & & & \\ & -a & & \\ & & -b & \\ & & & ab \end{pmatrix}$$

and hence

$$\operatorname{disc}(\mathrm{Nrd}_D) = a^2b^2 \equiv 1 \bmod (K^\times)^2.$$

Remark 3.6. *If $q \colon V \to K$ is a non-degenerate quadratic form, two vectors $x, y \in V$ are called* orthogonal *if*

$$\beta_q(x, y) = \frac{1}{2}(q(x+y) - q(x) - q(y)) = 0.$$

In this case we write $x \perp y$.

Proposition 3.7. *Let (V, q) be a quadratic form. Then V admits an* orthogonal basis *i.e. a basis $\{e_1, \ldots, e_n\}$ of V such that $e_i \perp e_j$ for all $i \neq j$. In this basis, q is* diagonal *— we have*

$$q(x_1e_1 + \cdots + x_ne_n) = a_1x_1^2 + \cdots + a_nx_n^2$$

for some $a_1, \ldots, a_n \in K$.

Proof. Exercise.

Definition 3.8 (Isotropy)**.**

- *A vector $0 \neq x \in V$ is called* isotropic *if $q(x) = 0$ (equivalently, if $x \perp x$).*

- *A subspace $0 \neq U \subset V$ is called* isotropic *if $q(u) = 0$ for all $u \in U$.*

- *A quadratic form (q, V) is called* isotropic *if there exists $0 \neq x \in V$ which is isotropic.*

Example 3.9. *Consider $V = K^2$ with the quadratic form*

$$q(x_1, x_2) = x_1x_2.$$

The vectors

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \textit{and} \quad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

are isotropic; hence the subspaces they each generate are isotropic. Hence a non-degenerate quadratic form (V, q) can admit a basis of V which consists of isotropic vectors.

Definition 3.10. *Let D be a quaternion algebra. The sub-K-vector space*

$$D^0 := \{x \in D \colon \operatorname{Tr}(x) = x + x^* = 0\}$$

is called the space of pure quaternions. *D^0 is the orthogonal complement of $K \subset D$ under the quadratic form Nrd_D. Moreover, the restriction of the norm map*

$$\mathrm{Nrd}_D|_{D^0} \colon D^0 \to K$$

is a non-degenerate quadratic form with discriminant 1 on a 3-dimensional K-vector space.

Proposition 3.11. *The map $D \mapsto \mathrm{Nrd}_D|_{D^0}$ induces a bijection*

$$\{\textit{quaternion algebras } D/K\}_{/\sim} \xrightarrow{\sim} \left\{ \begin{matrix} \textit{non-degenerate quadratic forms} \\ (V,q) \textit{ with } \dim_K V = 3, \\ \operatorname{disc} q = 1 \end{matrix} \right\}_{/\sim}.$$

Under this bijection, $\left(\frac{a,b}{K}\right)$ is sent to $-ax^2 - by^2 + abz^2$. In particular, $M_2(K)$ corresponds to the unique (up to isomorphism) isotropic form $x^2 - y^2 - z^2$.

Proof. The final uniqueness claim follows from the fact that every quaternion algebra is either a division algebra or isomorphic to $M_2(K)$, and $x \mapsto xx^*$ can only be isotropic in the latter case.

Clearly the map $D \mapsto (D^0, \mathrm{Nrd}_D|_{D^0})$ is surjective, since we can choose a diagonal quadratic form (K^3, q') in any equivalence class, where $q'(x, y, z) = -ax^2 - by^2 + cz^2$, and then see that $\left(\frac{a,b}{K}\right)$ is sent to (K^3, q'). To prove that the map is injective, assume D, D' are quaternion algebras over K such that there exists an isomorphism of quadratic forms

$$f \colon (D^0, \mathrm{Nrd}_D|_{D^0}) \xrightarrow{\sim} (D'^0, \mathrm{Nrd}_{D'}|_{D'^0}).$$

Assume without loss of generality that $D \simeq \left(\frac{a,b}{K}\right)$; then we can see that

$$\mathrm{Nrd}_{D'}(f(i)) = \mathrm{Nrd}_D(i) = -a^2$$

and

$$\mathrm{Nrd}_{D'}(f(i)) = f(i)f(i)^* = -f(i)^2$$

since $x^* = -x$ for all pure quaternions $x \in D'^0$. Hence we get that $\mathrm{Nrd}_{D'}(f(i)) = a^2$ and $\mathrm{Nrd}_{D'}(f(j)) = b^2$. We further observe that quadratic form isomorphisms preserve orthogonality, hence $f(i), f(j), f(ij)$ are all orthogonal. Moreover, we have

$$\mathrm{Nrd}_{D'}(f(i) + f(j)) = \mathrm{Nrd}_{D'}(f(i)) + \mathrm{Nrd}_{D'}(f(j));$$

expanding out the LHS yields

$$f(j)f(i)^* + f(i)f(j)^* = -f(j)f(i) - f(i)f(j) = 0,$$

so $f(i)f(j) = -f(j)f(i)$. We further can see that $f(i)f(j)$ must also be orthogonal to $f(i)$ and $f(j)$:

$$\begin{aligned}
\mathrm{Nrd}_{D'}(f(i)f(j) + f(i)) &= (f(i)f(j) + f(i))(f(j)f(i) - f(i)) \\
&= f(i)f(j)^2 f(i) - f(i)^2 + f(i)f(j)f(i) - f(i)f(j)f(i) \\
&= \mathrm{Nrd}_{D'}(f(i)f(j)) + \mathrm{Nrd}_{D'}(f(i)) \\
\mathrm{Nrd}_{D'}(f(i)f(j) + f(j)) &= \mathrm{Nrd}_{D'}((f(i)f(j) + f(j))^*) \\
&= (f(j)f(i) - f(j))(f(i)f(j) + f(j)) \\
&= f(j)f(i)^2 f(j) - f(j)^2 + f(j)f(i)f(j) - f(j)f(i)f(j) \\
&= \mathrm{Nrd}_{D'}(f(i)f(j)) + \mathrm{Nrd}_{D'}(f(j)).
\end{aligned}$$

As D'^0 has dimension three, the orthogonality conditions imply that $f(i)f(j) = \alpha f(ij)$ for some $\alpha \in K^\times$. Comparing reduced norms, we see that $\alpha^2 = 1$ so $\alpha \in \{\pm 1\}$.

If $\alpha = 1$ then setting $f(1) = 1$ we can extend f to a K-algebra isomorphism $D \xrightarrow{\sim} D'$.

If $\alpha = -1$ then $f(ij) = -f(i)f(j) = f(j)f(i)$, so setting $f(1) = 1$ and composing with the anti-involution $x \mapsto x^*$ gives a K-algebra isomorphism $D \xrightarrow{\sim} D'$.

Definition 3.12 (Hasse invariant and Hilbert symbol)**.** *Let D be a quaternion algebra. The* Hasse invariant *of D is defined as*

$$\varepsilon(D) := \begin{cases} 1 & \text{if } D \simeq M_2(K) \\ -1 & \text{if } D \text{ is a division algebra.} \end{cases}$$

Let $a, b \in K^\times$; we define the Hilbert symbol

$$(\cdot,\cdot)_K \colon K^\times \times K^\times \to \{\pm 1\}$$

by the condition that $(a,b)_K = 1$ if and only if the quaternion algebra $\left(\frac{a,b}{K}\right)$ is split.

Remark 3.13. *We have $(a,b)_K = 1$ if and only if the equation*

$$ax^2 + by^2 = 1$$

has a solution with $x, y \in K$. This is called Hilbert's criterion *for the splitting of a quaternion algebra.*

Lemma 3.14. *Let $a, b, c, x, y \in K^\times$; then*

1. $(ax^2, by^2)_K = (a,b)_K$
2. $(a,b)_K = (b,a)_K$

3. *If K is a (p-adic) local field, then* $(a,b)_K(a,c)_K = (a,bc)_K$
4. $(a,-a)_K = (a,1-a)_K = 1$
5. *If* $(a,b)_K = 1$ *for all* $b \in K^\times$*, then* $a \in (K^\times)^2$
6. $(a,b)_K = 1$ *if and only if* $a \in \mathrm{Nm}_{L/K}(L^\times)$ *for* $L = K(\sqrt{b})$.

Proof.

1. Clear from $\left(\frac{ax^2,b}{K}\right) \simeq \left(\frac{a,b}{K}\right)$.
2. Clear from $\left(\frac{a,b}{K}\right) \simeq \left(\frac{b,a}{K}\right)$.
3. This follows by using 2. and 6. and the fact that $\mathrm{Nm}_{L/K}(L^\times)$ has index 2 in $K^\times$ for L/K a quadratic extension of local fields. Local class field theory kills this last fact; alternatively we can directly compute this this for K with odd residue characteristic by taking advantage of the fact that the quadratic extensions are $K(\sqrt{\pi}), K(\sqrt{a}), K(\sqrt{\pi a})$ for π a uniformizer and $a \in \mathcal{O}_K^\times$ a lift of a non-square class modulo π.
4. $x = 1, y = -1$ is a solution to $ax^2 + (1-a)y^2 = 1$ and since $\mathrm{char}(K) \neq 2$ we can find $x, y \in K$ solving
$$\begin{aligned} x + y &= \frac{1}{a} \\ x - y &= 1 \end{aligned}$$
so that $a(x+y)(x-y) = ax^2 - ay^2 = 1$.
5. By 6. we see that $\mathrm{Nm}_{L/K}(L^\times) = K^\times$ for $L = K(\sqrt{a})$, but if L/K is a degree 2 extension then the norm image has index 2 per the computation in 3. Hence $L = K$ and $a \in (K^\times)^2$.
6. $L = K$ if and only if $b \in (K^\times)^2$, and in this case it is clear that $(a,b)_K = 1$ for all $a \in K^\times = \mathrm{Nm}_{L/K}(L^\times)$. On the other hand, if $b \notin (K^\times)^2$ we can rearrange $ax^2 + by^2 = 1$ to
$$a = \frac{1}{x^2} - b\frac{y^2}{x^2} = \mathrm{Nm}_{L/K}\left(\frac{1}{x} + \sqrt{b}\frac{y}{x}\right)$$
so $a \in \mathrm{Nm}_{L/K}$ if and only if $(a,b)_K = 1$, modulo the fact that we need $x \neq 0$ in Hilbert's criterion for the implication
$$(a,b)_K = 1 \implies a \in \mathrm{Nm}_{L/K}(L^\times).$$
In the remaining case that $x = 0$, however, we see that $b \in (K^\times)^2$ and we are back in the original case, so we are done.

Example 3.15. *Let $K/\mathbf{Q}_p$ be a finite extension with $p > 2$. We have*

$$K^\times/(K^\times)^2 = \{1, e, \pi, \pi e\}$$

where π is a uniformizer and $e \in \mathcal{O}_K^\times$ is such that $K(\sqrt{e})/K$ is unramified.

The table of Hilbert symbols $(a, b)_K$ is

$(a,b)_K$	1	e	π	πe
1	1	1	1	1
e	1	1	-1	-1
π	1	-1	ε	$-\varepsilon$
πe	1	-1	$-\varepsilon$	ε

where $\varepsilon \in \{\pm 1\}$ depends on the extension $K/\mathbf{Q}_p$.

Definition 3.16. *Let (V, q) be a quadratic form over K. Write q in diagonal form as*

$$q(x_1, \dots, x_n) = a_1 x_1^2 + \dots + a_n x_n^2.$$

The Hasse invariant *of q is defined as*

$$\varepsilon(q) := \prod_{i<j} (a_i, a_j)_K \in \{\pm 1\}.$$

Remark 3.17. *The Hasse invariant depends only on the equivalence class of q, not on the specific isomorphism $q \sim a_1 x_1^2 + \dots + a_n x_n^2$.*

Theorem 3.18 (Classification of quadratic forms over a local field)**.** *Let $K = \mathbf{C}$ or a finite extension of $\mathbf{Q}_p$. Then two non-degenerate quadratic forms (V, q) and (V', q') are isomorphic if and only if*

- $\dim V = \dim V'$
- $\operatorname{disc} q = \operatorname{disc} q'$
- $\varepsilon(q) = \varepsilon(q')$.

The proof of this is mostly encapsulated in the following

Theorem 3.19 (Representability for quadratic forms over a local field)**.** *Let $K = \mathbf{C}$ or a finite extension of $\mathbf{Q}_p$ and (V, q) a non-degenerate quadratic form. Then q represents $a \in K^\times/(K^\times)^2$ (i.e. there exists $v \in V$ such that $a = q(v)$) if and only if one of the following occurs:*

- $\dim V = 1$ *and* $a = \operatorname{disc} q$;
- $\dim V = 2$ *and* $(a, -\operatorname{disc} q)_K = \varepsilon(q)$;
- $\dim V = 3$ *and either* $a \neq -\operatorname{disc} q$ *or* $a = -\operatorname{disc} q$ *and* $(-1, -\operatorname{disc} q) = \varepsilon(q)$;
- $\dim V \geq 4$.

In particular, two quadratic forms represent the same elements of $K^\times$ if and only if they have the same dimension, discriminant, and Hasse invariant.

Proof of theorem 3.18. We induct on $n = \dim V$; the case $n = 1$ is trivial, so assume the theorem is true for dimension n and let $\dim V = n+1$. By theorem 3.19 there exists $a \in K^\times$ which is represented by both q and q'. As such we may write

$$q \sim ax^2 + p \quad \text{and} \quad q' \sim ax^2 + p'$$

where p, p' are non-degenerate quadratic forms of dimension n. We moreover have

$$\operatorname{disc} q = a \operatorname{disc} p = a \operatorname{disc} p' = \operatorname{disc} q'$$

and

$$\varepsilon(q) = (a, \operatorname{disc} p)\varepsilon(p) = (a, \operatorname{disc} p')\varepsilon(p') = \varepsilon(q'),$$

so

$$\operatorname{disc} p = \operatorname{disc} p' \quad \text{and} \quad \varepsilon(p) = \varepsilon(p').$$

Hence p and p' are isomorphic by the induction hypothesis, and it follows that q and q' are also isomorphic.

Proof of theorem 3.19. The theorem is tautological for $K = \mathbf{C}$ so assume K is a finite extension of $\mathbf{Q}_p$. We proceed by cases on $n = \dim V$:

$n = 1$: clear.

$n = 2$: write $q \sim a_1x_1^2 + a_2x_2^2$; then per Hilbert's criterion q represents a if and only if $(a_1a^{-1}, a_2a^{-1})_K = 1$. But

$$\begin{aligned}(a_1a^{-1}, a_2a^{-1})_K &= (a_1, a_2)_K(a_1, a^{-1})_K(a^{-1}, a_2)_K(a^{-1}, a^{-1})_K \\ &= \varepsilon(q)(a^{-1}, \operatorname{disc} q)_K(a^{-1}, -1)_K = \varepsilon(q)(a^{-1}, -\operatorname{disc} q)_K = \varepsilon(q)(a, -\operatorname{disc} q)_K.\end{aligned}$$

$n = 3$: write $q \sim a_1x_1^2 + a_2x_2^2 + a_3x_3^2$; q represents a if and only if there exists $b \in K^\times$ which is represented by both $a_1x_1^2 + a_2x_2^2$ and $ax_4^2 - a_3x_3^2$. By the previous case, this happens if and only if

$$(a_1, a_2)_K = (b, -a_1a_2)_K \quad \text{and} \quad (a, -a_3)_K = (b, aa_3)_K.$$

Defining

$$\begin{aligned} X &:= \left\{ b \in K^\times : (a_1, a_2)_K = (b, -a_1a_2)_K \right\} \\ Y &:= \left\{ b \in K^\times : (a, -a_3)_K = (b, aa_3)_K \right\}, \end{aligned}$$

we see q represents a if and only if $X \cap Y \neq \varnothing$. Evidently if $a_1a_2 = -aa_3$ and $(a_1, a_2)_K = -(a, -a_3)_K$ then $X \cap Y = \varnothing$. On the other hand, if $X \cap Y = \varnothing$ then upon consideration of the two homomorphisms

$$\begin{aligned} K^\times/(K^\times)^2 &\to \{\pm 1\} \\ x &\mapsto (x, -a_1a_2)_K \\ x &\mapsto (x, aa_3)_K \end{aligned}$$

and finiteness of $K^\times/(K^\times)^2$, we find that X and Y must be cosets of the respective kernels, which are equal and of index 2 in $K^\times/(K^\times)^2$. Hence $(a_1, a_2)_K = -(a, -a_3)_K$ and $(x, -a_1 a_2)_K = (x, a a_3)_K$ for each $x \in K^\times$. Nondegeneracy of the Hilbert symbol (lemma 3.14 5.) now implies $a_1 a_2 = -a a_3$. Summarising, we have that a is not represented by q if and only if $a_1 a_2 = -a a_3$ and $(a_1, a_2)_K = -(a, -a_3)_K$. The first condition is equivalent to $a = -\operatorname{disc} q$; if this holds then we have

$$\varepsilon(q) = (a_1, a_2)_K (a_1 a_2, a_3)_K = (a_1, a_2)_K (-a a_3, a_3)_K = (a_1, a_2)_K (a, a_3)_K.$$

Hence the second condition is equivalent to $\varepsilon(q) = -(-1, a) = -(-1, \operatorname{disc} q)$.

$n \geq 4$: clearly it is enough to show the case $n = 4$. As $[K^\times : (K^\times)^2] \geq 4$ we see from the $n = 2$ case of this theorem that a quadratic form of dimension ≥ 2 represents at least 2 classes in $K^\times/(K^\times)^2$. In particular, there exists $b \in K^\times$ distinct from $-a^{-1} \operatorname{disc} q$ which is represented by q. Writing $q \sim bx^2 + q'$, where q' has dimension 3, we see $\operatorname{disc} q' = b^{-1} \operatorname{disc} q \neq -a$. Hence by the previous case q' represents a, and so too does q.

4 The Global Picture

4.1 Group Cohomology

Recall: Let L/K be a finite Galois extension. There is an equivalence of categories

$$K\text{-Mod} \underset{-^G}{\overset{-\otimes_K L}{\rightleftarrows}} \left\{ \substack{W \in L\text{-Mod} \\ \text{with semilinear action of } \mathrm{Gal}(L/K)} \right\}$$

given by $V \mapsto V \otimes_K L$ and $W \mapsto W^G$. We want to use this to classify things defined in terms of linear algebra e.g. quadratic forms, quaternion algebras.

Fix the following set up: Let V be a K-vector space, $V^* = \mathrm{Hom}_K(V, K)$ the dual, and $x \in V^{\otimes p} \otimes V^{*\otimes q}$ for some $p, q \in \mathbf{N}$. Two such pairs (V, x) and (V', x') are called isomorphic over K if there exists a K-linear isomorphism $f\colon V \xrightarrow{\sim} V'$ such that $f(x) = x'$. We write this as $(V, x) \simeq_K (V', x')$. We get isomorphisms $V^{\otimes p} \xrightarrow{\sim} V'^{\otimes p}$ and $V'^{*\otimes q} \xrightarrow{\sim} V^{*\otimes q}$. Given (V, x) we are interested in

$$\mathcal{E}_{(V,x)}(L/K) \coloneqq \left\{(V', x')\colon (V'_L, x'_L) \simeq_L (V_L, x_L)\right\}_{/\simeq_K}.$$

where $V_L \coloneqq V \otimes_K L$ and $x_L \coloneqq x \otimes 1 \in V_L^{\otimes p} \otimes V_L^{*\otimes q} = (V^{\otimes p} \otimes V^{*\otimes q}) \otimes L$.

Example 4.1.

$p = q = 0$: *Then $\mathcal{E}_{(V,x)}(L/K)$ is the set of isomorphism classes of K-vector spaces that become isomorphic to V upon base change to L. This has precisely one element since vector spaces of the same dimension are isomorphic and base change preserves dimension.*

$p = 1, q = 2$: *Say D/K is a quaternion algebra (or more generally a central simple algebra). e.g. $D = M_2(K)$ or $M_n(K)$. We have a multiplication map $D \otimes_K D \to D$ i.e. $m \in \mathrm{Hom}_K(D^{\otimes 2}, D) = D \otimes_K D^{*\otimes 2}$. Then $\mathcal{E}_{(D,m)}$ is the isomorphism classes of quaternion algebras over K which become isomorphic to D upon base change to L.*

$p = 0, q = 2$: *Assume* $\mathrm{char}\, K \neq 2$. *Let $q\colon V \to K$ be a quadratic form with corresponding bilinear form $\beta\colon V \times V \to K$. Then $\beta \in V^{*\otimes 2}$. Then $\mathcal{E}_{(V,\beta)}(L/K)$ is equivalence classes of quadratic forms over K which become equivalent to q upon base change to L. Over $\bar{K}$, every nondegenerate quadratic form on an n-dimensional vector space, every quadratic form is isomorphic to $q_0 = x_1^2 + \cdots + x_n^2$. Then $\mathcal{E}_{(K^n, q_0)}(\bar{K}/K)$ is the isomorphism classes of nondegenerate quadratic forms of dimension n over K.*

Given (V, x) let us write

$$A_K \coloneqq \mathrm{Aut}_K(V, x) = \{f \in \mathrm{GL}(V)\colon f(x) = x\}$$

and similarly $A_L \coloneqq \mathrm{Aut}_L(V_L, x_L)$. Assume L/K is Galois; note that $G = \mathrm{Gal}(L/K)$ acts on A_L via $\sigma(f) \coloneqq (x \mapsto \sigma \circ f \circ \sigma^{-1}(x))$ for $f \in A_L$ and $\sigma \in G$.

Here σ acts on $V \otimes_K L$ by $\sigma(v \otimes \lambda) = v \otimes \sigma(\lambda)$. **Caution:** The action of G on A_L is an action of a possibly non-abelian group on a possibly non-abelian group. In particular, A_L is not necessarily a module over the group algebra $\mathbf{Z}[G]$.

Definition 4.2. *Let G be a group acting on a group A i.e. there is a group homomorphism $G \to \operatorname{Aut}(A)$.*

1. *A (1-)cocycle is a map $c\colon G \to A$ ($g \mapsto c_g$) such that $c_{gh} = c_g \cdot gc_h$.*
2. *Two cocycles c, c' are called equivalent if there exists $a \in A$ such that $c'_g = a^{-1} \cdot c_g \cdot (ga)$ for all $g \in G$.*

The set of equivalence classes of 1-cocycles is denoted $H^1(G, A)$. Note that $H^1(G, A)$ is a pointed set with distinguished element the class of $c\colon g \mapsto 1$.

If $\varphi\colon A \to B$ is a G-equivariant group homomorphism, we obtain a map of pointed sets $H^1(G, A) \to H^1(G, B)$.

Remark 4.3. *If A is an abelian group then $H^1(G, A)$ is also an abelian group (the distinguished point being 0) and it agrees with the usual derived functor definition of group cohomology.*

Proposition 4.4. *In the setup from earlier, there is a canonical bijection of pointed sets*

$$\mathcal{E}_{(V,x)}(L/K) \xrightarrow{\sim} H^1(\operatorname{Gal}(L/K), A_L).$$

Example 4.5.

$p = q = 0$: *$V = K^n$, then $A_L = \mathrm{GL}_n(L)$. $\operatorname{Gal}(L/K)$ acts on $\mathrm{GL}_n(L)$ by acting on the entries of the matrix. Then*

$$H^1(\operatorname{Gal}(L/K), \mathrm{GL}_n(L)) = \{*\}\,;$$

this can be shown directly by a generalized version of Hilbert's Satz 90.[24]

[24] We will use this fact in the proof of proposition 4.4, so it is important that this can be proven independently.

$p = 1, q = 2$: *let $V = D = M_2(K)$, $x = m$ multiplication. Then A_L is the set of L-algebra automorphisms of $M_2(L)$, and each of these automorphisms is inner, hence $A_L = \mathrm{PGL}_2(L)$. The Galois group again acts on A_L via the entries of the matrix. Thus the K-isomorphism classes of quaternion algebras over K are in bijection with*

$$H^1(\operatorname{Gal}_K, \mathrm{PGL}_2(\bar{K})).$$

Similarly, the isomorphism classes of quaternion algebras over K which become isomorphic to $M_2(L)$ upon base change to L are in bijection with $H^1(\operatorname{Gal}(L/K), \mathrm{PGL}_2(L))$. In the case $K = \mathbf{R}$ or a finite extension of $\mathbf{Q}_p$, we have

$$H^1(\operatorname{Gal}_K, \mathrm{PGL}_2(\bar{K})) = \{[M_2(K)], [D]\}$$

where D is a non-split quaternion algebra over K.

$p = 0, q = 2$: *let* $\beta\colon V \times V \to K$ *be a symmetric bilinear form corresponding to a quadratic form* $Q\colon V \to K$. *Then* $A_L = O_\beta(L)$ *is the orthogonal group over* L *defined by* β. *Then* $\mathrm{Gal}(L/K)$ *acts on the entries of the matrices. Then isomorphism classes of quadratic forms which become isomorphic to* Q *upon base change to* L *are in bijection with* $H^1(\mathrm{Gal}(L/K), O_\beta(L))$. *For* $L = \bar{K}$, *we have that isomorphism classes of nondegenerate* n*-dimensional bilinear forms are in bijection with* $H^1(\mathrm{Gal}_K, O_n)$ *where* O_n *is the "standard" orthogonal group on* $\bar{K}^n$.

Proof of proposition 4.4. We define the map

$$\vartheta\colon \mathcal{E}_{(V,x)}(L/K) \to H^1(\mathrm{Gal}(L/K), A_L)$$

as follows: let (V', x') represent an element of $\mathcal{E}_{(V,x)}(L/K)$. There exists an isomorphism $f\colon V_L \xrightarrow{\sim} V'_L$ such that $f(x_L) = x'_L$. Set $c_f\colon \mathrm{Gal}(L/K) \to A_L$ to be the cocycle

$$g \mapsto f^{-1} \circ (gf) = f^{-1} \circ g \circ f \circ g^{-1}.$$

Since x is defined over K, g preserves x_L, hence it follows that $c_f(g)$ preserves x_L. Moreover, c_f is a 1-cocycle. A direct computation shows that choosing a different representative of $[(V', x')]$ or a different isomorphism $f'\colon V_L \xrightarrow{\sim} V'_L$ gives an equivalent cocycle. Thus $\vartheta\colon [(V', x')] \mapsto [c_f]$ is a well-defined map

$$\mathcal{E}_{(V,x)}(L/K) \to H^1(\mathrm{Gal}(L/K), A_L).$$

To show ϑ is injective, assume we have (V', x') and (V'', x'') and isomorphisms $f'\colon V_L \xrightarrow{\sim} V'_L$ and $f''\colon V_L \xrightarrow{\sim} V''_L$ such that $c_{f'}$ is equivalent to $c_{f''}$. Then there exists $h \in A_L$ such that

$$c_{f''}(g) = h^{-1} \circ c_{f'}(g) \circ (gh).$$

After precomposition of f'' with h, we may assume $c_{f''} = c_{f'}$, i.e.

$$f'^{-1} \circ \sigma(f') = f''^{-1} \circ \sigma(f'')$$

for all $\sigma \in \mathrm{Gal}(L/K)$. This is equivalent to

$$f'' \circ f'^{-1} = \sigma(f'' \circ f'^{-1})$$

for all $\sigma \in \mathrm{Gal}(L/K)$, so the L-isomorphism $f'' \circ f'^{-1}\colon V'_L \xrightarrow{\sim} V''_L$ descends to a K-isomorphism $V' \xrightarrow{\sim} V''$ mapping $x' \mapsto x''$. Hence (V', x') and (V'', x'') are equivalent in $\mathcal{E}_{(V,x)}(L/K)$ and ϑ is injective.

For surjectivity, let $c\colon \mathrm{Gal}(L/K) \to A_L$ be a 1-cocycle. Then

$$c'\colon \mathrm{Gal}(L/K) \xrightarrow{c} A_L \hookrightarrow \mathrm{Aut}_L(V_L)$$

is a cocycle with values in $\mathrm{Aut}_L(V_L)$. By the generalized Satz 90 (the computation $H^1(\mathrm{Gal}(L/K), \mathrm{GL}_n(L)) = \{*\}$), c' is equivalent to the trivial cocycle. That is, there exists an L-automorphism f such that

$$c_\sigma = c'_\sigma = f^{-1} \circ \sigma(f).$$

Then $f\colon V_L \to V_L$ extends to

$$V_L^{\otimes p} \otimes_L V_L^{*\otimes q} \to V_L^{\otimes p} \otimes_L V_L^{*\otimes q}.$$

We claim that $x'_L := f(x_L) \in V_L^{\otimes p} \otimes_L V_L^{*\otimes q}$ is $\mathrm{Gal}(L/K)$-invariant. Then letting $x' = x'_L$, by construction we have $\vartheta((V, x')) = c$. Indeed,

$$\sigma(x'_L) = \sigma(f(x_L)) = \sigma(f)(\sigma(x_L)) = \sigma(f)(x_L) = f \circ c_\sigma(x_L) = f(x_L) = x'_L.$$

4.2 *Quaternion algebras over a global field*

Theorem 4.6 (Hilbert Reciprocity)**.** *Let F be a number field and $a, b \in F^\times$; then $(a,b)_v := (a,b)_{F_v} = 1$ for all but finitely many places $v \in V_F$. Hence the product*

$$\prod_{v \in V_F} (a,b)_v$$

is defined. Furthermore, the product is always equal to 1.

Proof for $F = \mathbf{Q}$. Write the Hilbert symbol $(\cdot,\cdot)_p$ for $\mathbf{Q}_p$ (p odd) in terms of the Legendre symbol $\left(\frac{\cdot}{p}\right)$. For $a = p^\alpha u$, $b = p^\beta v$ where $\alpha, \beta \in \mathbf{Z}$ and $u, v \in \mathbf{Z}_p^\times$, we have

$$(a,b)_p = (-1)^{\alpha\beta\frac{p-1}{2}} \left(\frac{u}{p}\right)^\beta \left(\frac{v}{p}\right)^\alpha.$$

To see this, we may assume $\alpha, \beta \in \{0, 1\}$ since the Hilbert symbol is invariant up to squares.

If $\alpha = \beta = 0$ then $(a,b)_p = (u,v)_p = 1$ since every quaternion algebra over $\mathbf{F}_p$ splits.

If $\alpha = 1, \beta = 0$ then $vy^2 = 1$ has a solution mod p if and only if $\left(\frac{v}{p}\right) = 1$, and any solution lifts to a solution of $pux^2 + vy^2 = 1$ in $\mathbf{Z}_p$.

If $\alpha = 0, \beta = 1$ this is the same as the previous case.

If $\alpha = \beta = 1$ the bilinearity of the Hilbert symbol and previous cases give us

$$(a,b)_p = (pu, pv)_p = (p,p)_p (u,p)_p (p,v)_p (u,v)_p = (p,p)_p \left(\frac{u}{p}\right)\left(\frac{v}{p}\right),$$

so it suffices to compute $(p,p)_p$. For this we note that Hilbert's criterion is equivalent to asking whether $p = x^2 + y^2$ has solutions in $\mathbf{Q}_p$, and this is further equivalent to when $x^2 + y^2$ is isotropic over $\mathbf{F}_p$. But this latter condition is classically equivalent to $p \equiv 1 \bmod 4$ or $(-1)^{(p-1)/2} = 1$.

Since $(\cdot,\cdot)_v$ is bilinear for all places v, it suffices for proving Hilbert reciprocity to assume a, b are either -1 or a prime.

If $a = b = -1$: then $(-1,-1)_p = 1$ for all $p \neq 2$ by the previous expression in Legendre symbols. Over $\mathbf{R}$, $(-1,-1)_\infty = -1$ since $(-1,-1)_\infty$ represents the non-split algebra $\mathbf{H}$ (the Hamilton quaternions). At 2 we can also check that $(-1,-1)_2 = -1$, as -1 fails to be a sum of two squares in $\mathbf{Q}_2$ by considerations modulo 4.

If $a = -1, b = l$: If $l = 2$ then it is clear that $(-1,2)_v = 1$ for all v. If $l \neq 2$ then $(-1,l)_v = 1$ for $v \neq 2, l$ and $(-1,l)_l = \left(\frac{-1}{l}\right) = (-1)^{\frac{l-1}{2}}$. At $v = 2$ we can check that $x^2 - ly^2 = -1$ has no solutions modulo 4 for $l \equiv 3 \bmod 4$. The case where $l \equiv 1 \bmod 4$ is trickier, but we will still be able to lift the solution mod 4 with the appropriate version of Hensel's lemma. Hence $(-1,l)_2 = (-1,l)_l = \left(\frac{-1}{l}\right)$.

If $a = l, b = p$ (l and p prime): If $l = p$ then

$$(-1,l)_v = (l,-1)_v = (l,-1)_v(l,-l)_v = (l,l)_v$$

for all v and we are in the previous case. Otherwise when $p = 2$ we have $(l,2)_v = 1$ for all $v \neq 2, l$ and $(l,2)_l = (l,2)_2 = \left(\frac{2}{l}\right)$. If $l \neq p$ are both odd, then $(l,p)_v = 1$ for $v \neq 2, l, p$ and $(l,p)_l = \left(\frac{p}{l}\right)$ and $(l,p)_p = \left(\frac{l}{p}\right)$. Moreover, we can do a (somewhat involved) computation to show that $(l,p)_2 = (-1)^{\frac{(l-1)(p-1)}{4}}$ – the key to this is that Hensel will still allow lifting solutions mod 8 to $\mathbf{Z}_2$. Hence the product formula here is equivalent to Quadratic Reciprocity.

Proof for F any number field. We locally describe the Hilbert symbol, however instead of Legendre symbols we use the Artin symbol:

$$(a,b)_v \sqrt{b} = \left(\frac{F_v(\sqrt{b})/F_v}{a}\right)\sqrt{b}.$$

For fixed $a, b \in F^\times$, a is a unit at almost all places $v \in V_F$ and $F(\sqrt{b})/F$ is unramified at almost all places, and for such v, we see $(a,b)_v = 1$ so the product of the Hilbert symbols is well-defined. Local-global compatibility of class field theory states that the global Artin map is the product of the local Artin symbols, hence

$$\sqrt{b}\prod_{v \in V_F}(a,b)_v = \sqrt{b}\left(\frac{F(\sqrt{b})/F}{(a)_v}\right).$$

But the global Artin map is trivial on F, so we obtain

$$\prod_{v \in V_F}(a,b)_v = 1.$$

Theorem 4.7 (Classification of quaternion algebras)**.** *Let F be a number field*

1. *Let D/F be a quaternion algebra. Then $\mathrm{Ram}(D) := \{v \in V_F : D_v = D \otimes_F F_v$ is non-split$\}$ is finite and even.*

2. *Let $D_1, D_2/F$ be quaternion algebras such that $D_{1,v} \simeq D_{2,v}$ for all $v \in V_F$; then $D_1 \simeq D_2$.*

3. *Let $S \subset V_F$ be a finite even subset which contains no complex places. Then there exists a quaternion algebra D/F with $\mathrm{Ram}(D) = S$ which is unique up to isomorphism.*

Equivalently *Let D_v/F_v be quaternion algebras such that $S := \{v \in V_F \colon D_v \not\simeq M_2(F_v)\}$ is finite and even. Then there exists a (unique up to isomorphism) quaternion algebra D/F such that $D \otimes_F F_v \simeq D_v$ for all v.*

Remark 4.8. *For $F = \mathbf{Q}$ and a prime p, this theorem says that there exists a unique quaternion algebra over $\mathbf{Q}$ which is ramified exactly at p and ∞. If $p = 2$ then this is $\left(\frac{-1,-1}{\mathbf{Q}}\right)$. If $p \equiv 3 \bmod 4$ this is $\left(\frac{-1,-p}{\mathbf{Q}}\right)$. If $p \equiv 5 \bmod 8$ this is $\left(\frac{-2,-p}{\mathbf{Q}}\right)$. For $p \equiv 1 \bmod 8$ this is $\left(\frac{-l,-p}{\mathbf{Q}}\right)$ where l is a prime which is inert in $\mathbf{Q}(\sqrt{-1})$ and $\mathbf{Q}(\sqrt{p})$. Showing this is left as an exercise.*

Proof for $F = \mathbf{Q}$. As $D \simeq \left(\frac{a,b}{F}\right)$ for some $a, b \in F^\times$, the first assertion is a restatement of Hilbert Reciprocity. The uniqueness in 3. is a consequence of 2. To prove the existence part of 3., let

$$N := \prod_{p \in S \setminus \{\infty\}} p$$

and define

$$\varepsilon := \begin{cases} 1 & \text{if } \infty \notin S \\ -1 & \text{if } \infty \in S. \end{cases}$$

Let Q be a rational prime such that εQ is not a square modulo all odd primes in S, and such that

$$\varepsilon Q \equiv \begin{cases} 1 \bmod 8 & \text{if } 2 \notin S \\ 5 \bmod 8 & \text{if } 2 \in S. \end{cases}$$

Set $D := \left(\frac{\varepsilon N, \varepsilon Q}{\mathbf{Q}}\right)$ and check that it has the correct ramification behaviour.

If $v = \infty$: then $(\varepsilon N, \varepsilon Q)_\infty = \varepsilon$.

If $v \nmid NQ$: then $N, Q \in \mathbf{Z}_v^\times$ so $(\varepsilon N, \varepsilon Q)_v = 1$.

If $v \in S$: then $Q \in \mathbf{Z}_v^\times$, $\sqrt{\varepsilon Q}$ generates an unramified quadratic extension of $\mathbf{Q}_v$, and $N \in \mathbf{Z}_v$ is a uniformizer, so $(\varepsilon N, \varepsilon Q)_v = -1$.

If $v = 2$: then we can use the Hilbert symbol computations from the proof of Hilbert reciprocity to show that $(\varepsilon N, \varepsilon Q) = 1$ if and only if $2 \notin S$.

If $v = Q$: we must have $(\varepsilon N, \varepsilon Q)_Q = 1$ by Hilbert reciprocity.

We are left to prove part 2. This is a special case of the Hasse–Minkowski theorem:

Theorem 4.9. *Let F be a number field*

1. *If q_1, q_2 are quadratic forms on F^n. Then $q_{1,v} \simeq q_{2,v}$ for all v implies $q_1 \simeq q_2$.*
2. *If q is a quadratic for on F^n such that q_v is isotropic for all v, then q is isotropic over F.*

Proof. See [Rav24, §2].

Proof for F any number field. We only need to prove the existence statement of 3. By weak approximation for $\mathbf{G}_m$ we can choose $a \in F^\times$ such that $E := F(\sqrt{a})$ is a quadratic extension of F ramified at all real places and inert at all finite places in S. For each finite $v \in S$ fix a uniformizer $\varpi_v \in \mathcal{O}_{F_v}$ and for each real $v \in S$ set $\varpi_v := -1$. As E_v/F_v is unramified, we know ϖ_v is not a norm in this extension; hence it is not an adèlic norm either i.e. $\varpi_v \notin \mathrm{Nm}_{E/F}(\mathbf{A}_E^\times)$. Moreover, we have $\varpi_v \notin F^\times \mathrm{Nm}_{E/F}(\mathbf{A}_E^\times)$; else there exists $x \in F^\times$ which is a local norm at all places other than v and not a local norm at v. The Hasse norm theorem for quadratic extensions then implies x is a global norm, hence a local norm at v which is a contradiction.

Now set
$$z := \prod_{v \in S} \varpi_v;$$
Artin reciprocity gives
$$[\mathbf{A}_F^\times : F^\times \mathrm{Nm}_{E/F}(\mathbf{A}_E^\times)] = [\mathbf{A}_F^\times / F^\times : \mathrm{Nm}_{E/F}(\mathbf{A}_E^\times / E^\times)] = 2,$$
so $z \in F^\times \mathrm{Nm}_{E/F}(\mathbf{A}_E^\times)$ as S is even. Writing $z = by$ for $b \in F^\times$ and $y \in \mathrm{Nm}_{E/F}(\mathbf{A}_E^\times)$, we define
$$D := \left(\frac{a,b}{F}\right)$$
and we claim $\mathrm{Ram}(D) = S$.

For $v \in S$ finite: by design, $F_v(\sqrt{a})$ is the unique unramified quadratic extension of F_v and $b = \varpi_v y_v^{-1} \notin \mathrm{Nm}_{E_v/F_v}(E_v^\times)$. Hence b has odd valuation, so $D_v \simeq \left(\frac{a,\varpi_v}{F_v}\right)$ is the unique non-split quaternion algebra over F_v.

For $v \in S$ real: a was chosen to be negative in F_v and $b = -y_v$ is also negative; hence D_v is the Hamilton quaternions.

For $v \notin S$: if v is ramified in E/F then $b \in \mathrm{Nm}_{E_v/F_v}(E_v^\times)$ so $(a,b)_v = 1$ and D_v splits. Otherwise $E_v \simeq F_v^2$ embeds into D_v, so D_v cannot be a division algebra.

A linear algebraic group G/F such that
$$\text{Ш}(F, G(\bar{F})) := \ker\left(H^1(F, G(\bar{F})) \xrightarrow{\text{res}} \prod_{v \in V_F} H^1(F_v, G(\bar{F}_v))\right) = 0$$

is said to satisfy the Hasse principle. This is not true for all G, but it is true for O_n (Hasse–Minkowski) and any semisimple G of adjoint type (such as PGL_n). Patching together this fact for PGL_n for all n, the cohomology long exact sequence induced by

$$1 \longrightarrow \mathbf{G}_m \longrightarrow \mathrm{GL}_n \longrightarrow \mathrm{PGL}_n \longrightarrow 1,$$

and Hilbert's Satz 90 gives

$$Ш^2(F, \bar{F}^\times) := \ker\left(H^2(F, \bar{F}^\times) \to \prod_{v \in V_F} H^2(F_v, \bar{F_v}^\times)\right) = 0.$$

Recalling that $H^2(F, \bar{F}^\times) \simeq \mathrm{Br}(F)$, we have a short exact sequence

$$0 \longrightarrow \mathrm{Br}(F) \longrightarrow \bigoplus_{v \in V_F} \mathrm{Br}(F_v) \xrightarrow{\sum \mathrm{inv}_v} \mathbf{Q}/\mathbf{Z} \longrightarrow 0.$$

Moreover, specializing to 2-torsion, we have

$$0 \longrightarrow \mathrm{Br}(F)[2] \longrightarrow \bigoplus_{v \in V_F} \mathrm{Br}(F_v)[2] \xrightarrow{\sum \mathrm{inv}_v} \tfrac{1}{2}\mathbf{Z}/\mathbf{Z} \longrightarrow 0,$$

and for local fields and hence number fields we can identify $\mathrm{Br}(K)[2]$ with the isomorphism classes of quaternion algebras over K.

4.3 *The group associated to a quaternion algebra and its adèlic points*

Let D/K be a quaternion algebra. Consider the functor $G = G_D \colon K\text{-}\mathsf{Alg} \to \mathsf{Grp}$ which maps $R \mapsto (D \otimes_K R)^\times$. e.g. if $D = M_2(K)$ then $D \otimes_K R = M_2(R)$, and hence $G_D(R) = \mathrm{GL}_2(R)$.

Lemma 4.10. *The functor G is representable by an affine group scheme of finite type/K i.e. G is a linear algebraic group. Moreover, $G \times_{\operatorname{Spec} K} \operatorname{Spec} \bar{K} \simeq \mathrm{GL}_{2,\bar{K}}$.*

Proof. The "moreover" part is clear since

$$(G \times_{\operatorname{Spec} K} \operatorname{Spec} \bar{K})(R) = (D \otimes_K R)^\times = D \otimes_K (\bar{K} \otimes_{\bar{K}} R))^\times = \mathrm{GL}_2(R),$$

and since the functors of points agree on all $\bar{K}$-algebras, the group schemes themselves are isomorphic by Yoneda. To show representability, note that $\mathbf{D} \colon K\text{-}\mathsf{Alg} \to \mathsf{Set}$ given by $R \mapsto D \otimes_K R$ is representable by $\mathbf{A}^4_K$ once we choose a K-vector space isomorphism $D \simeq K^4$. The reduced norm defines a map $\mathbf{D} \to \mathbf{A}^1_K$ which is a morphism of K-algebraic varieties. But then G is precisely the preimage of $\mathbf{G}_{m,K} = \mathbf{A}^1_K \setminus \{0\}$ i.e. $G \simeq \mathbf{D} \times_{\mathbf{A}^1_K} \mathbf{G}_{m,K}$.

Now let $K/\mathbf{Q}_p$ be a local field and D/K a quaternion algebra. Let $\mathcal{O}_D \subset D$ be a maximal order. Then we can consider $\mathcal{G} = \mathcal{G}_{\mathcal{O}_D} \colon \mathcal{O}_K\text{-}\mathsf{Alg} \to \mathsf{Grp}$ given by $R \mapsto (\mathcal{O}_D \otimes_{\mathcal{O}_K} R)^\times$. The same argument shows that $\mathcal{G}_{\mathcal{O}_D}$ is representable by an affine finite-type $\mathcal{O}_K$-group scheme. Moreover, this group scheme is smooth by a fibrewise argument. Evidently $\mathcal{G} \times_{\operatorname{Spec} \mathcal{O}_K} \operatorname{Spec} K = G$ and if D is split then $\mathcal{G}_D \simeq \mathrm{GL}_{2,\mathcal{O}_K}$. If D is not split then $\mathcal{G}_D \times_{\operatorname{Spec}\mathcal{O}_K} \operatorname{Spec}\mathcal{O}_{\breve{K}}$ is *not* isomorphic to $\mathrm{GL}_{2,\mathcal{O}_{\breve{K}}}$. So over $\mathcal{O}_{\breve{K}}$ (or $\mathcal{O}_L$ for any extension L/K such that D_L splits), $\mathcal{G} \times_{\operatorname{Spec}\mathcal{O}_K} \operatorname{Spec}\mathcal{O}_L$ and $\mathrm{GL}_{2,\mathcal{O}_L}$ are two different integral models of $\mathrm{GL}_{2,\breve{K}}$. In fact, $\mathcal{G} \times_{\operatorname{Spec}\mathcal{O}_K} \operatorname{Spec}\mathcal{O}_{\breve{K}}$ is the Iwahori group scheme.

4.3.1 *The Iwahori group scheme*

The Iwahori group scheme $\mathcal{I}/\mathcal{O}_K$ is the group of automorphisms of a nontrivial lattice chain in K^2. More specifically, let $\Lambda_1 = \mathcal{O}_K^2 = \mathcal{O}_K e_1 \oplus \mathcal{O}_K e_2$ and $\Lambda_2 := \varpi^{-1}\mathcal{O}_K e_1 \oplus \mathcal{O}_K e_2$, where $\varpi \in \mathcal{O}_K$ is a fixed uniformizer. We have a chain of embeddings

$$\Lambda_1 \xhookrightarrow{\iota_1} \Lambda_2 \xhookrightarrow{\iota_2} \varpi^{-1}\Lambda_1.$$

We define $\mathcal{I}\colon \mathcal{O}_K\text{-}\mathsf{Alg} \to \mathsf{Grp}$ by sending R to the subgroup of $\operatorname{Aut}_R(\Lambda_1 \otimes_{\mathcal{O}_K} R) \times \operatorname{Aut}_R(\Lambda_2 \otimes_{\mathcal{O}_K} R)$ consisting of (g_1, g_2) such that

1. $(\iota_1 \otimes \mathrm{id}_R) \circ g_1 = g_2 \circ (\iota_1 \otimes \mathrm{id}_R)$
2. $(\iota_2 \otimes \mathrm{id}_R) \circ g_2 = (\varpi^{-1} g_1) \circ (\iota_2 \otimes \mathrm{id}_R)$.

From the definition, we can see that $\mathcal{I}$ is representable by a closed subgroup scheme of $\mathrm{GL}_{2,\mathcal{O}_K} \times \mathrm{GL}_{2,\mathcal{O}_K}$. Moreover, if R is ϖ-torsionfree then 1. holds if and only if 2. holds. We can use this to check that

$$\mathcal{I}(\mathcal{O}_K) = \left\{ \begin{pmatrix} a & b \\ \varpi c & d \end{pmatrix} : a, b, c, d \in \mathcal{O}_K;\ a, d \in \mathcal{O}_K^\times \right\} \subset \mathrm{GL}_2(\mathcal{O}_K).$$

$\mathcal{I}(\mathcal{O}_K) \subset \mathrm{GL}_2(\mathcal{O}_K) \subset \mathrm{GL}_2(K)$ is a compact open subgroup called the *Iwahori subgroup*. It is the preimage of the upper triangular matrices in $\mathrm{GL}_2(\kappa)$ under the natural projection $\mathcal{O}_K \twoheadrightarrow \kappa := \mathcal{O}_K/\varpi$, hence finite index and open in $\mathrm{GL}_2(\mathcal{O}_K)$.

We can consider the special fibre of $\mathcal{I}$: $\bar{\mathcal{I}} := \mathcal{I} \times_{\operatorname{Spec}\mathcal{O}_K} \operatorname{Spec}\kappa$. On κ-algebras, we have

$$\bar{\mathcal{I}}(R) = \left\{ \left(\begin{pmatrix} a & b \\ 0 & d \end{pmatrix}, \begin{pmatrix} a & 0 \\ c & d \end{pmatrix} \right) \in \mathrm{GL}_2(R)^2 \colon a, d \in R^\times,\ b, c \in R \right\}.$$

Hence, we have a Cartesian diagram of κ-schemes

$$\begin{tikzcd} \bar{\mathcal{I}} \arrow[r] \arrow[d] & T \arrow[d, "\Delta"] \\ B \times B \arrow[r, "p_1 \times p_2"'] & T \times T \end{tikzcd}$$

where $B \subset \mathrm{GL}_{2,\kappa}$ is the Borel subgroup of upper triangular matrices, $T \subset B$ is the maximal torus (subgroup of diagonal matrices), and $p_1, p_2\colon B \to T$ are given by

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \mapsto \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} a' & b' \\ 0 & d' \end{pmatrix} \mapsto \begin{pmatrix} d' & 0 \\ 0 & a' \end{pmatrix}.$$

Thus $\bar{\mathcal{I}}$ is a 4-dimensional reduced (hence smooth) κ-group scheme; however it is not reductive – its maximal reductive quotient is T. The upshot here is that $\mathcal{I}$ is a smooth finite-type affine group scheme over $\mathcal{O}_K$ with generic fibre $\mathrm{GL}_{2,K}$. But we see from the special fibre that $\mathcal{I}$ is not isomorphic to $\mathrm{GL}_{2,\mathcal{O}_K}$!

Exercise 6. *Let D/K be a non-split quaternion algebra with maximal order $\mathcal{O}_D$. Show that $\mathcal{G}_{\mathcal{O}_D} \times_{\operatorname{Spec} \mathcal{O}_K} \operatorname{Spec} \mathcal{O}_L \simeq \mathcal{I}_{\mathcal{O}_L}$ for any (finite) extension L/K such that D_L is split. This implies that $\mathcal{G}_{\mathcal{O}_D}$ is a smooth $\mathcal{O}_K$-group scheme.*

4.3.2 *The adèlic points of G_D*

Let F be a number field and D/F a quaternion algebra; let $G = G_D$ be the corresponding group scheme as defined earlier. We want to talk about models of G over $\mathcal{O}_F$ and about $G(\mathbf{A}_F)$.

Definition 4.11.

1. *Let W be an F-vector space; an $\mathcal{O}_F$-lattice $\Lambda \subset W$ is a finitely generated $\mathcal{O}_F$-submodule Λ such that the canonical map $\Lambda \otimes_{\mathcal{O}_F} F \to W$ is an isomorphism. Note that in this case Λ is a finite projective $\mathcal{O}_F$-module of rank $\dim_F W$.*

2. *Let D/F be a quaternion algebra; an order $\mathcal{O}_D \subset D$ is an $\mathcal{O}_F$-lattice that is a subring.*

There is always an order $\mathcal{O}_D \subset D$; write $D \simeq \left(\frac{a,b}{F}\right)$ and scale so that $a, b \in \mathcal{O}_F \cap F^\times$. Then $\mathcal{O}_F \oplus \mathcal{O}_F i \oplus \mathcal{O}_F j \oplus \mathcal{O}_F k$ is an order of D.

Lemma 4.12. *Let W be an F-vector space and $\Lambda_0 \subset W$ be an $\mathcal{O}_F$-lattice. Then there is a bijection*

$$\{\Lambda \subset W\ \mathcal{O}_F\text{-lattices}\} \xrightarrow{\sim} \left\{(\Lambda_v)_{v \in V_f} : \begin{smallmatrix} \Lambda_v \subset W_v = W \otimes_F F_v \text{ is an } \mathcal{O}_{F_v}\text{-lattice,} \\ \Lambda_v = \Lambda_{0,v} \text{ for almost all } v \end{smallmatrix}\right\}$$

given by $\Lambda \mapsto (\Lambda_v = \Lambda \otimes_{\mathcal{O}_F} \mathcal{O}_{F_v})_v$.

Proof. Fix a basis $\{x_1, \ldots, x_n\}$ of $W \simeq F^n$ so that Λ_0 is identified with

$$\mathcal{O}_F x_1 \oplus \cdots \oplus \mathcal{O}_F x_n.$$

We construct an inverse to the map $\Lambda \mapsto (\Lambda_v)_v$ as follows. Given a sequence of lattices $(\Lambda_v)_{v \in V_f}$ satisfying the stated conditions, we note that in particular we must have $\Lambda_v = \mathcal{O}_{F_v}^n$ for almost all v. For such v which do not satisfy this, we can write

$$\Lambda_v = \pi_v^{e_{1,v}} \mathcal{O}_{F_v} x_1 \oplus \cdots \oplus \pi_v^{e_{n,v}} \mathcal{O}_{F_v} x_n$$

for $e_{i,v} \in \mathbf{Z}$ and π_v some uniformizer in $\mathcal{O}_{F_v}$. For each place v, let $\mathfrak{p}_v \subset \mathcal{O}_F$ be the unique prime ideal such that $|\cdot|_{\mathfrak{p}_v} \in v$. We define

$$\Lambda := \left(\bigoplus_v \mathfrak{p}_v^{e_{1,v}}\right) x_1 \oplus \cdots \oplus \left(\bigoplus_v \mathfrak{p}_v^{e_{n,v}}\right) x_n.$$

Then it is clear that this is a lattice in W and this assignment is inverse to the map given by tensoring.

Consider the group $G = G_D$ over F; let $\mathcal{O}_D \subset D$ be a maximal order. Then $\mathcal{G} = \mathcal{G}_{\mathcal{O}_D} \colon \mathcal{O}_F\text{-}\mathsf{Alg} \to \mathsf{Grp}$ given by $R \mapsto (R \otimes_{\mathcal{O}_F} \mathcal{O}_D)^\times$ is representable by a smooth group scheme over $\mathcal{O}_F$ such that $\mathcal{G} \times_{\operatorname{Spec} \mathcal{O}_F} \operatorname{Spec} F \simeq G$ and $\mathcal{G} \times_{\operatorname{Spec} \mathcal{O}_F} \operatorname{Spec} \mathcal{O}_{F_v} \simeq \mathcal{G}_{\mathcal{O}_{D,v}}$.

Definition 4.13. *The adelic points of G are*

$$G(\mathbf{A}_F) := \prod_{v \in V_F}{}' (G(F_v), \mathcal{G}(\mathcal{O}_{F_v})).$$

Remark 4.14.

1. *This does not depend on the choice of maximal order $\mathcal{O}_D$ used to define the model $\mathcal{G}$.*
2. *$G(\mathbf{A}_F) \hookrightarrow (D \otimes_F \mathbf{A}_F) \times \mathbf{A}_F$ by $g \in (D \otimes_F \mathbf{A}_F)^\times \mapsto (g, \operatorname{Nrd}(g)^{-1})$ is a closed embedding.*

Theorem 4.15. *Let D/F be a quaternion algebra that is a division algebra. Let $m < M \in \mathbf{R}_{>0}$,*

$$Y := \{x \in G(\mathbf{A}_F) \colon m \leq \|\operatorname{Nrd(x)}\| \leq M\}.$$

Then the image of Y in $G(F)\backslash G(\mathbf{A}_F)$ is compact. In particular, $G(F)\backslash G^1(\mathbf{A}_F)$ is compact, where $G^1(\mathbf{A}_F) = \{x \in G(\mathbf{A}_F) \colon \|\operatorname{Nrd}(x)\| = 1\}$.

Proof. We follow the proof of the analogous fact for $\mathbf{G}_m$ (theorem 1.31). Let μ denote the product Haar measure on $D \otimes_F \mathbf{A}_F$; as we have

$$D\backslash(D \otimes_F \mathbf{A}_F) \simeq (F\backslash\mathbf{A}_F)^4$$

as topological groups, it follows that $D\backslash(D \otimes_F \mathbf{A}_F)$ is compact and has finite measure under the pushforward of μ. Let $C_0 \subset D \otimes_F \mathbf{A}_F$ be a compact subset such that $\mu(C_0) > M\mu(D\backslash(D \otimes_F \mathbf{A}_F))$, and define

$$C_1 := \{y - z \colon y, z \in C_0\} \quad \text{and} \quad C := C_1 \cap Y.$$

Clearly C_1 is compact and by continuity of Nrd on D and lemma 1.34, Y is closed in $D \otimes_F \mathbf{A}_F$. Hence C is closed in C_1 and itself compact. Moreover, the subspace topologies from $D \otimes_F \mathbf{A}_F$ and $G(\mathbf{A}_F)$ agree on Y by *lemma* 1.34, so the natural map $C \to G(F)\backslash G(\mathbf{A}_F)$ is continuous. We claim that the image of C coincides with the image of Y under this map. For $x \in Y$, we clearly have (by checking locally and using that the Haar measure is a product of local Haar measures) $\mu(x^{-1}C_0) = |\operatorname{Nrd}(x)|^{-1}\mu(C_0)$; but by assumption

$$|\operatorname{Nrd}(x)|^{-1}\mu(C_0) \geq M^{-1}\mu(C_0) > \mu(D\backslash(D \otimes_F \mathbf{A}_F)).$$

Hence there exist distinct $y, z \in C_0$ so that

$$u := x^{-1}(y - z) \in D;$$

as D is a division algebra it follows that in fact $u \in D^\times$. Moreover,

$$|\operatorname{Nrd}(y - z)| = |\operatorname{Nrd}(u)|\,|\operatorname{Nrd}(x)| = |\operatorname{Nrd}(x)| \in [m, M]$$

and hence $y - z \in C$.

Theorem 4.16. *Let D be a quaternion algebra over a number field F and let $v \in V_{F,\infty}$ be an infinite place. Then there exists a compact $C \subset G(\mathbf{A}_F)$ such that $G(\mathbf{A}_F) = G(F)G(F_v)C$.*

Proof. There are two cases: when D is non-split the previous theorem applies. Consider

$$\{|\mathrm{Nrd}(g_v)|_v : g_v \in G(F_v)\} \subset \{\|\mathrm{Nrd}(g)\| : g \in G(\mathbf{A}_F)\} \subset \mathbf{R}_{>0}.$$

Since v is an infinite place, $G(F_v)$ is one of $\mathrm{GL}_2(\mathbf{C})$, $\mathrm{GL}_2(\mathbf{R})$, or $\mathbf{H}^\times$. We have $|\mathrm{Nrd}(g_v)|_v = |\det(g_v)|^2$, $|\det(g_v)|$, or

$$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 \quad \text{where} \quad g_v = \alpha + \beta i + \gamma j + \delta k \in \mathbf{H}^\times,$$

respectively. In all cases, we get

$$\{|\mathrm{Nrd}(g_v)|_v : g_v \in G(F_v)\} = \mathbf{R}_{>0},$$

hence $G(F_v)G^1(\mathbf{A}_F) = G(\mathbf{A}_F)$ and the claim follows from the preceding theorem.

In the second case, $G = \mathrm{GL}_{2,F}$. Let $B \subset \mathrm{GL}_2$ be the closed subgroup scheme of upper triangular matrices, let $N \subset B$ be the closed subgroup scheme of unipotent upper triangular matrices, and let $T \subset B$ be the closed subgroup scheme of diagonal matrices. Note that $N \simeq \mathbf{G}_a$ and $T \simeq \mathbf{G}_m^2$; we have already proven this statement for $\mathbf{G}_a$ and $\mathbf{G}_m$ and hence for N and T. Multiplication induces an isomorphism of schemes $T \times N \xrightarrow{\sim} B$ (this is not an isomorphism of group schemes!). In particular, $T(F_w) \times N(F_w) \to B(F_w)$ and $T(\mathbf{A}_F) \times N(\mathbf{A}_F) \to B(\mathbf{A}_F)$ are homeomorphisms. For any $w \in V_F$, we claim there exists a compact $C_w \subset G(F_w)$ such that $G(F_w) = B(F_w)C_w$. For w finite we may take $C_w = \mathrm{GL}_2(\mathcal{O}_{F_w})$, for w real we can take $C_w = O_2(\mathbf{R})$, and for w complex we can take $C_w = U_2$ the unitary matrices. Then $C := \prod_w C_w$ satisfies

$$G(\mathbf{A}_F) = B(\mathbf{A}_F)C = T(\mathbf{A}_F)N(\mathbf{A}_F)C.$$

By the corresponding statements for $\mathbf{G}_a$ and $\mathbf{G}_m$, there exist compact $C' \subset T(\mathbf{A}_F)$ and $C'' \subset N(\mathbf{A}_F)$ such that $T(\mathbf{A}_F) = T(F)T(F_v)C'$ and $N(\mathbf{A}_F) = N(F)N(F_v)C''$. Then

$$\begin{aligned} T(\mathbf{A}_F)N(\mathbf{A}_F) &= T(F)T(F_v)C'N(\mathbf{A}_F) = T(F)T(F_v)N(\mathbf{A}_F)C' \\ &= T(F)T(F_v)N(F)N(F_v)C''C' = T(F)N(F)T(F_v)N(F_v)C''C' = B(F)B(F_v)C''. \end{aligned}$$

Hence $G(\mathbf{A}_F) = B(F)B(F_v)C''C'C \subset G(F)G(F_v)C''C'C$.

Corollary 4.17 (Finiteness of the class number)**.** *Let D/F be a quaternion algebra, $S \subset V_F$ a finite set of places with $V_{F,\infty} \subset S$. Let $K \subset G(\mathbf{A}_F^S)$ be a compact open subgroup. Then $G(F)\backslash G(\mathbf{A}_F^S)/K$ is finite.*

Proof. Let $v \in S$ be infinite and let $C \subset G(\mathbf{A}_F)$ be compact such that $G(\mathbf{A}_F) = G(F)G(F_v)C$. Then

$$C \longrightarrow G(\mathbf{A}_F) \longrightarrow\!\!\!\!\rightarrow G(\mathbf{A}_F^S) \longrightarrow\!\!\!\!\rightarrow G(F)\backslash G(\mathbf{A}_F^S)$$

is surjective, hence $G(F)\backslash G(\mathbf{A}_F^S)/K$ is compact. As K is open the double quotient is also discrete and hence finite.

Note that $G(F)\backslash G(\mathbf{A}_F^S)$ might be non-Hausdorff, hence why we formulate the corollary about $G(F)\backslash G(\mathbf{A}_F^S)/K$ instead. Choosing a maximal order $\mathcal{O}_D \subset D$, we can write (similarly to the case of $\mathbf{G}_m$)

$$D^\times \backslash \prod_{v \in V_f} D_v^\times / \prod_{v \in V_f} \mathcal{O}_{D_v}^\times = G(F) \backslash G(\mathbf{A}_F^\infty)/K$$

where $K \coloneqq \prod_{v \in V_f} \mathcal{O}_{D_v}^\times \subset G(\mathbf{A}_F)$ is a maximal compact open subgroup. Moreover, defining invertible fractional ideals as $\mathcal{O}_F$-lattices $I \subset D$ which are right $\mathcal{O}_D$-modules, the LHS can be identified with invertible fractional ideals modulo the principal fractional ideals.

The most general form of the previous theorem is the following:

Theorem 4.18 (Borel–Prasad)**.** *Let G/F be a linear algebraic group, $S \subset V_F$ a finite set of places with $V_{F,\infty} \subset S$, and $K \subset G(\mathbf{A}_F^S)$ a compact open subgroup. Then $G(F)\backslash G(\mathbf{A}_F^S)/K$ is finite.*

Let D/F be a quaternion algebra. We can consider the closed subgroup scheme $H \subset G$ defined as a functor of points by

$$H(R) \coloneqq \{g \in G(R) \colon \mathrm{Nrd}(g) = 1\}.$$

Note if $D = M_2(F)$ then $H = \mathrm{SL}_{2,F}$. At a completion F_v, either $H(F_v) = \mathrm{SL}_2(F_v)$ or $H(F_v)$ is compact.

Theorem 4.19 (Strong approximation for H)**.** *Let $S \subset V_F$ be finite such that $S \cap V_{F,\infty} \neq \varnothing$. Assume $H(\mathbf{A}_{F,S}) = \prod_{v \in S} H(F_v)$ is not compact (equivalently, $S \not\subset \mathrm{Ram}(D)$). Then $H(F)H(\mathbf{A}_{F,S}) \subset H(\mathbf{A}_F)$ is dense.*

Note that if $H(\mathbf{A}_{F,S})$ is compact then $H(F)H(\mathbf{A}_{F,S}) \subsetneq H(\mathbf{A}_F)$ is closed, as $H(F)$ is closed. The corresponding statement for G in place of H is wrong: the theorem fails for $\mathbf{G}_m$, which is a closed subgroup of G.

Proof. We first note that we easily have weak approximation: for $\mathbf{D}$ and G_D this follows from weak approximation for $\mathbf{G}_a$. For H we note that $[G(E), G(E)] = H(E)$ for any field extension E/F; if $G(E) = \mathrm{GL}_2(E)$ we have $[\mathrm{GL}_2(E), \mathrm{GL}_2(E)] = [\mathrm{SL}_2(E), \mathrm{SL}_2(E)] = \mathrm{SL}_2(E)$ from simplicity of $\mathrm{PSL}_2(E)$. If D_E is non-split then given $z \in H(E)$ we can find $x \in E(z) \subset D_E$ such that $z = x\bar{x}^{-1}$ by Hilbert's Satz 90; the Skolem–Noether theorem gives $y \in D_E^\times = G(E)$ such that $\bar{x} = yxy^{-1}$ for all $x \in D_E$. But then $z = [x, y] \in [G(E), G(E)]$. Now given $T \subset V_F$ finite and $(x_v)_v \in H(\mathbf{A}_{F,T})$ we write $x_v = y_v z_v y_v^{-1} z_v^{-1}$ for some $y_v, z_v \in G(F_v)$ for each $v \in T$. Weak approximation for G allows us to approximate $(y_v)_v$ and $(z_v)_v$ by some $y, z \in G(F)$, and clearly we obtain $yzy^{-1}z^{-1} \to (x_v)_v$ by continuity of multiplication.

To prove strong approximation it clearly suffices to show that $H(F)$ is dense in $H(\mathbf{A}_F^S)$. To this end let Z denote the closure of $H(F)$ in $H(\mathbf{A}_F^S)$; it is enough to establish that $H(F_v) \subset Z$ for almost all $v \notin S$. If $H = \mathrm{SL}_2$ this is easy: $\mathrm{SL}_2(F_v)$ is generated by the subgroups of unipotent upper and lower triangular matrices, $U(F_v)$ and $L(F_v)$. As U and L are closed subgroups of SL_2, it is enough to prove the same fact for U and L, but now since $U \simeq L \simeq \mathbf{G}_a$ we are done by strong approximation for $\mathbf{G}_a$.

5 Automorphic Forms

5.1 Modular Forms

Recall the following: let $k \in \mathbf{Z}$ and $\Gamma \subset \mathrm{SL}_2(\mathbf{Z})$ be a congruence subgroup e.g. $\Gamma = \Gamma_1(N)$ or $\Gamma = \Gamma_0(N)$ where

$$\Gamma_1(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbf{Z}) \colon a \equiv d \equiv 1 \bmod N,\, c \equiv 0 \bmod N \right\}$$
$$\Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbf{Z}) \colon c \equiv 0 \bmod N \right\}.$$

Let $\mathcal{M}_k(\Gamma)$ denote the space of (holomorphic) modular forms of weight k and level Γ and $\mathcal{S}_k(\Gamma) \subset \mathcal{M}_k(\Gamma)$ the subspace of cusp forms. These are finite-dimensional $\mathbf{C}$-vector spaces. We also have the following variant: let χ be a Dirichlet character mod N; let

$$\mathcal{M}_k(\Gamma_0(N), \chi) := \left\{ f \colon \mathfrak{h} \to \mathbf{C} \text{ holomorphic} \colon \forall \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N),\, f\left(\frac{az+b}{cz+d}\right) = \chi(d)(cz+d)^k f(z) \right\}.$$

For $l \nmid N$ where l is prime, we have the Hecke operators T_l on $\mathcal{M}_k(\Gamma_0(N), \chi)$ which satisfy $T_l T_{l'} = T_{l'} T_l$ for all $l, l' \nmid N$ and are normal with respect to the Petersson inner product on $\mathcal{M}_k(\Gamma_0(N), \chi)$. In particular, the Hecke operators are simultaneously orthogonally diagonalizable.

Definition 5.1. *A modular/cusp form in $\mathcal{M}_k(\Gamma_0(N), \chi)$ (resp. $\mathcal{S}_k(\Gamma_0(N), \chi)$) is called an eigenform if it is a simultaneous eigenvector for all T_l, $l \nmid N$.*

Note that an eigenform f defines a character $\lambda_f \colon \mathcal{H}_N \to \mathbf{C}$ such that $T_l f = \lambda_f(T_l) f$ for all $l \nmid N$. Here $\mathcal{H}_N$ is either $\mathbf{Z}[T_l \colon l \nmid N]$ or the subring of $\mathrm{End}(\mathcal{M}_k(\Gamma_0(N), \chi))$ generated by the Hecke operators.

For $N' \mid N$ there are two canonical maps $\mathcal{M}_k(\Gamma_1(N')) \to \mathcal{M}_k(\Gamma_1(N))$:

1. The canonical inclusion ι_1
2. For $N = dN'$ let $\gamma_d = \begin{pmatrix} d & 0 \\ 0 & 1 \end{pmatrix}$; then $\Gamma_1(N) \subset \gamma_d^{-1} \Gamma_1(N') \gamma_d$. Map $f \in \mathcal{M}_k(\Gamma_1(N'))$ to $\iota_2(f) := (z \mapsto \det \gamma_d^{k/2} f(\gamma_d z))$.

The space of old forms is

$$\mathcal{M}_k(\Gamma_1(N))^{\mathrm{old}} := \sum_{N' \mid N} \mathrm{im}(\iota_{1,N',N}) \oplus \mathrm{im}(\iota_{2,N',N}).$$

The space of newforms is the orthogonal complement of $\mathcal{M}_k(\Gamma_1(N))^{\mathrm{old}}$ with respect to the Petersson inner product. We have similar definitions for $\mathcal{M}_k(\Gamma_0(N), \chi), \mathcal{S}_k(\Gamma_1(N)), \mathcal{S}_k(\Gamma_0(N), \chi)$.

Lemma 5.2. *$\mathcal{M}_k(\Gamma_1(N))^{\mathrm{old}}$ and $\mathcal{M}_k(\Gamma_1(N))^{\mathrm{new}}$ are stable under the Hecke operators T_l for all $l \nmid N$.*

Proof. ι_1, ι_2 commute with T_l so $\mathcal{M}_k(\Gamma_1(N))^{\text{old}}$ is stable under the Hecke action. Normality of the T_l with respect to the inner product means $\mathcal{M}_k(\Gamma_1(N))^{\text{new}}$ is also stable.

Proposition 5.3 (Multiplicity one)**.** *Let $N \in \mathbf{Z}$, $\lambda \colon \mathcal{H}_N \to \mathbf{C}$ be a character such that*

$$\mathcal{S}_k(\Gamma_1(N))^{\text{new}}[\lambda] := \{f \in \mathcal{S}_k(\Gamma_1(N))^{\text{new}} \colon T_l f = \lambda(T_l) f \, \forall l \nmid N\}$$

is nonzero. Then $\dim_{\mathbf{C}} \mathcal{S}_k(\Gamma_1(N))[\lambda] = 1$.

5.2 *From modular forms to automorphic forms*

Let $\chi \colon (\mathbf{Z}/N)^\times \to \mathbf{C}^\times$ be a Dirichlet character; we can view χ as a character $\chi \colon \hat{\mathbf{Z}}^\times \to \mathbf{C}$ by passing through the quotient $\hat{\mathbf{Z}}^\times \twoheadrightarrow (\mathbf{Z}/N\mathbf{Z})^\times$. We can write

$$\mathbf{A}^\times = \mathbf{A}_{\mathbf{Q}}^\times = \mathbf{Q}^\times \cdot \mathbf{R}_{>0} \cdot \hat{\mathbf{Z}}^\times$$

as

$$\left| \mathbf{Q}^\times \backslash \mathbf{A}_f^\times / \hat{\mathbf{Z}}^\times \right| = h_{\mathbf{Q}} = 1;$$

hence we can extend a Dirichlet character to an idèle class character

$$\chi \colon \mathbf{Q}^\times \backslash \mathbf{A}^\times \to \mathbf{C}^\times$$

given by $a \cdot x \cdot y \mapsto \chi(y)$ for $a \in \mathbf{Q}^\times, x \in \mathbf{R}_{>0}, y \in \hat{\mathbf{Z}}^\times$.

Now that we have seen how to upgrade a Dirichlet character $\chi \colon (\mathbf{Z}/N\mathbf{Z})^\times \to \mathbf{C}^\times$ to a function $\mathbf{Q}^\times \backslash \mathbf{A}^\times \to \mathbf{C}$, let us look at how to relate modular forms to functions $\mathrm{GL}_2(\mathbf{Q}) \backslash \mathrm{GL}_2(\mathbf{A}) \to \mathbf{C}$. Set

$$\mathrm{GL}_2(\mathbf{R})^+ := \{g \in \mathrm{GL}_2(\mathbf{R}) \colon \det g > 0\}.$$

Lemma 5.4. *Let $K \subset \mathrm{GL}_2(\mathbf{A}_f)$ be a compact open subgroup such that* $\det K = \hat{\mathbf{Z}}^\times$. *Then* $\mathrm{GL}_2(\mathbf{A}) = \mathrm{GL}_2(\mathbf{Q}) \cdot \mathrm{GL}_2(\mathbf{R})^+ \cdot K$.

Proof. Exercise.

Given K as in the lemma we can define

$$\Gamma = \Gamma_K := \mathrm{GL}_2(\mathbf{Q}) \cap (\mathrm{GL}_2(\mathbf{R})^+ \cdot K) \subset \mathrm{SL}_2(\mathbf{Z}).$$

This is a congruence subgroup, e.g. $\Gamma_{K_0(N)} = \Gamma_0(N)$ where $K_0(N)$ is the preimage of the upper triangular matrices under the map $\mathrm{GL}_2(\hat{Z}) \twoheadrightarrow \mathrm{GL}_2(\mathbf{Z}/N\mathbf{Z})$. Similarly $\Gamma_{K_1(N)} = \Gamma_1(N)$ where $K_1(N)$ is the preimage of the unipotent upper triangular matrices under the same map. We get a bijection

$$\mathrm{GL}_2(\mathbf{Q}) \backslash \mathrm{GL}_2(\mathbf{A}) / K \xleftarrow{\sim} \Gamma_K \backslash \mathrm{GL}_2(\mathbf{R})^+.$$

Note that we can view modular forms of level Γ_K as functions on $\Gamma_K \backslash \mathrm{GL}_2(\mathbf{R})^+$ since $\mathrm{GL}_2(\mathbf{R})^+$ surjects on $\mathfrak{h}$ by $g_\infty \mapsto g_\infty \cdot i$. For $g \in \mathrm{GL}_2(\mathbf{R})^+, z \in \mathfrak{h}$ we set

$j(g,z) := \det(g)^{-1/2}(cz+d)$ where $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then j is a 1-cocycle for the $\mathrm{GL}_2(\mathbf{R})^+$-action on $\mathfrak{h}$.

Let $f \in \mathcal{S}_k(\Gamma_0(N), \chi)$ and define $\varphi_f \colon \mathrm{GL}_2(\mathbf{R})^+ \to \mathbf{C}$ by

$$g_\infty \mapsto f(g_\infty i) j(g_\infty, i)^{-k}.$$

We extend φ_f to $\varphi_f \colon \mathrm{GL}_2(\mathbf{A}) \to \mathbf{C}$ by

$$\gamma \cdot g_\infty \cdot u \mapsto \varphi_f(g_\infty)\chi(d),$$

where $\gamma \in \mathrm{GL}_2(\mathbf{Q}), g_\infty \in \mathrm{GL}_2(\mathbf{R})^+, u = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in K_0(N)$.

Note that $\varphi_f(\gamma g_\infty) = \varphi_f(g_\infty) \cdot \chi(d)$ for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N), g_\infty \in \mathrm{GL}_2(\mathbf{R})^+$. Also

$$\varphi_f(g_\infty z u_\infty) = \varphi_f(g_\infty)\mathrm{sgn}(z)^k e^{ki\vartheta}$$

where $z \in Z_\infty = \mathbf{R}^\times = Z\,\mathrm{GL}_2(\mathbf{R})$ and $u_\infty = \begin{pmatrix} \cos\vartheta & \sin\vartheta \\ -\sin\vartheta & \cos\vartheta \end{pmatrix} \in \mathrm{SO}_2(\mathbf{R}) \simeq S^1$.

Definition 5.5. *Let $N \in \mathbf{Z}$, $\chi \colon (\mathbf{Z}/N\mathbf{Z})^\times \to \mathbf{C}^\times$ a Dirichlet character. The space of automorphic cusp forms of level $K_0(N)$ and weight k is the subspace*

$$\mathcal{A}_0(\mathrm{GL}_2, k, K_0(N), \chi) \subset \mathbf{C}^{\mathrm{GL}_2(\mathbf{A})}$$

satisfying

i. *$\varphi(\gamma g u) = \chi(u)\varphi(g)$ for all $\gamma \in \mathrm{GL}_2(\mathbf{Q}), u \in K_0(N)$*

ii. *For all $g_f \in \mathrm{GL}_2(\mathbf{A}_f)$, the function $\varphi_1 \colon g \mapsto \varphi(g g_f)$ is C^∞ on $\mathrm{GL}_2(\mathbf{R})^+$ and $\varphi_1(zg) = \varphi_1(g)$ for all $z \in Z_\infty$ and $\varphi_1(g u_\infty) = e^{ki\vartheta}\varphi_1(g)$ for all*

$$u_\infty = \begin{pmatrix} \cos\vartheta & \sin\vartheta \\ -\sin\vartheta & \cos\vartheta \end{pmatrix} \in \mathrm{SO}_2(\mathbf{R}).$$

iii. *(Cuspidality)*

$$\int_{\mathbf{Q}\backslash\mathbf{A}} \varphi\left(\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \cdot g\right) dx = 0$$

for all $g \in \mathrm{GL}_2(\mathbf{A})$.

Proposition 5.6. *The map $f \mapsto \varphi_f$ defined above induces an isomorphism $\mathcal{S}_k(\Gamma_1(N), \chi) \to \mathcal{A}_0(\mathrm{GL}_2, k, K_0(N), \chi)$.*

5.3 Automorphic Forms on Quaternion Algebras

Let $D/\mathbf{Q}$ be a non-split quaternion algebra. Let $G = G_D$ be the reductive group associated to D. Let $Z = ZG$ denote the centre, which is isomorphic to $\mathbf{G}_m$.

Note that G is not split i.e. there is no split maximal torus (subgroup isomorphic to $\mathbf{G}_m^n$) defined over $\mathbf{Q}$. However there exists $E \hookrightarrow D$ where $E/\mathbf{Q}$ is a quadratic field extension such that $\operatorname{Res}_{E/\mathbf{Q}} \mathbf{G}_m \hookrightarrow G$ is a maximal torus. G is also not quasi-split so there is no Borel subgroup defined over $\mathbf{Q}$.

Proposition 5.7. *$G(\mathbf{Q})Z(\mathbf{A})\backslash G(\mathbf{A})$ is compact.*

Proof. For $m < M$, $m, M \in \mathbf{R}_{>0}$ the set

$$\{g \in G(\mathbf{A}) \colon m \leq |\operatorname{Nrd} g| \leq M\}$$

has compact image in $G(\mathbf{Q})\backslash G(\mathbf{A})$ and $\mathbf{A}^\times = Z(\mathbf{A}) \hookrightarrow G(\mathbf{A}) \stackrel{\mathrm{Nrd}}{\to} \mathbf{A}^\times$ is given by $x \mapsto x^2$. Hence the compact image surjects onto $G(\mathbf{Q})Z(\mathbf{A})\backslash G(\mathbf{A})$.

Let $D/\mathbf{Q}$ be a definite quaternion algebra i.e. a quaternion algebra such that $D \otimes_{\mathbf{Q}} \mathbf{R} \simeq \mathbf{H}$. Let G be the algebraic group attached to D and $\mathbf{G}_m \simeq Z \subset G$ the center. Let $\omega = \omega_\infty \otimes \omega_f \colon Z(\mathbf{A}) = \mathbf{A}^\times \to \mathbf{C}^\times$ be a unitary idèle class character. Here $\omega_\infty \colon \mathbf{R}^\times \to \mathbf{C}^\times$ and $\omega_f \colon \mathbf{A}_f^\times \to \mathbf{C}^\times$ are the infinite and finite parts of ω. Let $K \subset G(\mathbf{A}_f)$ be a compact open subgroup and define

$$L^2\left(G(\mathbf{Q})\backslash G(\mathbf{A}), \omega\right) := \left\{f \in L^2(G(\mathbf{Q})\backslash G(\mathbf{A})) \colon f(zg) = \omega(z)f(g)\, \forall g \in G(\mathbf{A}), z \in Z(\mathbf{A})\right\}.$$

$G(\mathbf{A})$ acts on this space by right translation. We can define the subspace

$$L^2\left(G(\mathbf{Q})\backslash G(\mathbf{A}), \omega, K\right) = \left\{f \in L^2\left(G(\mathbf{Q})\backslash G(\mathbf{A}), \omega\right) : f(gu) = f(g)\, \forall g \in G(\mathbf{A}), u \in K\right\};$$

$G(\mathbf{R})$ acts via right translation on this subspace. We want to understand the $G(\mathbf{A})$-representation $L^2(G(\mathbf{Q})\backslash G(\mathbf{A}), \omega)$ by decomposing as

$$L^2(G(\mathbf{Q})\backslash G(\mathbf{A}), \omega) = \widehat{\bigoplus_{\pi = \pi_\infty \otimes \pi_f}} (\pi_\infty \otimes \pi_f)^{m(\pi)}$$

where π_∞ is an irreducible representation of $G(\mathbf{R})$ with central character ω_∞. Since $Z(\mathbf{R})\backslash G(\mathbf{R})$ is compact, all π_∞ are finite dimensional over $\mathbf{C}$. π_f is an irreducible, smooth, admissible representation of $G(\mathbf{A}_f)$ and $m(\pi) \in \mathbf{Z}_{\geq 0}$.

Step 1: we can decompose $L^2(G(\mathbf{Q})\backslash G(\mathbf{A}), \omega, K)$ as a $G(\mathbf{R})$-representation:

$$L^2(G(\mathbf{Q})\backslash G(\mathbf{A}), \omega, K) = \widehat{\bigoplus_{\pi_\infty}} \pi_\infty \otimes \operatorname{Hom}_{G(\mathbf{R})}(\pi_\infty, L^2(G(\mathbf{Q})\backslash G(\mathbf{A}), \omega, K))$$

following the method for representations of finite groups. Since $G(\mathbf{R})$ is compact modulo its centre its representation theory behaves like that of finite groups. We have

$$\begin{aligned} &\operatorname{Hom}_{G(\mathbf{R})}(\pi_\infty, L^2(G(\mathbf{Q})\backslash G(\mathbf{A}), \omega, K)) \\ &= \{f \colon G(\mathbf{A}) \to \pi_\infty^* \colon f(\gamma z x u) = \omega(z) f(x)\, \forall \gamma \in G(\mathbf{Q}), z \in Z(\mathbf{A}), x \in G(\mathbf{A}), u \in K\}^{G(\mathbf{R})} \end{aligned}$$

where $G(\mathbf{R})$ acts as $(y \cdot f)(x) = y f(xy)$. This is also isomorphic to

$$S_{\pi_\infty}(G(\mathbf{A}_f), \omega_f, K) := \left\{f \colon G(\mathbf{A}_f) \to \pi_\infty^* \colon f(\gamma z x u) = \omega_f(z)\gamma^{-1} f(x)\, \forall z \in Z(\mathbf{A}_f), x \in G(\mathbf{A}_f), u \in K, \gamma \in G(\mathbf{Q})\right\},$$

which is a finite-dimensional $\mathbf{C}$-vector space since $G(\mathbf{Q})\backslash G(\mathbf{A}_f)/K$ is finite.

Step 2: Consider

$$\varinjlim_K L^2(G(\mathbf{Q})\backslash G(\mathbf{A}),\omega,K) \subset L^2(G(\mathbf{Q})\backslash G(\mathbf{A}),\omega);$$

this is the $G(\mathbf{A})$-subrepresentation consisting of all f such that $f(x,-)\colon G(\mathbf{A}_f) \to \mathbf{C}$ is locally constant. This is a dense subspace by usual measure-theoretic considerations. We have the decomposition

$$\varinjlim_K L^2(G(\mathbf{Q})\backslash G(\mathbf{A}),\omega,K) = \varinjlim_K \widehat{\bigoplus_{\pi_\infty}} \pi_\infty \otimes S_{\pi_\infty}(G(\mathbf{A}_f),\omega_f,K).$$

This contains as a dense subspace

$$\varinjlim_K \bigoplus_{\pi_\infty} \pi_\infty \otimes S_{\pi_\infty}(G(\mathbf{A}_f),\omega_f,K) = \bigoplus_{\pi_\infty} \pi_\infty \otimes \varinjlim_K S_{\pi_\infty}(G(\mathbf{A}_f),\omega_f,K)$$

where $\varinjlim_K S_{\pi_\infty}(G(\mathbf{A}_f),\omega_f,K)$ is a smooth admissible $G(\mathbf{A}_f)$-representation. If this decomposes as a direct sum of irreducible $G(\mathbf{A}_f)$-representation with finite multiplicities then

$$\bigoplus_{\pi=\pi_\infty\otimes\pi_f} (\pi_\infty \otimes \pi_f)^{m(\pi)}$$

is a dense subspace of $L^2(G(\mathbf{Q})\backslash G(\mathbf{A}),\omega)$ and we are done.

We assumed $D_{\mathbf{R}} \simeq \mathbf{H}$; what about other D? If D is non-split and $D_{\mathbf{R}} \simeq M_2(\mathbf{R})$ then there is still an analogue of the decomposition i.e. $L^2(G,\omega)$ decomposes as a completed direct sum where the $G(\mathbf{R})$-representations π_∞ are replaced by (g,K)-modules. If $D = M_2(\mathbf{Q})$ then the situation is much more complicated:

$$L^2(G(\mathbf{Q})\backslash G(\mathbf{A}),\omega) = L^2_{\text{cont}}(G(\mathbf{Q})\backslash G(\mathbf{A}),\omega) \oplus L^2_{\text{disc}}(G(\mathbf{Q})\backslash G(\mathbf{A}),\omega)$$

where L^2_{disc} is the maximal subrepresentation which has a decomposition as above and L^2_{cont} is mysterious.

5.4 *Smooth representations of locally profinite groups*

Definition 5.8. *A topological group H is called locally profinite if it contains an open profinite subgroup.*

Example 5.9. *For $K/\mathbf{Q}_p$ a finite extension,*

i. the additive group K contains $\mathcal{O}_K$;

ii. $K^\times$ contains $\mathcal{O}_K^\times$;

iii. $\mathrm{GL}_2(K) \supset \mathrm{GL}_2(\mathcal{O}_K)$;

iv. $D^\times \supset \mathcal{O}_D^\times$ for D/K a quaternion algebra.

For F a number field,

i. the finite adèles $\mathbf{A}_{F,f}$ *contain* $\prod_{v \in V_f} \mathcal{O}_{F_v}$*;*

ii. The finite idèles $\mathbf{A}_F^{\infty\times}$ *contain* $\prod_{v \in V_f} \mathcal{O}_{F_v}^\times$*;*

iii. $\mathrm{GL}_2(\mathbf{A}_{F,f}) \supset \prod_{v \in V_f} \mathrm{GL}_2(\mathcal{O}_{F_v})$*;*

iv. $G_D(\mathbf{A}_{F,f}) \supset \prod_{v \in V_f} (\mathcal{O}_D \otimes_{\mathcal{O}_F} \mathcal{O}_{F_v})^\times$ *for* $D/\mathbf{F}$ *a quaternion algebra.*

Definition 5.10. *Let* H *be a locally profinite group; a complex representation* $\pi\colon H \to \mathrm{GL}(V)$ *is called smooth if for all* $v \in V$*, the stabiliser* $\mathrm{Stab}_H\, v \subset H$ *is open.*

π *is called smooth admissible if in addition* $\dim_{\mathbf{C}} V^K < \infty$ *for all compact open subgroups* $K \subset H$*.*

Example 5.11. $C_c^\infty(H) = \{f\colon H \to \mathbf{C}$ *locally constant with compact support*$\}$ *with translation action by* H *is a smooth representation. For* $K/\mathbf{Q}_p$ *finite,* $\delta\colon K^\times \to \mathbf{C}^\times$ *is smooth if and only if* $\delta_{\mathcal{O}_K^\times}$ *factors through a finite quotient.*

In general it is hard to explicitly write down interesting smooth representations but

$$S_{\pi_\infty}(G(\mathbf{A}_f), \omega) := \varinjlim_{K \subset G(\mathbf{A}_f)} S_{\pi_\infty}(G(\mathbf{A}_f), \omega_f, K)$$

is a naturally occurring smooth admissible representation of $G(\mathbf{A}_f)$ as

$$S_{\pi_\infty}(G(\mathbf{A}_f), \omega)^K = S_{\pi_\infty}(G(\mathbf{A}_f), \omega_f, K)$$

is finite dimensional.

Given a locally profinite group H we can consider the representation $C_c^\infty(H)$; note that $f \in C_c^\infty(H)$ if and only if $f = \sum_{i=1}^n a_i \mathbf{1}_{A_i}$ where $a_i \in \mathbf{C}$ and $A_i \subset H$ are compact open. Fix a Haar measure μ on H; then

Lemma 5.12.

1. *The convolution* $C_c^\infty(H) \times C_c^\infty(H) \to C_c^\infty(H)$ *given by*

$$(f_1, f_2) \mapsto f_1 * f_2 := \left(g \mapsto \int_H f_1(h) f_2(gh^{-1})\, dh\right)$$

makes $C_c^\infty(H)$ *into a not necessarily commutative (unless* H *is commutative) and not necessarily unital (unless* H *is compact)* $\mathbf{C}$*-algebra.*

2. *Let* (π, V) *be a smooth representation of* H*. Then* $f \cdot v := \int_H f(h)\pi(h) \cdot v\, dh$ *makes* V *into a* $C_c^\infty(H)$*-module.*

Proof. Direct computation.

Let $K \subset H$ be a compact open subgroup. Then $e_K := \frac{1}{\mu(K)} \mathbf{1}_K \in C_c^\infty(H)$ is an idempotent since

$$e_K * e_K \colon g \mapsto \frac{1}{\mu(K)} \int_K e_K(gh^{-1})\, dh = \frac{1}{\mu(K)} \int_K \frac{1}{\mu(K)} \mathbf{1}_{gK}\, dh = \begin{cases} \frac{1}{\mu(K)} & \text{if } gK = K; \\ 0 & \text{if } gK \neq K \end{cases} = e_K.$$

If (π, V) is a smooth representation of H then $e_K \cdot V = V^K$. In particular, the unital $\mathbf{C}$-algebra $\mathcal{H}(H,K) := e_K C_c^\infty(H) e_K$ is precisely the subspace

$$\{f \in C_c^\infty(H) \colon f(k_1 h k_2) = f(h)\,\forall h \in H, k_1, k_2 \in K\}.$$

Given a smooth representation (π, V) the subspace $V^K = e_K V$ is naturally an $\mathcal{H}(H,K)$-module. This construction induces an equivalence of categories between the smooth representations of H and non-degenerate $C_c^\infty(H)$-modules. Here a $C_c^\infty(H)$-module M is non-degenerate if $M = \bigcup_{K \subset H} e_K M$ where K runs over compact open subgroups of H.

Proposition 5.13. *The map $(\pi, V) \mapsto V^K$ induces a bijection between irreducible smooth representations of H with $V^K \neq 0$ and simple non-degenerate $\mathcal{H}(H,K)$-modules.*

Let $\mathcal{A}_0(k, K, \omega_f)$ be the space of cuspidal automorphic forms defined earlier. $\mathcal{A}_0(k, \omega_f) := \varinjlim_K \mathcal{A}_0(k, K, \omega_f)$ is a smooth admissible representation of $\mathrm{GL}_2(\mathbf{A}_f)$. In order to understand the $\mathrm{GL}_2(\mathbf{A}_f)$-representation $S_{\pi_\infty}(\mathrm{GL}_2(\mathbf{A}_f), \omega_f)$ we have to understand the $\mathcal{H}(\mathrm{GL}_2(\mathbf{A}_f), K)$-module structure on $S_{\pi_\infty}(\mathrm{GL}_2(\mathbf{A}_f), \omega_f, K)$. Hence we need to understand the $\mathbf{C}$-algebras $\mathcal{H}(\mathrm{GL}_2(\mathbf{A}_f), K)$.

Theorem 5.14 (Satake isomorphism for $\mathrm{GL}_2/\mathbf{Q}_p$)**.** *Let $K = \mathrm{GL}_2(\mathbf{Z}_p) \subset \mathrm{GL}_2(\mathbf{Q}_p)$; then $\varphi\colon \mathbf{C}[X_1, X_2^{\pm 1}] \to \mathcal{H}(\mathrm{GL}_2(\mathbf{Q}_p), K)$ defined by*

$$X_1 \mapsto \mathbf{1}_{K \operatorname{diag}\{p,1\} K} \quad \text{and} \quad X_2 \mapsto \mathbf{1}_{K \operatorname{diag}\{p,p\} K}$$

is a $\mathbf{C}$-algebra isomorphism. In particular $\mathcal{H}(\mathrm{GL}_2(\mathbf{Q}_p), K)$ is a commutative finite-type $\mathbf{C}$-algebra and hence every simple $\mathcal{H}(\mathrm{GL}_2(\mathbf{Q}_p), K)$-module is a 1-dimensional $\mathbf{C}$-vector space.

Proof Sketch. $\mathcal{H}(\mathrm{GL}_2(\mathbf{Q}_p), K)$ has a $\mathbf{C}$-basis given by $\{\mathbf{1}_{KgK} \colon g \in \mathrm{GL}_2(\mathbf{Q}_p)\}$ and $K \backslash \mathrm{GL}_2(\mathbf{Q}_p)/K$ is in bijection with $\{(\lambda_1, \lambda_2) \in \mathbf{Z}^2 \colon \lambda_1 \geq \lambda_2\}$ by

$$(\lambda_1, \lambda_2) \mapsto K \begin{pmatrix} p^{\lambda_1} & \\ & p^{\lambda_2} \end{pmatrix}.$$

$\mathbf{C}[X_1, X_2^{\pm 1}]$ has a $\mathbf{C}$-basis $\left\{X_1^a X_2^b\right\}$ for $(a,b) \in \mathbf{Z}_{\geq 0} \times \mathbf{Z}$; we only need to show that

$$X_1^a X_2^b \mapsto \mathbf{1}_{K \operatorname{diag}\{p,1\} K}^a \mathbf{1}_{K \operatorname{diag}\{p,p\} K}^b$$

is an isomorphism of $\mathbf{C}$-vector spaces. In order to do so, we compute

$$\mathbf{1}_{K \operatorname{diag}\{p,1\} K}^a = \mathbf{1}_{K \operatorname{diag}\{p^a,1\} K} + \text{ linear combinations of } \mathbf{1}_{K \operatorname{diag}\{p^{a'},1\} K} \mathbf{1}_{K \operatorname{diag}\{p^b,p^b\} K}$$

for $a' < a$ and then deduce what we want inductively.

We want to now write the "global" algebra $\mathcal{H}(G(\mathbf{A}_f), K)$, $K = \prod_p K_p$ in terms of the "local" algebras $\mathcal{H}(G(\mathbf{Q}_p), K_p)$.

Definition 5.15. *Let I be a countable index set and $\{W_i\}_{i\in I}$ a set of $\mathbf{C}$-vector spaces. Let $J \subset I$ be finite such that for each $i \notin J$ we are given an element $w_{i,0} \in W_i \setminus \{0\}$. The restricted tensor product of the W_i with respect to the $w_{i,0}$ is*

$$\bigotimes_{i\in I}{}' W_i := \varinjlim_{\substack{J\subset S\subset I \\ \text{finite}}} \left(\bigotimes_{i\in S} W_i\right) \otimes_{\mathbf{C}} \bigotimes \mathbf{C} w_{i,0}.$$

Lemma 5.16. *Let I, J as above and assume that the W_i are $\mathbf{C}$-algebras and that the $w_{i,0} \in W_i$ are idempotent for $i \notin J$. Then $\bigotimes_{i\in I}{}' W_i$ naturally becomes a $\mathbf{C}$-algebra.*

Example 5.17. *The key example is*

$$C_c^\infty(G(\mathbf{A}_f)) = \bigotimes_p{}' C_c^\infty(G(\mathbf{Q}_p))$$

with respect to the elements $e_p = \mathbf{1}_{\mathrm{GL}_2(\mathbf{Z}_p)}$ at the primes p such that $G(\mathbf{Q}_p) \simeq \mathrm{GL}_2(\mathbf{Q}_p)$.

If $K = \prod_p K_p \subset G(\mathbf{A}_f)$ for $K_p \subset G(\mathbf{Q}_p)$ compact open subgroups, then

$$\mathcal{H}(G(\mathbf{A}_f), K) = \bigotimes_p{}' \mathcal{H}(G(\mathbf{Q}_p), K_p)$$

computed with respect to the unit element $1 = e_K$.

$\mathcal{H}(\mathrm{GL}_2(\mathbf{A}_f), K_1(N))$ acts on $\mathcal{A}_0(k, K_0(N), \omega_f) \simeq S_k(\Gamma_0(N), \omega_f)$. We have

$$K_1(N) = \prod_{p\mid N} K_p \times \prod_{p\nmid N} \mathrm{GL}_2(\mathbf{Z}_p)$$

where

$$K_p = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbf{Z}_p) \colon a, d \equiv 1 \bmod N,\ c \equiv 0 \bmod N \right\}.$$

Hence the Hecke algebra decomposes as a restricted tensor product

$$\mathcal{H}(G(\mathbf{A}_f), K_1(N)) = \bigotimes_p{}' \mathcal{H}(\mathrm{GL}_2(\mathbf{Q}_p), K_p).$$

This RHS contains the factors $\mathcal{H}(\mathrm{GL}_2(\mathbf{Q}_l), \mathrm{GL}_2(\mathbf{Z}_l))$ for $l \nmid N$, which by the Satake isomorphism are isomorphic to

$$\mathbf{C}\left[\mathbf{1}_{\mathrm{GL}_2(\mathbf{Z}_l)\,\mathrm{diag}\{l,1\}\,\mathrm{GL}_2(\mathbf{Z}_l)}, \mathbf{1}^{\pm 1}_{\mathrm{GL}_2(\mathbf{Z}_l)\,\mathrm{diag}\{l,l\}\,\mathrm{GL}_2(\mathbf{Z}_l)}\right].$$

Proposition 5.18. *$l^{\frac{k-2}{2}} \mathbf{1}_{\mathrm{GL}_2(\mathbf{Z}_l)\,\mathrm{diag}\{l,1\}\,\mathrm{GL}_2(\mathbf{Z}_l)}$ acts on $S_k(\Gamma_0(N), \omega_f)$ as the Hecke operator T_l!*

Proof. Letting $K = \mathrm{GL}_2(\mathbf{Z}_l)$, note that

$$K \begin{pmatrix} l & \\ & 1 \end{pmatrix} K = \begin{pmatrix} l & \\ & 1 \end{pmatrix} K \sqcup \bigsqcup_{i=0}^{l-1} \begin{pmatrix} 1 & 0 \\ i & l \end{pmatrix} K.$$

Then for a smooth $\mathrm{GL}_2(\mathbf{Q}_l)$-representation π and $v \in \pi^K$, we have

$$\begin{aligned}
\mathbf{1}_{K\,\mathrm{diag}\{l,1\}K} \cdot v &:= \int_G \mathbf{1}_{K\,\mathrm{diag}\{l,1\}K}(x) xv\, dx \\
&= \int_G \mathbf{1}_{\mathrm{diag}\{l,1\}K}(x) xv\, dx + \sum_{i=0}^{l-1} \int_G \mathbf{1}_{\begin{pmatrix} 1 & 0 \\ i & l \end{pmatrix} K}(x) xv\, dx \\
&= \mu(\mathrm{diag}\,\{l,1\}\,K) \begin{pmatrix} l & \\ & 1 \end{pmatrix} v + \sum_{i=0}^{l-1} \mu\left(\begin{pmatrix} 1 & 0 \\ i & l \end{pmatrix} K\right) \begin{pmatrix} 1 & 0 \\ i & l \end{pmatrix} v \\
&= \begin{pmatrix} l & \\ & 1 \end{pmatrix} v + \sum_{i=0}^{l-1} \begin{pmatrix} 1 & 0 \\ i & l \end{pmatrix} v.
\end{aligned}$$

Recall that the Hecke operator T_l is defined by

$$T_l f \colon \tau \mapsto l^{\frac{k-2}{2}} \sum_{\nu=0}^{l} f|_k x_\nu(\tau)$$

where

$$x_\nu = \begin{pmatrix} 1 & 0 \\ \nu & l \end{pmatrix},\ \nu = 0, \ldots, l-1, \quad x_l = \begin{pmatrix} l & 0 \\ 0 & 1 \end{pmatrix}, \quad f|_k x_\nu(\tau) = f(x_\nu \tau) j(x_\nu, \tau)^{-k}.$$

Under the isomorphism $S_k(\Gamma_0(N), \omega_f) \xrightarrow{\sim} \mathcal{A}_0(k, K_0(N), \omega_f)$, $T_l f$ is sent to

$$\varphi_{T_l f} \colon \gamma g_\infty k \mapsto T_l f(g_\infty i) j(g_\infty, i)^{-k}$$

and the RHS is

$$\begin{aligned}
l^{\frac{k-2}{2}} \sum_{\nu=0}^{l} f(x_\nu g_\infty i) j(x_\nu, g_\infty i)^{-k} j(g_\infty, i)^{-k} &= l^{\frac{k-2}{2}} \sum_{\nu=0}^{l} \varphi_f(\gamma x_\nu g_\infty k) \\
&= l^{\frac{k-2}{2}} (\mathbf{1}_{\mathrm{GL}_2(\mathbf{Z}_l)\,\mathrm{diag}\{l,1\}\,\mathrm{GL}_2(\mathbf{Z}_l)} \varphi_f)(\gamma g_\infty k).
\end{aligned}$$

Hence we can see that the action of $l^{\frac{k-2}{2}} \mathbf{1}_{\mathrm{GL}_2(\mathbf{Z}_l)\,\mathrm{diag}\{l,1\}\,\mathrm{GL}_2(\mathbf{Z}_l)}$ on $S_k(\Gamma_0(N), \omega_f)$ is the Hecke operator T_l as claimed.

Let $D/\mathbf{Q}$ be a definite quaternion algebra and G the group scheme associated to D. Let $k \in \mathbf{Z}_{\geq 2}$ and ω_f a Dirichlet character. Note that $\pi_\infty = \mathrm{Sym}^{k-2}\mathbf{C}^2$ produces all the irreducible representations of $G(\mathbf{R}) \subset \mathrm{GL}_2(\mathbf{C})$. We want to compare $\mathcal{A}_0(k, K_0(N), \omega_f)$ with $S_{\pi_\infty}(G(\mathbf{A}_f), K', \omega_f)$. Set $d := \prod_{p \in \mathrm{Ram}(D)} p$. Assume $(d, N) = 1$. Note that

$$K_0(N) = \prod_p K_p$$

where $K_p \subset \mathrm{GL}_2(\mathbf{Q}_p)$ and $K_p = \mathrm{GL}_2(\mathbf{Z}_p)$ if $p \nmid N$. We write

$$K' = \prod_p K'_p$$

where $K'_p := K_p \subset G(\mathbf{Q}_p) = \mathrm{GL}_2(\mathbf{Q}_p)$ if $p \nmid d$ and $K_p = \mathcal{O}^\times_{D \otimes_{\mathbf{Q}} \mathbf{Q}_p}$ for $p \mid d$. Note that if $p \nmid Nd$ then $\mathcal{H}(\mathrm{GL}_2(\mathbf{Q}_p), \mathrm{GL}_2(\mathbf{Z}_p))$ acts on both $\mathcal{A}_0(k, K_0(N), \omega_f)$ and $S_{\pi_\infty}(G(\mathbf{A}_f), K', \omega_f)$.

Theorem 5.19 (Jacquet–Langlands). *In the above situation, if $k > 2$ there exists an embedding of $\mathcal{H}(G(\mathbf{A}_f^{Nd}), \prod_{p \nmid Nd} \mathrm{GL}_2(\mathbf{Z}_p))$-modules*

$$S_{\mathrm{Sym}^{k-2}\mathbf{C}^2}(G(\mathbf{A}_f), K_1(N)) \hookrightarrow S_k(\Gamma(N)) \simeq \mathcal{A}_0(k, K_1(N)).$$

If $k = 2$ then we have an embedding

$$S_{\mathrm{trivial}}(G(\mathbf{A}_f), K_1(N)) / \{\textit{constant functions}\} \hookrightarrow S_2(\Gamma_1(N)).$$

In both cases, the image of this embedding is the space of d-newforms.

Recall the multiplicity one theorem for modular forms tells us that if $f \in S_k(\Gamma_1(N))$ is a newform and $\chi_f \colon \mathbf{C}[T_l \colon l \nmid N] \to \mathbf{C}$ is the corresponding character, then the eigenspace generated by f is one dimensional.

In representation-theoretic terms, we have the non-degenerate $C_c^\infty(\mathrm{GL}_2(\mathbf{A}_f))$-module

$$\mathcal{A}_0(k, \omega_f) = \varinjlim_K \mathcal{A}_0(k, K, \omega_f),$$

and this decomposes as a countable direct sum of irreducible smooth representations π_f of $\mathrm{GL}_2(\mathbf{A}_f)$, each occurring with multiplicity 1. Indeed, let $f \in \mathcal{A}_0(k, K_0(N), \omega_f) \simeq S_k(\Gamma_0(N), \omega_f) \subset S_k(\Gamma_1(N))$ be a newform. Then the $\mathcal{H}(G(\mathbf{A}_f), K_1(N))$-submodule spanned by f is a one dimensional $\mathbf{C}$-vector space since

$$T_l(hf) = h(T_l f) = h(\chi_f(T_l) f) = \chi_f(T_l) f$$

for all $l \nmid N$.

Using the Jacquet–Langlands theorem we use the same argument to obtain

Theorem 5.20. *Let $D/\mathbf{Q}$ be a definite quaternion algebra and $k \geq 2$. Then $S_{\mathrm{Sym}^{k-2}\mathbf{C}^2}(G_D(\mathbf{A}_f), \omega_f)$ decomposes as a direct sum of irreducible smooth $G_D(\mathbf{A}_f)$-representations, each of which occurs with multiplicity 1.*

As $G(\mathbf{A}_f) = \prod_p{}' G(\mathbf{Q}_p)$ and $C_c^\infty(G(\mathbf{A}_f)) = \bigotimes_p{}' C_c^\infty(G(\mathbf{Q}_p))$, we might hope that every irreducible smooth $G(\mathbf{A}_f)$-representation (resp. simple non-degenerate $C_c^\infty(G(\mathbf{A}_f))$-module) π decomposes as a tensor product of $G(\mathbf{Q}_p)$-representations.

Definition 5.21. *A $C_c^\infty(G(\mathbf{A}_f)) = \bigotimes_p{}' C_c^\infty(G(\mathbf{Q}_p))$-module π is called factorizable if*

$$\pi = \bigotimes_p{}' \pi_p$$

for $C_c^\infty(G(\mathbf{Q}_p))$-modules π_p such that $e_{K_p}\pi_p = \pi_p^{K_p}$ is one-dimensional for almost all p.

Theorem 5.22. *Every non-degenerate simple $C_c^\infty(G(\mathbf{A}_f))$-module is factorizable.*

6 *Fairytales*[25] *about Galois representations*

6.1 *The Local Story*

Let $K/\mathbf{Q}_p$ be finite. From local class field theory there is a continuous homomorphism

$$\mathrm{Art}_K\colon K^\times \to \mathrm{Gal}(\bar{K}/K)^{\mathrm{ab}}$$

called the *Artin map* which has dense image. We can make this an isomorphism of topological groups by replacing $\mathrm{Gal}_K^{\mathrm{ab}}$ with the Weil group W_K. This is the preimage of $\mathrm{Frob}_q^{\mathbf{Z}} \subset \mathrm{Gal}(\bar{\kappa}/\kappa)$ where $\mathbf{F}_q = \kappa := \mathcal{O}_K/\pi_K$ in

$$1 \longrightarrow I_K \longrightarrow \mathrm{Gal}(\bar{K}/K) \longrightarrow \mathrm{Gal}(\bar{\kappa}/\kappa) \longrightarrow 1.$$

Here to make the isomorphism one of topological groups we modify the topology on W_K so that I_K is an open subgroup (this corresponds to giving the quotient $\mathbf{Z}$ the discrete topology instead of the subspace topology from $\hat{\mathbf{Z}}$). Given this construction, Art_K induces an isomorphism of topological groups $K^\times \simeq W_K^{\mathrm{ab}}$. From this we get that (for any algebraically closed characteristic 0 coefficient field C) there is a bijection

$$\left\{\underset{\chi\colon K^\times \to C^\times}{\text{continuous/smooth characters}}\right\}_{/\sim} \iff \left\{\underset{W_K^{\mathrm{ab}} \to C^\times}{\text{continuous/smooth characters}}\right\}_{/\sim}.$$

Since $C^\times$ is commutative, the characters of W_K are the same as characters of W_K^{ab}. Since C is algebraically closed the characters of $K^\times$ are the irreducible representations of $\mathrm{GL}_1(K)$ and characters of W_K are 1-dimensional representations of W_K. Note that any character of $\mathrm{Gal}(\bar{K}/K)$ is determined by its values on W_K, but not all characters of W_K extend to Gal_K. Hence we can also formulate a correspondence between irreducible representations of $\mathrm{GL}_1(K)$ and 1-dimensional representations of Gal_K; however, this would not be a bijection.

We can upgrade the previous correspondence to the following:

Theorem 6.1 (Local Langlands Correspondence for GL_n/K; Harris–Taylor)**.** *There is a natural bijection*

$$\left\{\begin{matrix}\text{smooth, irreducible } C\text{-}\\ \text{representations of } \mathrm{GL}_n(K)\end{matrix}\right\}_{/\sim} \iff \left\{\begin{matrix}n\text{-dimensional Frobenius semi-simple}\\ \text{Weil–Deligne } C\text{-representations } (\rho, N)\end{matrix}\right\}_{/\sim}$$

Where the RHS is representations $\rho\colon W_K \to \mathrm{GL}(V)$ such that $\rho|_{I_K}$ factors through a finite quotient and $\rho(g)$ is semisimple for all $g \notin I_K$. Moreover $N \in \mathrm{End}_C(V)$ must satisfy $\rho(g)N\rho(g)^{-1} = q^{\|g\|}N$ where $q^{\|g\|}$ is the image of g in $W_K \twoheadrightarrow q^{\mathbf{Z}}$.

Note that the RHS here is precisely the RHS in the previous formulation from local class field theory i.e. in one dimension the Frobenius semi-simple Weil–Deligne representations are precisely the one dimensional representations of W_K.

[25] The word 'fairytales' here means that we will largely be asserting things without attempting to prove them. This should not be construed as meaning the things described in this section are fictitious.

Weil–Deligne representations arise naturally when studying l-adic representations of p-adic fields. Let $K/\mathbf{Q}_p$ be finite and $l \neq p$ a prime. We have the short exact sequences

$$1 \longrightarrow I_K \longrightarrow \mathrm{Gal}_K \longrightarrow \mathrm{Gal}_\kappa \longrightarrow 1$$

$$1 \longrightarrow P_K \longrightarrow I_K \longrightarrow \prod_{q \neq p} \mathbf{Z}_q \longrightarrow 1$$

where P_K is the wild inertia subgroup and we have

$$\mathrm{Gal}_K / P_K \simeq \left(\prod_{q \neq p} \mathbf{Z}_q \right) \rtimes \mathrm{Gal}_\kappa$$

where the Frobenius acts on the product of $\mathbf{Z}_q$ via the cyclotomic character. Let $r\colon \mathrm{Gal}_K \to \mathrm{GL}_n(\mathbf{Q}_l)$ be a continuous representation. As P_K is a pro-p group, it follows that $r|_{P_K}$ must factor through a finite quotient. Similarly,

$$\ker(I_K \twoheadrightarrow \prod_{q \neq p} \mathbf{Z}_q \twoheadrightarrow \mathbf{Z}_l)$$

must be finite index. What do continuous representations $\mathbf{Z}_l \to \mathrm{GL}_n(\mathbf{Q}_l)$ look like? A continuous representation $\rho\colon \mathbf{Z}_l \to \mathrm{GL}_n(\mathbf{Q}_l)$ is potentially unipotent i.e. $\rho|_{l^m \mathbf{Z}_l}\colon S \mapsto u^S = \exp(SN)$ for some $m \gg 0$, where $u = \exp(N)$ for some nilpotent $N \in M_n(\mathbf{Q}_l)$. If we change ρ to $\rho \cdot \exp(- \cdot N)$ we get a smooth representation! Now putting things together, given a continuous representation $r\colon \mathrm{Gal}_K \to \mathrm{GL}_n(\mathbf{Q}_l)$ we can construct a smooth representation $\rho\colon W_K \to \mathrm{GL}_n(\mathbf{Q}_l)$ together with a nilpotent $N \in M_n(\mathbf{Q}_l)$ such that $\rho(g)N\rho(g)^{-1} = q^{\|g\|}N$. This construction constitutes a fully faithful embedding of l-adic Galois representations into Weil–Deligne representations.

Theorem 6.2 (Local Jacquet–Langlands Correspondence)**.** *Let D/K be a central division algebra of dimension n^2. The smooth irreducible representations of $D^\times$ embeds into the smooth, irreducible C-representations of $\mathrm{GL}_n(K)$. Under the local Langlands Correspondence, these representations of $D^\times$ correspond to the indecomposable n-dimensional Frobenius semi-simple Weil–Deligne representations. Consequently for any two central division algebras D_1, D_2 of the same dimension, there is a canonical bijection between the smooth irreducible representations of $D_1^\times$ and $D_2^\times$.*

6.2 The Global Story

Let F be a number field; given our "understanding" of Galois representations in the local setting, can we understand representations of Gal_F? Yes to some extent and no to some bigger extent. Given a completion F_v of F we can identify Gal_{F_v} as a subgroup of Gal_F by choosing embeddings

$$\begin{array}{ccc} F & \hookrightarrow & F_v \\ \downarrow & & \downarrow \\ \bar{F} & \hookrightarrow & \overline{F_v}; \end{array}$$

we then have

Theorem 6.3 (Čebotarev Density). *Let $S \subset V_{F,f}$ be finite. For each $v \in V_{F,f} \setminus S$ let $g_v \in \mathrm{Gal}_F$ be the image of a lift $\widetilde{\mathrm{Frob}_v} \in \mathrm{Gal}_{F_v}$ of Frob_v. Then the subgroup of Gal_F generated by $\left\{g_v \colon v \in V_{F,f} \setminus S\right\}$ is dense in Gal_F.*

In light of this theorem, given a continuous representation $r \colon \mathrm{Gal}_F \to \mathrm{GL}_n(C)$, r is completely determined by $r|_{\mathrm{Gal}_{F_v}}$ for almost all v. However, it is mysterious whether given local representations $r_v \colon \mathrm{Gal}_{F_v} \to \mathrm{GL}_n(C)$ for almost all v, there is a representation $r \colon \mathrm{Gal}_F \to \mathrm{GL}_n(C)$ such that $r|_{\mathrm{Gal}_{F_v}} = r_v$. A related mystery is: given a representation r of Gal_F, how is $r|_{\mathrm{Gal}_{F_w}}$ determined by $r|_{\mathrm{Gal}_{F_v}}$ for $v \neq w$?

Let $F = \mathbf{Q}$ and $G/\mathbf{Q}$ be the group attached to a (indefinite) quaternion algebra (e.g. $G = \mathrm{GL}_2$). For p a prime a (nice) representation $r_p \colon \mathrm{Gal}_{\mathbf{Q}_p} \to \mathrm{GL}_2(\overline{\mathbf{Q}_l})$ corresponds under local Langlands to a Weil–Deligne representation on a 2-dimensional $\overline{\mathbf{Q}_l}$-vector space. Such representations also correspond to irreducible smooth representations π_p of $G(\mathbf{Q}_p)$ (for almost all p). Ideally we could say on one hand that there exists a global Galois representation $r \colon \mathrm{Gal}_{\mathbf{Q}} \to \mathrm{GL}_2(\overline{\mathbf{Q}_l})$ such that $r|_{\mathrm{Gal}_{\mathbf{Q}_p}} = r_p$ for (almost) all p. On the other this would say that the restricted tensor product $\bigotimes_p{}' \pi_p$ embeds into $L^2(G(\mathbf{Q})\backslash G(\mathbf{A}))$. A correspondence between these two worlds would constitute a "global Langlands Correspondence."

The easier (but not easy!) direction of this (still conjectural) correspondence is automorphic $\implies$ Galois: Let

$$\pi = \pi_\infty \otimes \pi_f \subset L^2(G(\mathbf{Q})\backslash G(\mathbf{A}))$$

e.g. π is associated to a weight k cuspidal modular form. We know that $\pi_f = \bigotimes_p{}' \pi_p$ and $\pi_p^{\mathrm{GL}_2(\mathbf{Z}_p)} \neq 0$ is 1-dimensional for almost all p. Moreover the Hecke algebra $\mathcal{H}(\mathrm{GL}_2(\mathbf{Q}_p), \mathrm{GL}_2(\mathbf{Z}_p))$ acts via the character $\chi_p \colon \mathbf{C}[X_1, X_2^{\pm 1}] \to \mathbf{C}$.

As an aside: if we have π_p then the corresponding local Weil–Deligne representation (ρ_p, N_p) is given by $N_p = 0$ and $\rho_p|_{I_{\mathbf{Q}_p}} = 0$ and is completely determined by the characteristic polynomial of Frobenius $\mathrm{Char}(\rho_p(\mathrm{Frob}_p)) = T^2 \pm \chi_p(X_1)T + \chi_p(X_2)$.

Theorem 6.4 (Deligne). *There exists a number field $L/\mathbf{Q}$ such that the χ_p are defined over L i.e. $\chi_p(X_i) \in L$. Fixing an embedding $L \hookrightarrow \overline{\mathbf{Q}_l}$ for some l there exists a continuous representation $r_\pi \colon \mathrm{Gal}_{\mathbf{Q}} \to \mathrm{GL}_2(\overline{\mathbf{Q}_l})$ such that for almost all $p \neq l$, $\pi_p^{\mathrm{GL}_2(\mathbf{Z}_p)} \neq 0$ and $r_\pi|_{\mathrm{Gal}_{\mathbf{Q}_p}}$ is unramified (trivial on $I_{\mathbf{Q}_p}$) and $\mathrm{Tr}(r_\pi(\mathrm{Frob}_p)) = \chi_p(X_1)$.*

In the other direction, we have partial results such as the following:

Theorem 6.5 (Modularity theorem; Wiles, Taylor–Wiles, Breuil–Conrad–Diamond–Taylor). *For $E/\mathbf{Q}$ an elliptic curve and l a rational prime, define the l-adic Tate module*

$$T_l E := \varprojlim_n E[l^n] \simeq \mathbf{Z}_l^2.$$

Let $\rho_{E,l} \colon \mathrm{Gal}_{\mathbf{Q}} \to \mathrm{GL}_2(\mathbf{Z}_l)$ denote the associated l-adic Galois representation via the action of $\mathrm{Gal}_{\mathbf{Q}}$ on E. Then there exists a weight 2 newform of level N with associated character $\chi_f \colon \mathbf{Z}[T_p \colon p \nmid lN] \to \mathbf{Z}_l$ such that $\mathrm{Tr}(\rho_{E,l}(\mathrm{Frob}_p)) = \chi_f(T_p)$ for all $p \nmid lN$.

Theorem 6.6 (Serre's modularity conjecture; Khare–Wintenberger)**.** *Let κ be a finite field and let $\rho\colon \mathrm{Gal}_{\mathbf{Q}} \to \mathrm{GL}_2(\kappa)$ be an absolutely irreducible Galois representation which is odd i.e. the image of complex conjugation has trace 0. Then there exists a modular form of (predictable) weight k and level N with associated character $\chi_f\colon \mathbf{Z}[T_p\colon p \nmid N] \to \kappa$ such that $\mathrm{Tr}(\rho_{E,l}(\mathrm{Frob}_p)) = \chi_f(T_p)$ for all $p \nmid N$.*

Despite these monumental results, however, even for GL_2 the global Langlands correspondence is unresolved in full!

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